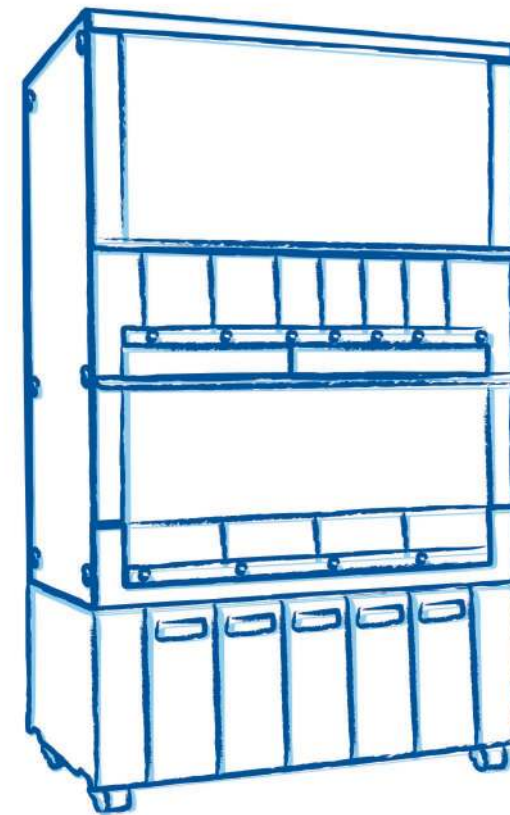


erytra®

Instructions for Use



Erytra

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Instructions for Use

GRIFOLS

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11/2022

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Erytra

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PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING TO WORK WITH THE ERYTRA ANALYZER

The relevant Safety information for this equipment can be found in Section 2.

The following symbols are used in these instructions:



WARNING: Indicates a hazardous situation for persons (including patients), which has protective devices or systems. Nonetheless, it is advisable to notify the Supervisor or the Operator.



CAUTION: Indicates a hazardous situation that could result in material damages to the equipment or other property. This situation can be avoided by following these instructions or using other types of protection.



NOTE: Used for clarifications or additional or emphasized information.

Definitions

Supervisor:	Person or group responsible for using and maintaining the equipment and for adequate training of the Operators.
Operator:	Person who uses the equipment for the purpose for which it was designed. The Operator must have laboratory technician-level training and knowledge of immunohematology. He or she must also be familiar with the tests performed by the Erytra analyzer, and must have completed a training course in order to use the analyzer.
Qualified Technician:	Person responsible for the installation, repair and special maintenance of the equipment, who has been specifically trained for this purpose.

All restrictions that refer to the Supervisor shall also be valid for the Operator.

The design and specifications for this equipment are subject to modification without prior notice.

This instrument is protected by international patents affecting the entire unit and its parts.

This document is available in several languages. The translations have been done using the master document in **English**. In the event of any doubts or disputes, the text of the master document in **English** shall prevail.

Changes with respect to the previous revision are shaded in grey. Deletion of text is indicated as **Δ**.

Revision History

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
Part # 3044446 SEPT16	• The User's Manual has been reorganized in order to have a clearer structure. • The information included in Section 2. Analyzer Specifications has been updated. • The packaging and product labels (Section 2) have been modified, according to the new regulations. • It has been added information about incubators performance in Section 4.2.2.4. • It has been added information about centrifuges performance in Section 4.2.2.5. • Section 4.2.4 has been updated with information about manual opening of lower level containers. • Samples racks and Reagents racks have been changed to Samples holders and Reagents holders in the User's Manual text. Table 3 of Section 4.3 has been also updated. • It has been added information about OS installation in Section 4.2.4. • Software version has been updated in Section 6. • It has been added information about new Manual Prime icon in Sections 6.5, 11.1 and 11.2.4. • A note and a warning regarding the use of ISBT128 barcode have been included in Section 7.3.1.1. • A note regarding high titre serum samples has been included in Section 7.3.2.1. • It has been added information about new Query by Sample performance in Section 7.3.2.1. • The User's Manual has been modified to change distilled water for purified water. • It has been added information about Autoexportation and Autovalidation option in Section 7.6.1. • Information about Overincubation incidence has been updated in Section 7.6.3. • It has been added information about Autoprint option in Section 7.6.7. • Information about new QC options has been included in Sections 9.1 and 9.3. • Several images have been modified to included some language error changes. • It has been modified information about surface decontamination procedure in Section 11.8.5. • DG Wash has been replaced for DG Clean in the User's Manual text.

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
Part # 3052906 JUNE18	• A warning about disconnecting the instrument from main has been added in Section 4.1.1. • Information about labelling has been added in Section 2.2. • Safety information has been updated and included in Sections 2.1 and 2.4. • A note about the effect of bubbles in the vials during the dispensation of reagents has been added in Section 7.5.1. • Information about fluidic actions has been updated in Sections 11.1 and 11.2.4. • Table 1 has been updated in Section 3.1. • Table 2 has been updated in Section 3.2. • Information about acoustic alarm has been added in Sections 7.5, 7.5.1, 7.5.2, 7.5.3, 7.5.5, 7.5.6, 7.5.7. • Information about sample status viewing has been updated in Section 7.3.1.4. • Information about programming samples has been updated in Section 7.3.2.1. • Information about results validation has been updated in Section 7.6.1. • User roles permissions are updated in Section 10.1. • Ethyl alcohol is replaced for Denatured alcohol in the User's Manual text. • Isopropyl alcohol 50% is added in the information Section 11.7.2. • Diagnostic program information updated in Section 12. • Wash solution is replaced for System solution in the User's Manual text. • Some images are updated in Section 7.6. • Probe replacement information has been updated in Section 11.10. • Information about Cybersecurity and Data Protection has been added in the new Section 15. • Some title sections are renamed in Section 1. • Incident message about clots has been updated in the Section 17. • Software version has been updated in Section 6.

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
Part # 0024783/1.0-GB MAY19	• Update the date and code of this Instructions for Use. • Typology is reviewed in all the document. • Decontamination container term is unified in all the document. A note about decontamination container is added in Section 11.8.2. • Information about LIS and remote connection is updated in Section 3.3. • Update the Table 2 in Section 3.2. • Update the information about doors opening in Sections 4.1.4 and 17.2. • Update information about special buttons in Section 7.6.2. • Update information about installation requirements in Section 5.2. • Update a note about software version in Section 6. • Update information in Section 1.2. • Update of the Table 3 in Section 4.3. • Others tab is replaced by Maintenance tab in all the document. • Update information about over incubation time in Section 7.6.3.2. Information of Sections 5.2.4, 5.2.5, 7.3.2.1, 7.6.9, 7.6.10, 7.6.11, 10.1, 11.2, 11.2.5, 11.2.7, 11.3, 11.4, 11.5 and 15.2 is updated according to General Data Protection Regulation. • A warning about cleaner solution use is added in Section 7.5.1. • Information is updated in Sections 6.3, 7.5 and 7.5.1.
Part # 0024783/2.0-GB APRIL20	• Update the date and code of this Instructions for Use. • Typology is reviewed in all the document. • A typo error has been corrected in Table 1 in Section 3.1. • Replace soap for mild detergent in all the document. • Image of new lot QC warning has been added in Section 9.3. • Troubleshooting information in Section 17.1.2 is updated. • Information in Section 11.8.6 has been updated. • Information about user configuration overview has been updated in Section 10. • Diagnostic information has been updated in Sections 10.1 and 12. • Privacy information has been updated in Section 15.2. • Decontamination Certificate information has been updated in Section 18.1. • Information about closing the drawers has been added in Sections 7.3.1, 7.3.2, 7.5.1 and 7.5.3.

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
Part # 0024783/3.0-GB MAR21	• Date and code of this Instructions for Use has been updated. • A revision history of changes has been added and located at the beginning of this document. • Information about computer operating system of the instrument has been modified in Section 5.2.4. • Information about the web browser used for remote connection has been modified at Sections 3.3 and 5.2.5. • The pediatric samples holder reference code has been updated in Table 3 at Section 4.3. • The samples holder for tubes > 16 mm reference code has been added in Table 3 at Section 4.3. • The blue labels for pediatric holder reference code have been added in Table 3 at Section 4.3. • Reference codes of external handheld barcode reader and its holder have been added in Table 3 at Section 4.3. • "Accessories" has been replaced by "Complementary material" in all the document. • Pediatric samples holder image of Figure 13 has been updated at Section 7.3.1.3. • Information about sample tubes with a diameter slightly larger than 16 mm has been added at Section 7.3.1.3. • Information about external handheld barcode reader and its holder has been added at Section 4.3. • "Grossly" has been added as a nuance of haemolysed samples in a warning of Section 7.3.1.2. • A note referring to the software version to which these Instructions for Use apply has been removed from Section 6. • Information about a validation message and its description has been removed from Table 9 at Section 17.1.1. • A warning about expired reagents has been added at Section 7.5.1. • Labelling information (about CE and UKCA mark) and packaging and product labels have been updated at Section 2.2. • For professional use only has been added at Section 1. • A note about incident reporting has been added at Section 2.1. • UKCA mark has been added at the beginning of the document. • Performance Studies information has been added at the new Section 19. • Information about the cards holder lid has been removed in all document. • Result "0" has been replaced by "-" at Table 4 at Section 7.6.2. • A warning about electrostatic discharges has been added at Section 2.1. • "Range" has been removed from pollution degree field at Table 1 at Section 3.1.

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
Part # 0024783/4.0-GB OCT21 Part # 0024783/5.0-GB NOV22	- Section 3.4 Regulations has been removed from the document. - BeyondTrust has been added in remote connection information at Sections 3.2, 3.3 and 5.2.5. "D:\Backup folder" has been replaced by "E:\Backup folder" at Section 11.3. - A troubleshooting message about missing reagent incubation step has been removed from Section 7.6.3.2. - TeamViewer has been removed in remote connection information at Sections 3.2, 3.3, 5.2.5 and 12. - A note about continuous use of reagent vials placed on the instrument has been added in Section 7.5.1. - Date and code of this Instructions for Use has been updated. - The explanation of how changes are marked throughout the manual has been added at the beginning. - Reference to FreeImage has been removed from this document. - The symbol ® has been removed through all the document. "Well" has been replaced by "Microtube" throughout the Instructions for Use. - Grammatical mistakes are corrected throughout the Instructions for Use. "BeyondTrust" has been replaced by "Secure Remote Connectivity" through this document. - A warning has been created in Section 2.1. - A note about the expected useful life of Erytra has been added in Section 2.1. - Sample container and Remote connection specifications have been reworded in Table 2 in Section 3.2. - A warning about manual cards removal has been added in Section 4.2.2.3. - Table 3 has been updated in Section 4.3. - T-connector tool has been removed as a component from Sections 4.3 and 11.8.1. - Sections 5.2.1 and 5.2.3 have been described in more detail. - Section 5.2.2 has been created. - Figure 8 and its description has been updated in Section 6. - Description of the About screen has been created in Section 6.4. - Rewording of Section 7.3.1.1. - A note related to pediatric sample holders has been added in Section 7.3.1.1. - A caution has been added in Section 7.3.1.3. - A note establishing the maximum sample titer to avoid carry-over effects has been updated in Section 7.3.1.3. - Figure 17 and its description has been updated.

PART NUMBER/ REVISION DATE	DESCRIPTION OF CHANGES
	- Icons have been included in Section 7.3.2. - On-board stability tracking has been explained in Section 7.3.2. - Cautions and notes about the tracking of the new on-board stability functionality have been added in Section 7.3.2. - A note remembering to remove and store the consumables has been added in Sections 7.3.2 and 7.7. - A note about continuous use of reagent red blood cells vials placed on the instrument has been updated in Sections 7.3.2 and 7.5.1. - Information regarding missing components has been widened in Section 7.5. - A caution about the on-board expiration has been added in Section 7.5. - On-board stability tracking has been mentioned at Section 7.5.1. - A caution about artefacts or drops found has been added in Section 7.5.2. - A note about antibody titer threshold has been added in Section 7.6.1. - Traceability information has been restructured and updated with new information in Section 7.6.9. - Password limitations has been described at Section 10.3. "Daily Maintenance" has been changed to "Basic Maintenance" in Section 11. - Information regarding the use of isopropyl alcohol pads has been added in Section 11.7.2. - Decontaminating the Fluidic System has been restructured and updated with new information at Section 11.8.2. - A warning related to Diagnostic program has been added in Section 12.

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1 Introduction

THIS ANALYZER IS INTENDED FOR *IN VITRO* DIAGNOSTIC USE AND PROFESSIONAL USE ONLY.

DIAGNOSTIC GRIFOLS, S.A., HAS VALIDATED TECHNIQUES/PROFILES WITH ITS REAGENTS. THESE ARE DISPLAYED LOCKED AND CAN NOT BE MODIFIED.

DIAGNOSTIC GRIFOLS, S.A., HAS VALIDATED THE PERFORMANCE OF THIS EQUIPMENT ACCORDING TO THE STATED INTENDED USE, USING ITS OWN REAGENTS AND ASSOCIATED TESTS. DIAGNOSTIC GRIFOLS, S.A. DOES NOT ASSUME ANY RESPONSIBILITY FOR THE CORRECT PERFORMANCE, IF IT IS USED WITH REAGENTS MANUFACTURED BY OTHERS.

These usage instructions, addressed to Operators of the Erytra analyzer, contain all information **necessary** to work safely and appropriately with the instrument.

Read all information contained herein carefully before starting to work with the equipment, particularly the topics on maintenance, the suggestions for proper use, and the information on contaminating products.

When in doubt, contact the manufacturer or the corresponding distributor before starting any operations.

This manual must be accessible at all times to the persons working with the instrument.

1.1 Intended Use

Erytra is a fully automatic high-performance analyzer designed to automate *in vitro* immunohaematological tests on human blood, using Grifols gel card technology, including blood group and antigen typing, antibody screening, compatibility tests and direct Coombs test.

1.2 Principle of Operation

The Erytra analyzer is a modular instrument that automates the gel immunohematology tests indicated below:

- ABO red cell and serum grouping.
- Rh(D) typing.
- Antigen typing.
- Antibody detection.
- Antibody identification.

- Antibody titration.
- Direct antiglobulin test.
- Compatibility tests (crossmatching).

The tests listed are basically applied at blood banks to study compatibility between donors and recipients for blood transfusions, and to diagnose mother-fetus blood type incompatibilities.

The basis of all tests performed using this equipment is the reaction between the blood antigen (sample or reagent) and the antibody (present in the serum/plasma or reagent). This causes red blood cell agglutination, which is viewed after filtering through the gel microtubes on the Grifols gel cards.

1.3 Purpose, Limitations and Operator Training

1.3.1 Purpose

The purpose of the Erytra analyzer is to automate all necessary operations and procedures while using gel immunohematology tests in order to:

1. Absorb the entire workload involved in creating test profiles, optimizing profile implementation in the shortest time and with the most accurate results possible.
2. Increase process safety, eliminating possible identification and transcription errors.
3. Increase analytical reliability by standardizing all steps, eliminating possible handling and processing errors, and interpreting the results with objective criteria.
4. Decrease the danger of contamination for Operators by reducing Operators interaction with the samples and reagents during the analytical process. Operator intervention is limited to loading and unloading the analyzer.

In addition, the Erytra analyzer adapts to the needs and *modus operandi* of the different workflows in immunohematology laboratories, donation centres, transfusion units and clinical testing laboratories, as well as different work rhythms: Routine, emergency, and the flow of samples produced over different shifts.

1.3.2 Limitations

- The Erytra analyzer is only designed to automate the processing and reading of the immunohematology tests described in Section 1.1, and cannot perform any other type of tests except those for which it was intended.
- The Erytra analyzer can only use microtube cards with Grifols gel and reagents authorized by Diagnostic Grifols, S.A.
- The analyzer can only work with profiles containing tests validated by Diagnostic Grifols, S.A.
- The Erytra analyzer can only use complementary material designed and provided by Diagnostic Grifols, S.A.

- The positive identification of samples and reagents is dependent on the tubes of samples, diluents and reagents being identified by barcodes. To ensure utmost security in the identification of samples, the use of barcodes with checksums is recommended.
- Result processing includes a review and validation step (see Section 7.6). Special attention must be paid to any result containing a warning (processing incident, special result, or doubtful or discrepant reactions), which must be reviewed individually by the Operator.



CAUTION: The use of the Erytra analyzer for any purpose other than those specified by the manufacturer shall automatically invalidate any type of warranty.

1.3.3 Training

Use of the Erytra analyzer is reserved to duly skilled personnel, who have received specific training on the use of this product. The Operator must also have Laboratory Technician-level training and knowledge of immunohematology.

1.4 Basic Operating Principle

To fulfil the purpose for which it was designed, the Erytra analyzer automatically performs the steps described in the procedures for most gel immunohematology tests:

- Positive sample identification. Identification is performed in a way that ensures that the results will be correlated with the barcodes used by the sample at all times.
- Positive reagent identification. Reagents are understood to be: Reagent diluents, antisera and red blood cells.
- Positive card identification.
- System for transporting cards between the different modules in the unit.
- Sample dilution.
- Dispensation of serum and/or plasma in the microtubes on the cards.
- Dispensation of sample dilution in  the microtubes on the cards.
- Dispensation of reagents in the microtubes on the cards.
- Incubation of the cards.
- Centrifugation of the cards.
- Reading and semi-quantification of the reaction grade in the card microtubes.
- Interpretation of the results.

The data analyzed can be stored, displayed, validated and printed. In addition, the analyzer provides integrated features such as STAT analysis and a **Quality Control** module.

1.4.1 General Description of Operation

The analysis begins after putting the sample into the analyzer and the **Worksheet** is configured. To do this, the sample identification and test parameter information must be entered.

1.4.1.1 Programming the Workload

There are two ways of programming the workload on the analyzer:

- Sample identification and automatic request petition allocation through a connection to the laboratory's LIS. The analyzer executes the petitions according to the sample identification numbers read by the barcode reader.
- Sample identification and manual petition allocation. The petition information is entered manually on the **Worksheet** in the Erytra analyzer software.

1.4.1.2 Test Execution

Once the samples have been introduced in the analyzer and the tests/profiles have been assigned, the dilution and sample and reagent dispensation steps are carried out in the corresponding microtubes on the Grifols gel cards, in accordance with the procedure described for each test. If necessary, the cards are incubated and finally centrifuged. The cards are transported to the reader, where a camera takes photographs that will later be processed using a vision algorithm in order to quantify reaction grade. The cards used for the analysis are disposed of automatically or stored in the Service Racks for later review.

1.4.1.3 Mathematical Processing of the Basic Reading Values

The analyzer has a reading algorithm that applies a particular mathematical process to the digitalization of the image in order to obtain parameters that characterize it and make it possible to quantify the reaction grade.

1.4.1.4 Interpretation

In addition, the analyzer software is equipped with interpretation matrices for each test. These combine the different results from quantifying the reaction grade in the microtubes for the test, in order to provide an analytical interpretation of the results.

The results are automatically updated on the **Results** screen.

1.4.1.5 Review and Validation of the Results

The Erytra analyzer software provides a detailed analysis of the results and makes it possible to display the images for each test, the reaction grades and the interpretations.

In addition, the software provides information at all times on the status of the quality policy configured for each test in the analyzer, and the traceability of the reagents used.

The acceptance of a result during the validation step must consider, in addition to the result itself, any incident deriving from the condition of the sample, the analyzer, the result itself, the quality policy, and any additional clinical information. Such incidents require acceptance prior to the validation of the result.

Validation of the results by the Operator implies acceptance, and can be done individually, by sample, by test, by profile or in multiple ways, as long as this does not produce results with incidents or special results.

1.4.1.6 Managing and Printing Results

Independently from the connection to the LIS, the Erytra analyzer software is equipped with a **Database** where all results are stored, including primary images, and are accessible to the Operator.

The Erytra analyzer software provides different reports, which can be selected by the Operator.

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2 Safety Information

2.1 Important Safety Instructions

The following instructions have direct implications for equipment safety:



WARNING: Inappropriate use of this equipment can damage the protection that it provides.



WARNING: To avoid the risk of electric shock, this equipment must only be connected to an electrical network equipped with protective grounding.



WARNING: In order to disconnect the instrument from mains, unplug the power cord either from the power outlet or the connector located in the rear part of the analyzer.



WARNING: Some substances used with this equipment represent a chemical or biological risk. Established standards for laboratory work and appropriate safety precautions, such as the use of gloves and other means of protection, must be followed when managing the analyzer. This can include the following circumstances:

- When handling samples and reagents.
- When working with liquids in and around the equipment.
- When handling waste and waste containers. They may contain biohazardous waste.
- When performing maintenance activities.



WARNING: All areas of the instrument must be considered potentially biohazardous and handled with the appropriate cares as for the laboratory's Standard Operating Procedures (SOPs).



WARNING: According to EN61010-2-020, the centrifuge has not been designed as a biosafe system. Thus it should not be used for centrifuging hazardous materials which are known to be contaminated with toxic, radioactive or pathogenic micro-organisms.



WARNING: If liquid is spilled inside the unit, it must be unplugged from the electrical outlet, cleaned and decontaminated. Under no circumstances may it be disassembled by personnel other than Qualified Technicians, and the equipment must not be plugged in again until it is checked by a Qualified Technician.



WARNING: This equipment may only be used by the Supervisor, Operators or Qualified Technicians.



WARNING: The electromagnetic compatibility and electrical safety tests have been conducted using the cable supplied by the manufacturer. However, if you must necessarily use another cable, make sure that it meets the following specifications:

Type of cable: Flexible cable with PVC insulation and coating and 3 conductors (neutral, phase and ground).

Length: 2 m.

Conductor cross-section: 1 mm² or greater.

Type of connector: For IEC outlets. Connector for class I equipment.

Type of plug: The type of plug used in the country where the equipment is installed, with a ground connection.

The connector, cable and plug must meet the electrical safety requirements applicable in your country.



WARNING: All basic safety precautions must always be taken, including those indicated in Section 2.4. In addition, please read the installation rules specified in Section 5.2 before installing the equipment.



WARNING: The use of non-specified complementary material or cables can cause an increase in electromagnetic emissions and/or a reduction in the immunity of the instrument.



WARNING: This equipment may experience interferences from adjacent equipment, or other units located nearby. Observe the unit to verify that it operates properly with the configuration used normally.



WARNING: In extremely dry environments and floors covered of synthetic material, high level electrostatic discharge can occur. Although not demonstrated, discharges of high levels may damage the equipment but it is improbable that this will have any effect on the results provided by the instrument.



NOTE: Any serious incident occurring in relation to the device in the European Union should be reported to the manufacturer and the competent authority of the Member State in which the user and/or the patient is established.



NOTE: The expected life is defined as the period of time where the failure rate remains constant, if appropriate usage, care and maintenance of the equipment are performed, including:

- Utilization of the instrument according to its intended use.
- Utilization of the instrument under average conditions (as listed in Section 3.1).
- Utilization of the instrument under normal and customary use.
- Operation by trained professional staff.
- Regular cleaning, decontamination and preventive maintenance performed at planned intervals as specified in the Instructions for Use and the Technical Manual.
- Repair and servicing work performed by authorized and trained service personnel, in accordance to relevant instructions.
- Use of original spare parts only.

Once this period has expired the equipment can still be used safely according to the specified intended use but, in addition to regular cleaning, decontamination, care and repair, regular (at least yearly) functional inspections shall be performed as preventive maintenance by a Qualified Technician as the failure rate may progressively increase along the years. Contact your local Grifols service representative to schedule it.

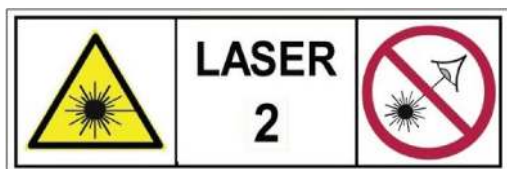
2.2 Equipment Markings

The equipment is marked with the following labels:

- (1) Laser product label.
- (2) Biological risk from probes.
- (3) Risk of injury from probes.
- (4) Moving part.
- (5) Identification of the containers.
- (6) Identification of the positions for sample holders.
- (7) Identification of the positions for reagent holders.
- (8) Identification of the positions for Grifols gel card racks and orientation of the barcode.
- (9) Biological risk from the Service Racks and their contents.
- (10) Biological risk from the connection for emptying into the laboratory drain.
- (11) Equipment packaging.
- (12) Identification label showing the basic technical characteristics of the unit.
- (13) Switch and connection to the electrical network.
- (14) Electrical specifications.
- (15) Electrostatic discharges (ESD).
- (16) RS 232 connection.

(1) Laser product label

Located in the head of the upper and lower robotic arms, this label warns of the danger posed by the barcode laser. The label also contains the visual information required by standard EN 60825-1: *Laser product safety. Part 1: Equipment classification, requirements and safety guide.*



(2) and (3) Labels warning of biological risk and risk of injury from the probes

These are located on the front of the upper robotic arm, where the dispensation probes are located. These probes pose a biological risk because they are in contact with samples, reagents, etc., and a risk of injury due to the sharpness of their ends.

**(4) Moving parts label**

Located on the bottom door and in the centrifuge pots, this label warns of the danger of opening these elements because both provide access to moving parts that could cause injuries to the Operator while in motion.

**(5) Container identification labels**

These labels are located on the front of the unit, on the lids of the system and waste solution containers, and on the sides of the card disposal container. They are also located on the vertical part of the container drawer housings, where they are visible when the drawers are open, and at the base of the drawer housing the card disposal container. Their function is to identify the contents and location of each container.



Identification label for the System Solution A container.



Identification label for the lid of the System Solution A container.



Identification label for the System Solution B container.



Identification label for the lid of the System Solution B container.



Identification label for the waste drawers (for system solution and cards).
This label also warns of the biological risk of the contents.



Identification label for the lid of the waste solution container.



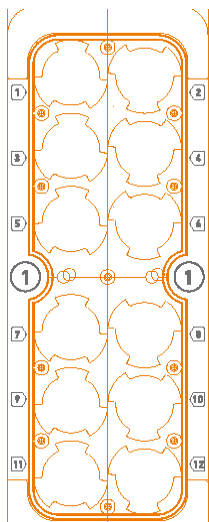
Decontamination container identification label.



Decontamination container lid identification label.

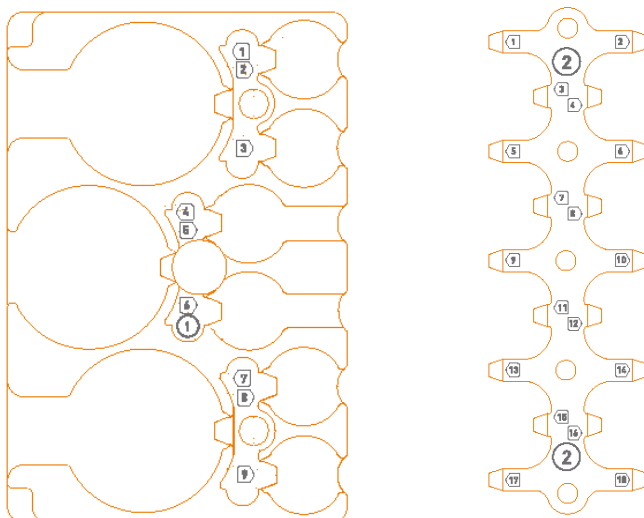
(6) Identification labels for sample holder positions

Located at the base of the corresponding sample drawers, these identify the relative position of the sample tube positions on the holders, and of the holders within the drawer.



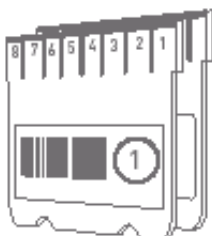
(7) Identification labels for the positions on the reagent holders

Located on the reagent holders themselves, these identify the relative position of the vial positions on the holders, and of the holders within the drawer.



(8) Identification labels for the Grifols gel card rack positions

Located at the base of the corresponding card drawers, these identify the relative position of the card racks within the drawer and indicate how the Operator should orient the barcode.



(9) Label warning of biological risk from the Service Racks and their contents

Located at position 4 of card drawer 3 and 4, it identifies and warns of the biological risk posed by the cards contained in the Service Racks.



(10) Biological risk from the connection for emptying into the laboratory drain

Label located on the bottom left and back of the analyzer, warning of the biological risk of the connection for direct emptying of waste solutions into the laboratory drain.



(11) Equipment packaging label

The following symbols are used:



Serial number



Catalogue number



Permitted temperature range during storage and transport



Permitted relative humidity range during storage and transport



Medical product for *in vitro* diagnostics



The Erytra analyzer meets the requirements of Regulation (EU) 2017/746



The Erytra analyzer meets the requirements of UK Medical Device Regulation 2002

(12) Identification label showing the basic technical characteristics of the unit

Label located on the right wall of the second level, next to the lower robotic arm, containing technical information about the unit (network voltage, frequency, power, fuses), as well as the model, serial number, date of manufacture, name and address of the manufacturer, and compliance information.



The following symbols are used:



Year of manufacture



Serial number



Medical product for *in vitro* diagnostics



The Erytra analyzer meets the requirements of Regulation (EU) 2017/746



The Erytra analyzer meets the requirements of UK Medical Device Regulation 2002



Indicates that the Erytra analyzer is subject to selective waste collection at the end of its useful life, as established by Directive 2012/19/EU on Waste Electrical and Electronic Equipment. See Section 14.



The Unique Device Identifier, which contains the following information:

(01) GTIN number

(11) Manufacturing year

(21) Serial number of the instrument

This information is also included in the adjacent DataMatrix barcode



Catalogue number



Manufacturer



The Erytra analyzer is in accordance with the ETL requirements listed by Intertek

(13) Label for the switch and connection to the electrical network

Located on the bottom right, on the backside of the analyzer, this label indicates the position of the main switch and the connector for the electrical network cable.



(14) Electrical specifications label

Located on the bottom right, on the backside of the analyzer, there is located the main supply label, which indicates the electrical specifications at which the analyzer can work with.

3051177	100 - 240 V~
	50 - 60 Hz
	600 W
	2 x T 6.3A L 250 V, 5x20 mm

(15) Electrostatic discharge (ESD) label

Located next to the connections (RS 232, USB and LAN), this label warns of the risk of electrostatic discharges that could damage the analyzer, and indicates that the pins on such connections must not be touched with the hands to avoid fortuitous discharges.



(16) RS 232 connection label

Located on the back of the analyzer, this label identifies the connector for the RS 232 connection cable.



2.3 Complementary Material Markings

(1) Biological risk from the card disposal container bags and their contents

Symbol printed on the bag itself, warning of the biological risk from the cards contained in the bag. The bags are single-use.



2.4 Read Before Using

- Use the equipment only for the purpose described in these Instructions for Use.
- Do not insert objects in any opening on the equipment, unless specified in these Instructions for Use.
- Do not use the equipment or the complementary materials if they do not operate properly or have suffered any damage. Some examples of typical defects are:
 - Visible damage due to a fall.
 - Visible damage due to spilled liquids.
 - Visible damage due to storage in unfavourable conditions for a long time, or due to inadequate transport conditions.
 - Damage to the electrical cable or its connector.
 - Pieces or parts of the analyzer improperly assembled.
- Do not use the equipment in a dangerous atmosphere, or with hazardous materials for which it was not designed.
- Do not use complementary material other than those supplied or recommended by the manufacturer.
- Do not introduce flammable or explosive materials, or chemicals that interact vigorously, in the centrifuge.
- Keep the air outlets free of any dirt or obstruction, such as threads, hair, dust, etc.
- The network fuses must only be replaced by Qualified Technician.
- The unit may only be disassembled by Qualified Technician, for either internal cleaning and/or repair purposes. Before proceeding to disassemble the unit, unplug it from the electrical outlet.
- Disconnect the unit or leave it in an inactive state after use.

Erytra

3 Analyzer Specifications

3.1 Environmental Requirements and Dimensions

Table 1. Technical Specifications

FEATURE	TECHNICAL SPECIFICATIONS	
MODEL	Erytra	
POWER SUPPLY	Voltage:	100 - 240 V ~
	Frequency:	50 - 60 Hz
	Maximum input power consumption:	600 W
	Consumption in inactive state:	30 VA
	Fuses:	2 x T 6.3 A L 250 V, 5x20 mm
PROTECTION AGAINST ELECTRIC SHOCKS	Class:	I
INSTALLATION CATEGORY	Overvoltage category II (local levels, instruments, portable equipment, etc.)	
DIMENSIONS	700 mm (depth) x 1060 mm (width) x 1785 mm (height)	
WEIGHT	Approximately 350 kg	

FEATURE	TECHNICAL SPECIFICATIONS	
OPERATING CONDITIONS	Indoor use	
	Temperature:	15 °C to 30 °C
	Maximum relative humidity non condensing:	85%
	Maximum altitude:	3000 m
	Mains supply maximum voltage fluctuations:	±10% of nominal voltage
	Pollution degree:	2
TRANSPORT AND STORAGE CONDITIONS	Temperature:	-29 °C to 60 °C
	Maximum relative humidity non condensing:	85%



CAUTION: The operating conditions stated in the preceding table, relate to the instrument only. Refer to the Instructions for Use of the reagents to be used with the Erytra analyzer, for additional information regarding their environmental conditions of use. Do not use the Erytra analyzer outside the specified operating conditions.



WARNING: Do not use this device in close proximity to sources of strong electromagnetic radiation, as these can interfere with the proper operation.

3.2 Functional Specifications

Table 2. Functional Specifications

Sample Station	
Sample loading capacity	96 tubes simultaneously
Sample loading system	8 removable holders with 12 positions
Continuous sample loading	Yes
Emergency sample management	Yes
Sample containers	Generically, plastic or glass tubes with the following dimensions: - Diameter: $9 \leq \varnothing \leq 16$ mm (samples holder) or $9 \leq \varnothing < 17$ mm (samples holder > 16 mm). - Height: ≤ 100 mm.
Positioning of the samples	Continuous and random
Sample identification	- Automatic by barcode reading. - Manual with identification.
Positive sample identification	Yes
Number of probes	2, able to pipette in parallel
Minimum sample volume	250 μL.
Sample dilution	Yes, by using the Grifols diluent in the Dilution Station

Sample barcode type	• Code 2 of 5 interleaved. • Code 3 of 9. • Codabar. • EAN8 / EAN13 / JAN8 / JAN13. • Codes 128 A, B & C (ISBT 128). • Others under configuration.
Reagent Station	
Reagent positions	54 positions
Reagent agitation	Yes, 36 positions
Reagent loading system	Through 4 removable holders in two independent, random-access drawers
Authorized reagents	• Reagents authorized by Diagnostic Grifols, S.A. • From the laboratory itself, labeled using the BCC code. To use this configuration, contact your local Grifols service representative.
Reagent vials	$21 \leq \varnothing \leq 22.5$ mm diameter
Automatic reagent identification	Yes, by barcode reading
Positive reagent identification	Yes
Minimum reagent detection volume	5% of the volume of the reagent in the vial, with a minimum of 225 μ L
Dead reagent volume	• Grifols reagents: 225 μL. • Grifols diluent: 6.9 mL. • Others: 1 mm over the bottom of the tube.
Card Station	
Grifols gel card loading capacity	A maximum of 400 cards

Grifols gel card loading system	Grifols gel card racks, directly in the 4 independently opening drawers
Continuous Grifols gel card loading	Yes
Number of different Grifols gel card types on board	Unlimited
Incubator	
Incubation temperature	24 °C 37 °C
Incubation time	10 - 15 minutes (if appropriate)
Number of incubators	4 independent incubators
Incubator capacity	12 cards
Centrifuges	
Number of centrifuges	2 independent centrifuges
Centrifugal capacity	12 cards
Centrifugation time	9 minutes
Reading Station	
Reading system	High-resolution color reading using a colour CCD camera
System Solutions and Waste Solutions	
System solutions	- System Solution A diluted for the fluidic system. - System Solution B diluted for external washing and final washing of the fluidic system.

Containers	• 1 container for System Solution A. • 1 container for System Solution B. • 2 containers for waste solutions. • 1 container for processed Grifols gel card disposal. • Option to double the capacity of System Solution A and B with the liquid waste drainage configuration activated.
Waste liquid drainage	Option to drain waste liquid directly to the laboratory sink
Container capacity	8 liters (each)
Container weight (when full)	≤ 9 kg
Container monitoring	Yes, by monitoring the volume by weight
Capacity of the card disposal container	400 Grifols gel cards
Others	
Dilutions	Yes, using a Dilution Station with 2 independent, self-washing cups
Level detection	Yes, on both probes
Correct aspiration detection	Yes, on both probes. Automatic and in real time, with obstruction detection
Total speed	100 Grifols gel cards/hour
Useful life	7 years

Outputs to peripherals	Standard interfaces: • 1 RS 232 connection. • 6 USB ports. • 1 LAN connection. Note: Only peripherals aimed to be connected to the described connector types are allowed
Remote connection	Yes. Under configuration, it is possible for authorized Operators by using the website application. It provides access to: • Results. • Database. • Quality Control. • User Management. To Qualified Technician by using the Secure Remote Connectivity for: • Support and maintenance purposes.

3.3 Program Specifications

The Erytra analyzer presents an integrated computer with the software preinstalled. The main characteristics of the Erytra analyzer software are:

- Execution of immunohematology tests.
- Digitalization and image processing to obtain the reaction grade of the reaction.
- Validation and report making, including the exporting of results to the LIS (Laboratory Information System).
- **Database.**

Other characteristics and services offered by the whole system:

- A color-coded user-friendly interface which aids and improves the understanding of the information and its analysis.
- Grouping of tests by profile and the simultaneous programming of the profile to a sample.
- Management of requests through the **Worksheet**.
- Downloading and management of sample test requests and demographic data from LIS, if configured connection to LIS.
- Validation by profiles of tests.

- Exporting of results to the **Database** and/or to the LIS (Laboratory Information System).
- Availability of alarm algorithms that allow the detection of incidents when reading and/or in the later results.
- Display of analyzer status and autonomy (system and waste solutions, Grifols gel cards, reagents, etc.) and of the sample petitions.
- Real-time display of the incubation and centrifugation steps.
- Management of reagent batches, allowing you to work with different reagent batches simultaneously.
- User access control.
- Total traceability of results: Users, analyzer, batch of reagents, reader, etc.
- Bidirectional interface with ASTM protocol standard NCCLS LIS2-A2 or NCCLS LIS1-A2 (compatible with LIS).
- Remote connection via HTTP standard and Microsoft Internet Explorer 11 web browser or alternative Grifols customized Chrome browser which allows the access to **Database** results remotely.
- Remote connection by means of the **Secure Remote Connectivity application** which allows to establish a connection from the system for the remote support.

4 Description of the Erytra Analyzer

4.1 Overview of the Erytra Analyzer

The Erytra analyzer was designed vertically to save laboratory space. For this purpose, the Erytra analyzer can be divided into three well differentiated levels.

- Upper level: Area designed to hold the Reagent Station, the Sample Station and the Dilution Station. The dispensing of reagents and samples in the microtubes of the cards is done directly in the incubators at the rear part of this level by means of a robotic arm with 2 probes.
- Middle level: Area designed to hold the Card Station and its management (selection, load, read, configuration and storage of processed cards for review). The movement of the Grifols gel cards is done by means of a robotic arm fitted with a clamp.
- Lower level: Stores the system solutions, the waste solutions and the card waste drawer.



Figure 1. Overview of the Erytra Analyzer

- (1) On/Off switch (at rear).
- (2) **Start** button.
- (3) Upper door of the analyzer which allows access to the first level for service access.
- (4) Lower door of the analyzer for access to the second level.
- (5) Push-buttons for opening drawers.
- (6) Touch screen.
- (7) Worktable.
- (8) Traffic light.
- (9) Connection for the emptying of waste solutions into the laboratory drain.
- (10) Syringes.

4.1.1 On/Off Switch

The master switch is on the back right (Figure 1, no. 1) of the instrument and controls the electrical mains power input. It has two positions:

1. **I**: On (to the left when facing the front of the instrument).
2. **O**: Off (to the right when facing the front of the instrument).

It is advisable to turn off the analyzer at this switch during long periods of inactivity.



WARNING: In order to disconnect the instrument from mains, unplug the power cord either from the power outlet or the connector located in the rear part of the analyzer.

4.1.2 Start Button

The **Start** button is found on the front right-hand side of the analyzer (Figure 1, no. 2) and it is in charge of turning on the Erytra analyzer instrument. It has a green light indicator which shows if the instrument is on or off (ON/OFF).

4.1.3 Upper Door

The Erytra analyzer has a top door (Figure 1, no. 3) which slides up to permit access to the whole work surface: Upper robotic arm, Dilution Station, Sample Station and Reagent Station. Its use is reserved for servicing, such as cleaning its inside.

4.1.4 Lower Door

The Erytra analyzer has a lower folding door (Figure 1, no. 4) which allows total access to the Card Station, centrifuges, incubators and reader. It is only used for servicing and its protective screw must be removed in order to open it. For further information consult Section 17.2.



WARNING: The Operator's normal interaction with the analyzer should be done using the drawers. The use of the top and bottom doors is limited to:

- Resolving analyzer incidents.
- Fully unloading the analyzer.
- Maintenance purposes (cleaning or repair).



CAUTION: The general doors should only be used in the cases described earlier and should not be used under normal working conditions of the analyzer.

Opening the upper door during the execution of a workload interrupts and cuts off the power to the different modules on that level, cancelling the tests in progress and resulting in loss of information and possible unexpected analyzer behaviour.



CAUTION: With the aim of maintaining positive identification of reagents and samples, the opening of the top door will also cause the loss of this identification information. Reagents and samples would have to be re-identified by opening their respective drawers.

4.1.5 Push-Buttons for Opening Drawers

The analyzer has push-buttons (Figure 1, no. 5) on the front of the instrument which, once the software is running, allow the drawers in the Reagent, Sample and Card Stations to be opened.



Figure 2. Push-buttons for the Upper Drawers

Buttons 1 and 2: Provide access to the Reagent Station and buttons 3, 4, 5 and 6 provide access to the Sample Station.



Figure 3. Push-buttons for the Middle Drawers

Buttons 1, 2, 3 and 4: Open the 4 drawers for accessing the Card Station.

4.1.6 Push-Button to Open and Load STAT Samples

STAT or urgent samples can be loaded at any time by pressing the **STAT Loading** button



, located to the right of the push-buttons for the drawers on the upper level (Figures 2 and 4).

4.1.7 Touch Screen

The Erytra analyzer is supplied with an integrated computer and touch screen. To access the touch screen (Figure 1, no. 6), press the front of the touch screen to release it and then pull it out. The touch screen viewing angle is adjustable to suit every user.

4.1.8 Worktable

The Erytra analyzer includes a worktable (Figure 1, no. 7) which can be used when loading and unloading the analyzer. The worktable can be accessed by pressing the front of the worktable to release it and then pulling out.

4.1.9 Traffic Light

The Erytra analyzer has a traffic light which allows monitoring its status from a distance. When the traffic light is green it means that the analyzer is working correctly and has all of the necessary resources to execute the programmed tests. When the traffic light turns yellow, there is a system abnormality that requires Operator attention. If the abnormality is not corrected, the traffic light will eventually turn red, indicating that the analyzer has stopped. In this case, an acoustic alarm will also be activated.

4.1.10 Blue LEDs on Sample Drawers

Each sample drawers of the Erytra analyzer has blue LEDs. These LEDs are off when the samples inserted into the holders are required by the analyzer. When the LED is on, they can be removed from the analyzer.

4.2 Modular Description of the Instrument

The Erytra analyzer is designed to permit complete Operator flexibility to interact with it at any time. This is why the analyzer has different drawers. They can be opened by using the analyzer software or by using the corresponding push-buttons which are located below them (Figure 1, no. 5).

Each level has the following parts:

4.2.1 Erytra: Upper Level



Figure 4. Overview of the Upper Level of the Erytra Analyzer

- (1) Upper robotic arm.
- (2) Sample Dilution Station.
- (3) Reagent Station with 2 separate drawers.
- (4) Sample Station with 4 separate drawers.

4.2.1.1 Upper Robotic Arm

The Erytra analyzer has one upper robotic arm with two probes for simultaneous aspiration and dispensing of samples and reagents into two Grifols gel cards. The probes have a dual function: First, they perforate the protective film of the microtubes that are going to be used and, second, in parallel, they dispense the correct volume of reagents and samples into the microtubes. Depending on the test to be processed, the system may dilute the sample in the Dilution Station before dispensing.

The arm is also equipped with a sensor which detects the presence of tubes, vials and bottles, and a barcode reader for their identification.

4.2.1.2 Dilution Station

The Dilution Station is located on the upper level of the Erytra analyzer and is where any necessary dilution of the samples with the Grifols diluent is done, before being dispensed into the microtubes of the Grifols gel cards.

The Dilution Station has two chambers so that two dilutions can be prepared simultaneously. It is also self-cleaning and does not use any disposable components.

4.2.1.3 Reagent Station

The analyzer allows the continuous loading and unloading of reagents through the 2 drawers that are in the Reagent Station (Figure 4, no. 3). Each drawer can hold 2 removable holders. The small holder is for the storage of reagents and diluents that do not require re-suspension while the large holder is for the storage of reagents that do require re-suspension.

The Reagent Station can hold up to 48 reagents vials (12 positions for reagents that do not require re-suspension and 36 for reagents that do require re-suspension) and 6 diluents bottles.

4.2.1.4 Sample Station

The analyzer allows the continuous loading and unloading of samples through the 4 sample drawers that are in the Sample Station (Figure 4, no. 4). Each drawer can contain 2 removable holders which hold 12 tubes each. The total capacity of the Sample Station is 96 tubes.

4.2.2 Erytra: Middle Level



Figure 5. Overview of the Middle Level of the Erytra Analyzer

- (1) Lower robotic arm.
- (2) Grifols gel Card Station with 4 separate drawers.
- (3) Service Racks.
- (4) 4 incubators.

4.2.2.1 Lower Robotic Arm

The Erytra analyzer includes a lower robotic arm (Figure 5, no. 1) that is equipped with a clamp for the transport of Grifols gel cards. The arm is also equipped with a sensor which detects the presence of Grifols gel cards and a barcode reader for Grifols gel card identification.

4.2.2.2 Grifols Gel Cards Station

The analyzer allows the continuous loading and unloading of Grifols gel cards through the 4 drawers of the Grifols gel Cards Station (Figure 5, no. 2). Each drawer can hold 4 Grifols gel card racks. The total capacity of the Grifols gel Cards Station is 16 racks, including the Service Racks (see Section 4.2.2.3). The Grifols gel card racks can be loaded directly into the Erytra analyzer in their original packaging, exactly as they are delivered by the manufacturer Diagnostic Grifols, S.A.



CAUTION: To speed up the identification process, the instrument reads the first Grifols gel card in the rack and then assumes that the rest of the Grifols gel card in the rack are of the same type (during test processing each Grifols gel card will be identified individually). If the instrument cannot read the first Grifols gel card, it then attempts to read the next two Grifols gel cards. If it cannot read any of the first three Grifols gel cards, the Grifols gel cards in that rack appear in red on the **Status > Cards** screen and cannot be used for test processing until they are identified by the instrument.

4.2.2.3 Service Racks

The Erytra analyzer has two Service Racks (Figure 5, no. 3) to store Grifols gel cards that have a process incident or a result incident associated with them, so that they may be physically reviewed by the Operator.

The analyzer also stores partially used Grifols gel cards in the Service Rack so that they are available for the instrument to reuse.



All the cards left in the Service Rack will have the following icon associated , once their detailed results are displayed on the **Results List**.

Each time a Service Rack drawer is opened and closed, the instrument attempts to identify all of the Grifols gel cards in the Service Racks.

The number of Service Racks in the analyzer can be configured. For more information, please contact your local Grifols service representative.



WARNING: Although the instrument will be in charge of removing already reviewed and completely used Grifols gel cards from the Service Rack, under certain circumstances, cards must be manually removed from the Service Rack.

4.2.2.4 Incubators

The Erytra analyzer has 4 independent incubators (Figure 5, no. 4) located on the middle level. Each incubator is made up of a block of 12 positions for Grifols gel cards. These incubators can be set to incubate at either 24 °C or 37 °C, depending on the need at the time and on the tests to be performed.

The incubators are mobile and move from the middle level to the upper level for the card dispensing step. Once the dispensing step is complete, the incubation step is performed also in this upper floor (if the test requires incubation) and then, they return to the middle level for the centrifugation step. The incubation will be done at the temperature and for the amount of time defined by the test to be performed.

It is possible to configure that none of the incubators is reserved for the processing of the STAT samples. For more information, contact your local Grifols service representative.

Should any of the incubators become disabled, the Erytra software can continue functioning with the rest of the incubators. To activate a deactivated incubator, consult Section 11.2.6 of this manual.

The loading of the incubators with the non-processed Grifols gel cards is performed when the instrument is able to pipette them. Therefore, there can be some delay between the moment when the sample is queued and the moment when the cards are loaded, which corresponds with the expected behaviour of the instrument.

4.2.2.5 Centrifuges

The Erytra analyzer has 2 independent centrifuges situated under the middle level. Each centrifuge can hold 12 Grifols gel cards and is programmed to centrifuge Grifols gel cards according to the parameters defined by the test.

The lower robotic arm positions the Grifols gel cards in a precise manner in each of the baskets of the head of the centrifuge. When the number of Grifols gel cards to be centrifuged is not even, the instrument takes a supplementary unprocessed card from one of the racks of available unprocessed Grifols gel cards to balance out the load in the centrifuge. This balance card will be returned to its original position when the centrifugation step has ended.



CAUTION: The centrifuges within this instrument should not be used independently. They are designed only for the automatic loading and unloading of the materials described in these instructions and should not be used for any other purpose.



CAUTION: The centrifuges are not designed to be handled by the Operator under any circumstances.

Should one of the centrifuges become disabled, the Erytra software can continue functioning with the remaining centrifuge. To activate a deactivated centrifuge, consult Section 11.2.6 of this manual.

Cards pipetted in different incubators can be loaded in the same centrifuge, if the instrument considers so, according to the routine of the moment.

4.2.3 Reader for Image Processing

The reader for image processing is situated on the middle level of the Erytra analyzer and uses a color CCD camera to capture and process Grifols gel card images. This allows the instrument to determine the reaction grade in each microtube and to apply the necessary algorithms in order to ensure a correct result.

The reader also captures the images of the Grifols gel card **microtubes** before the card is processed, to check the initial state of the gel in the microtube.

4.2.4 Erytra: Lower Level



Figure 6. Overview of the Lower Level of the Erytra Analyzer

- (1) 1 drawer for the system solution container of System Solution A.
- (2) 1 drawer for the system solution container of System Solution B.
- (3) 1 drawer for the card waste drawer.
- (4) 2 drawers for the solution waste drawer.

Unlike the rest of the drawers of the analyzer, the opening of the lower level drawers must be done exclusively by the analyzer software when the analyzer is switched on. For more information, see Section 6.1.

The lower drawers can be manually open by pushing the latch located below the container drawer to slide the drawer out (Section 17.3). This manual opening of the lower drawers must be only used for maintenance purposes, assuring that the analyzer is unplugged.

It is possible to configure the Erytra to drain waste liquids directly to the laboratory sink. This configuration allows placing system solution containers in the waste containers, which doubles the total capacity of System Solution A and System Solution B system solutions. For more information, contact your local Grifols service representative.

4.3 Complementary Material

Table 3. Complementary Material

CODE	NAME	DESCRIPTION	QUANTITY
			
213713	Samples holder	Holder to load samples into the analyzer	1
210714	Pediatric samples holder	Holder to load pediatric samples into the analyzer	2
210715	Samples holder > 16 mm	Holder to load sample tubes with a diameter slightly larger than 16 mm into the analyzer	2
213715	Erytra holder for mixed reagents	Holder to load reagents which require re-suspension into the Erytra	1
213716	Erytra holder for diluents	Holder to load diluent and reagents which do not require re-suspension into the Erytra	1
233738	Erytra solution container	Container designated to hold diluted System Solution (A or B)	1
213711	Bags Erytra	Disposable bags for card waste	100
213778	Diana BCC	Barcode printer	1
213776	BCC Labels	Blank labels for use with the Diana BCC printer	1000
213796	Erytra adapter insert for DG Clean	Adaptor that has to be placed in the diluent positions of the non-stirring reagent holder in order to allocate the DG Clean reagent	1

In addition, to the complementary material listed above, the following components can be used with the Erytra analyzer:

- A handheld external barcode reader can be attached to the analyzer for the identification of samples and cards.
- A holder for the handheld barcode reader can be attached to the analyzer to sustain and hang the barcode reader (see Figure 7).



- A printer, to print copies of the results (see Section 5.2.5).



Figure 7. Handheld Barcode Reader Hung in the Analyzer with its Holder

For more information of any complementary material or how to acquire them, please contact your local Grifols service representative.

5 Installation

5.1 Intervention on Delivery

This instrument must always be installed by a Qualified Technician.

The Qualified Technician will give the appropriate instructions to the Supervisor so that the instrument can be turned on.

When you receive the analyzer, you must check that all of the elements listed on the attached *Packing list* are present.

5.2 Installation Requirements

The place where you plan to install the instrument must meet the following requirements:

- If the instrument has been stored under environmental conditions different to those of the working area, it must remain under the working area environmental conditions for no less than an hour before being plugged in.
- Environmental conditions: See Section 3.1.
- Space requirements: It is recommended to have a solid horizontal surface that can support more than 350 kg of weight and that has a space at least 100 cm deep and 150 cm wide.
- The drawers when opened require a frontal space of 46 cm.
- Do not place the instrument outdoors.
- Do not place the instrument in a way that would hinder its access for maintenance purposes or for Technical Service work.
- Do not block the air vents.
- Do not place the instrument where it could get wet.
- Do not place the instrument where it would be exposed to direct sunlight. Avoid intense light sources.
- Do not position the instrument in a way that would make it difficult to use the general mains switch and ensure that appliance coupler or plug is easily accessible.
- Do not allow the instrument or its network cable to come into contact with surfaces that are too hot to touch.
- Do not place any object on the instrument.
- Check that the electrical installation the instrument will be connected to meets the rules and regulations of electrical input and use (including the earthed connection).
- Make sure that the instrument is completely level and with no obvious slant.

5.2.1 Electrical Outlet (Socket)

You need an electrical socket outlet for the Erytra analyzer. Consult the characteristics shown on the corresponding label of characteristics (see Section 2.2, no. 12).

The maximum electrical consumption, measured heat levels, are the following:

- Stand By: 0.006 kWh.
- Start Up/Operation: 0.126 kWh.
- Ready for Workload: 0.121 kWh.

5.2.2 Noise Level

The instrument does not create sound pressure level above 58.9 dBA in working mode. No additional hearing protection is required.

5.2.3 Water Intake Tapping Point

Water intake tapping point it is not necessary, as all the solutions that are used by the Erytra analyzer are stored within the instrument in removable containers with a capacity of 8 liters.

5.2.4 Drain

All of the generated waste is poured into two removable containers, each with a capacity of 8 liters. These containers should be emptied manually into a sewer or into an appropriate deposit.

It is possible to configure the instrument so that the liquid waste generated can be drained directly to the laboratory sink. To do this, please contact your local Grifols service representative.

5.2.5 Computer Equipment

The analyzer includes a computer and a touch screen monitor. The keyboard is virtual and located on the touch screen.

The computer runs with Microsoft Windows 10 Enterprise 2019 LTSC Operating System. The operating system and the Erytra analyzer software come already installed and preinstalled respectively.

If you wish to install a printer, contact your local Grifols service representative.

If necessary, other peripheral devices can be connected via USB such as: Keyboard, barcode reader, external hard drive, CD drive or memory devices.



CAUTION: To ensure the correct management of traceability and expirations, make sure that the Hour and Date configuration match your geographical location. For more information, contact your local Grifols service representative.

5.2.6 Remote Connection

Thanks to the communication characteristics of the Erytra analyzer, which include a wide range of standard interfaces, system information can be obtained remotely. The analyzers can be integrated into an existing PC network, through the Ethernet ports and also over the Internet using HTTP standard and Microsoft Internet Explorer 11 web browser or alternative Grifols customized Chrome browser (remote client). This connection allows viewing, and exporting of Results, as well as **Database** searches done by authorized users, virtually, from any remote desktop.

It is also possible to connect remotely to the analyzer via **Secure Remote Connectivity** for support and maintenance purposes.

For more information about the remote configuration of the analyzer, contact your local Grifols service representative.

5.3 Installation Procedure

The installation procedure is described in the following document *Installation Qualification (IQ)* in the Technical Service manual.



WARNING: The use of non-specified complementary material or cables can cause an increase in electromagnetic emissions and/or a reduction in the immunity of the instrument.

Erytra

6 Description of the Erytra Program



CAUTION: The profiles, reagents and results that appear on the screen shots of these Instructions for Use have been included as an example only.

When the **On** button of the analyzer is pressed (Figure 1, no. 2), the system automatically starts up the software. Once the Operator has logged in, the instrument will show the main screen:

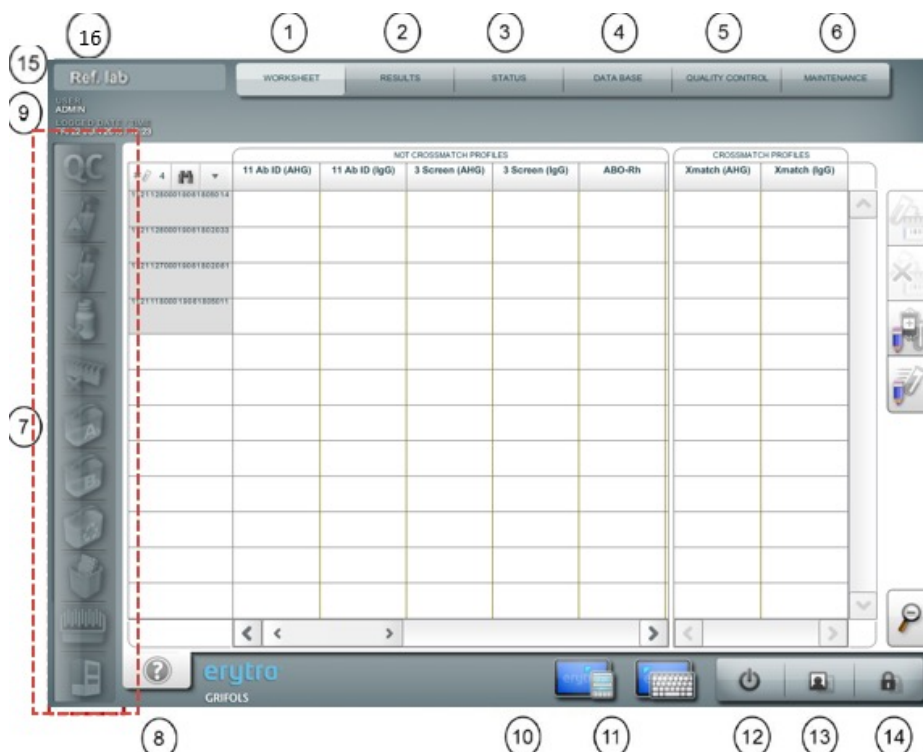


Figure 8. Overview of the Erytra Software

- (1) **Worksheet** tab.
- (2) **Results** tab.
- (3) **Status** tab.
- (4) **Database** tab.
- (5) **Quality Control** tab.
- (6) **Maintenance** tab.
- (7) **Warning Area**.
- (8) Help.
- (9) Identification of the active user.
- (10) Map of the analyzer.
- (11) **Virtual Keyboard** button.
- (12) **Shut Down** button of the Erytra analyzer.
- (13) **User Access Control** button.
- (14) **Password Change** button.
- (15) Configurable Instrument name.
- (16) **About**.

6.1 Opening Drawers



As well as the push-buttons for opening the drawers (Figure 1, no. 5), the button (Figure 8, no. 10) unfolds a map of the analyzer which allows the opening of the different analyzer drawers by using the Erytra software.

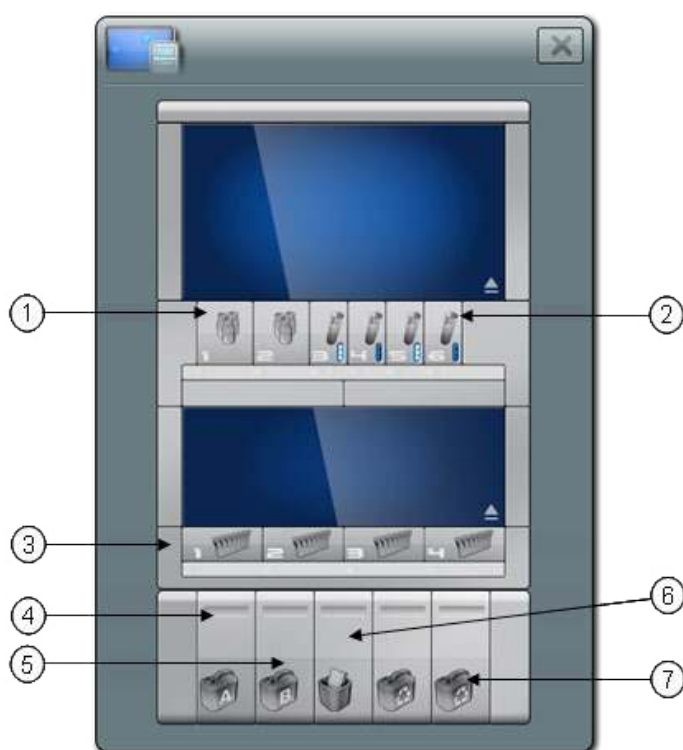


Figure 9. Map of the Analyzer that Allows the Opening of All of the Analyzer Drawers

- (1) Drawers 1 and 2: Access to the Reagent Station.
- (2) Drawers 3, 4, 5 and 6: Access to the Sample Station.
- (3) Drawers 1, 2, 3 and 4: Access to the Grifols gel Card Station.
- (4) Solution A drawers: Access to System Solution A.
- (5) Solution B drawer: Access to System Solution B.
- (6) Access drawer to the card waste drawer.
- (7) Access drawer to the waste solution container.

6.2 Virtual Keyboard

The application automatically opens a virtual keyboard for the introduction of alphanumeric characters in the programs fields, for the manual identification of a sample or the identification of

the Operator. The application also offers access to the virtual keyboard by pressing  situated at the bottom of the Erytra software (Figure 8, no. 11).

6.3 Description of the Warning Area

The **Warnings Area** is permanently shown on the left-hand side of the Erytra software screen. Its function is to let the Operator know about analyzer requirements or incidents that can be solved by his/her intervention. If these requirements are not resolved in reasonable amount of time they can become high grade incidents and could cause the interruption of the process due to lack of resources. When in normal status, these warnings remain deactivated and only appear when the software detects a need in one of the following resources or components for the completion of the workload:



Quality Control Warning: Derived from the quality policy.

- Pending: The analyzer is pending the analysis of a **Quality Control** for one or more profiles according to the quality policy and/or the programmed rate.
- Failed: Some processed **Quality Control** has not met the programmed acceptance criterion.



Sample Incident Warning: There are issues with the sample tubes (identification issues, lack of volume, clots, etc.) and therefore the Operator must solve them so that the analyzer can finish the execution of all of the programmed profiles for these samples.



Missing Samples Warning: Some sample tubes are not available within the analyzer for the loading of the assigned job.



Missing Reagents, Diluent and/or Cleaner Solution Warning: The analyzer shows that a determined reagent, diluent and/or cleaner solution are not available, or that the volume is not sufficient in order to finish the whole programmed workload.



Missing Grifols gel Cards Warning: The analyzer does not have enough cards available to finish the programmed workload.



System Solution A Container Warning: The analyzer requests that the System Solution A deposit be refilled so that it can finish the programmed workload.



System Solution B Container Warning: The analyzer requests that the System Solution B deposit be refilled so that it can finish the programmed workload.



Waste Solution Container Warning: The analyzer requests that the waste solution containers are emptied so that it can finish the programmed workload.



Cards Waste Container Warning: The analyzer requests that the card waste drawer is emptied so that it can finish the programmed workload.



Service Racks Warning: The analyzer requests that the Service Racks are emptied or changed so that it can finish the programmed workload.



Maintenance Warning: This warning is activated under the following conditions:

- When there are one or more disabled modules.
- When there are one or more actions (decontamination procedure, Technical Service inspection, system restart, etc.) that have expired (these appear in red in the **Summary** screen).
- When there is an action that is about to expire (these appear in orange in the **Summary** screen).

To resolve these issues, see Section 11.2 in the **Maintenance > Analyzer** screen on Section 11.

By pressing on each of the icons, the Erytra software will show the **Status** screen which will show the information that is needed to resolve the incident.



NOTE: The activation of these warnings coincides with the changing from green to amber of the analyzer's traffic light (Figure 1, no. 8). Solving these incidents will normalize the analyzer state and the traffic light will turn back to green.

6.4 Description of the About Screen

The **About** screen is the part of the software in charge to inform the Erytra system information:

- Major Version.
- User Software Version.
- Serial Number.
- Product Reference.
- Manufacturer Contact Details.
- CE Mark.
- UKCA Mark.
- *In vitro* use.
- Consult the Instructions for Use.

To access **About** screen (Figure 10), click on the top left of the main screen (Figure 8, no. 16).



Figure 10. About Information

6.5 Activation of Acoustic Alarm

When an error occurs in the Erytra analyzer, the system sounds an acoustic signal, for the purpose of alerting the nearest Operator of the error. Simultaneously, the  icon is activated on the screen of the Erytra software.



Figure 11. Icons Available in the Bottom Part of the Erytra Software

It is possible to stop the acoustic alarm by clicking anywhere within the error message or by clicking on the  icon itself.

6.6 Activation of the Fluidic Icon

When the instrument is performing a prime or a rinse, the fluidic action is shown as being currently performed by the activation of the fluidic icon  on the screen of the Erytra software.

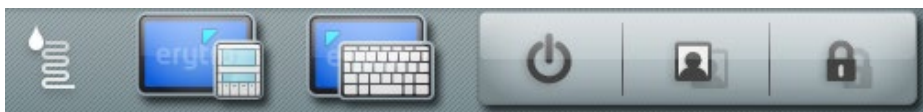


Figure 12. Fluidic Icon Activated in the Bottom Part of the Erytra Software

Erytra

7 Work Procedure

7.1 Analyzer Flow Diagram

Test processing consists of the following steps:

Prior to Testing

- Operator prepares reagents and samples.
- Operator starts the analyzer.
- Operator performs initial checks and required maintenance (as needed).
- Operator loads reagents, Grifols gel cards, and samples.
- Operator programs the **Worksheet** (as needed).
- Operator starts test execution.

During Testing

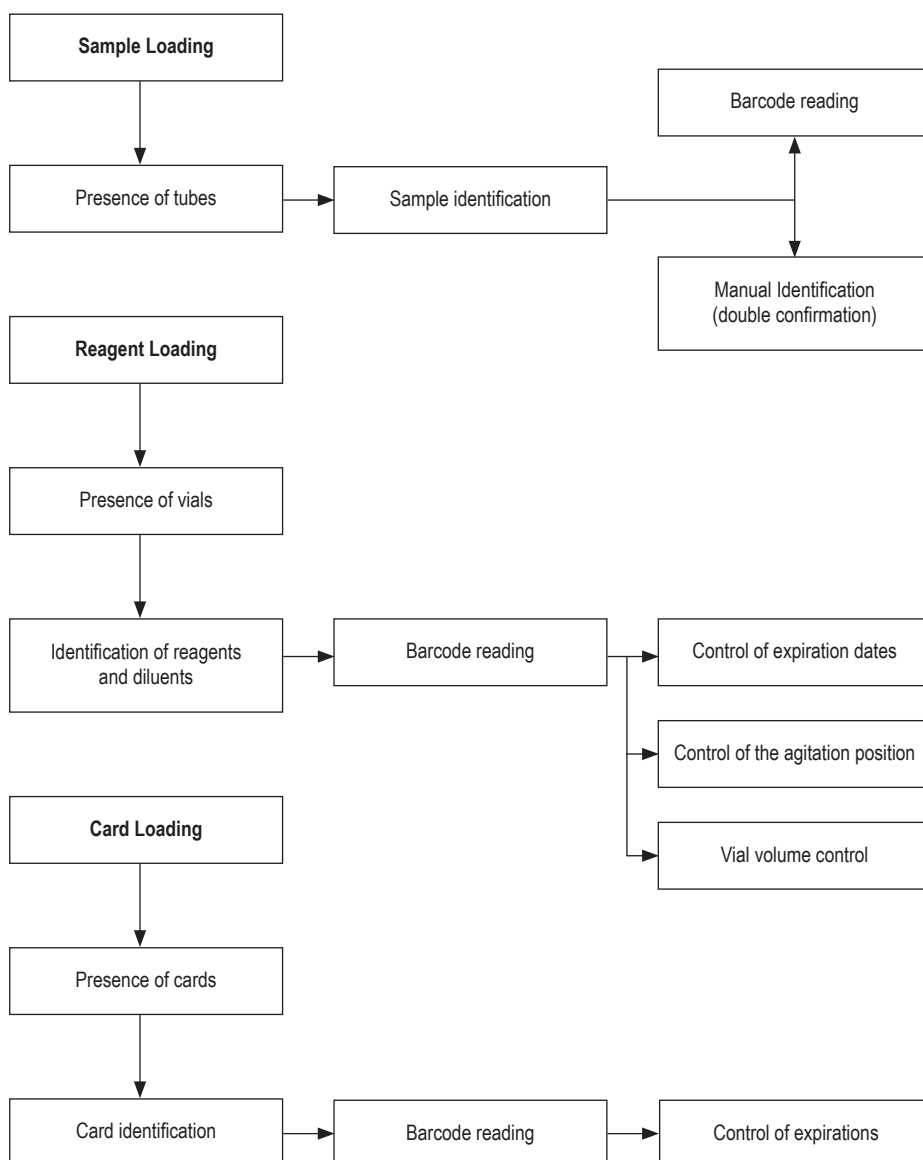
The following steps are performed to execute tests:

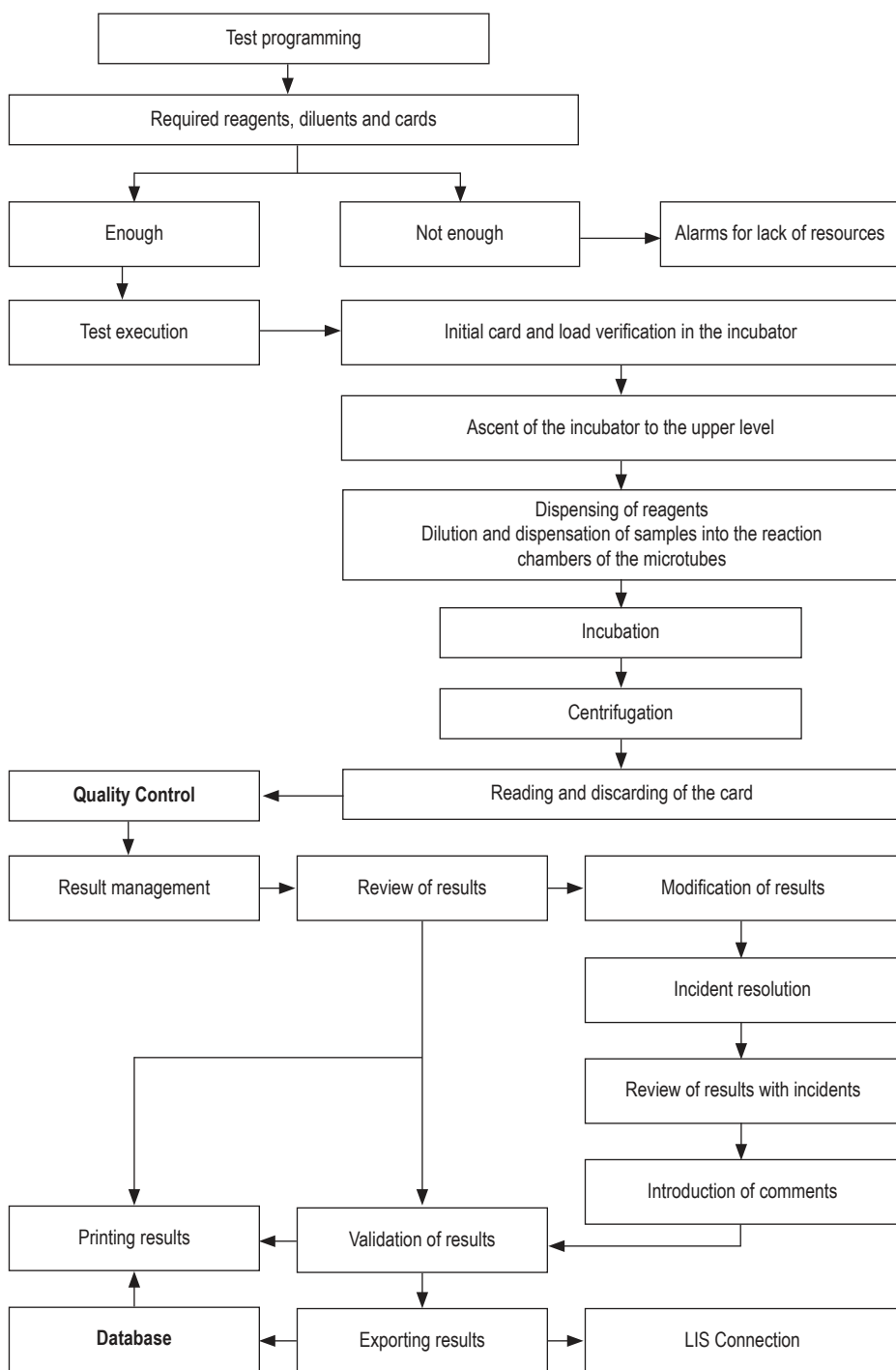
- The Erytra checks to see if there is a valid QC.
- The Erytra checks to see if all the required materials are present for test execution and alerts the Operator if any resources need attention.
- Operator addresses any resource requirements.
- The Grifols gel cards are identified and photographed, and then moved to the appropriate incubator.
- Reagents are dispensed and samples are diluted (if necessary) and dispensed into the corresponding microtubes of the Grifols gel cards, in accordance with the procedure described for each test.
- The Grifols gel cards are incubated (if necessary).
- The Grifols gel cards are centrifuged.
- The Grifols gel cards are transported to the reader, where the camera takes photographs of the microtubes.
- The analyzer applies a reading algorithm to the digitalization of the image in order to obtain parameters that characterize it and make it possible to quantify the reaction grade.
- The analyzer applies an interpretation matrix for each test. These matrices combine the different results from the microtube reaction grades for the test to provide an interpretation of the results.
- The results are updated on the **Results** screen.

- The Grifols gel cards used for the analysis are either disposed of or stored in the Service Racks for later review (if the Grifols gel card has an associated incident or special result) or future use (if the Grifols gel card has unused **microtubes**).

Following Testing

- Operator reviews and validates results.
- Operator saves and prints results (as required).
- Operator unloads and shuts down the analyzer at the end of the workday.





7.2 Preparing for Work

7.2.1 Start Up

7.2.1.1 Turned On

- Check that the analyzer is connected to the main power supply and that the mains switch is turned on (Figure 1, no. 1).
- Press the **On** button (Figure 1, no. 2).

During start up, which will take 4 minutes, the instrument will restart the various modules and will verify the communication between them along with the integrity of the system.

7.2.1.2 Identification of the Operator

If the enable passwords option is activated in the **Access Control** window, then:

- Introduce your **Username** and **Password**.
- Select mode **Erytra Program**.
- Press  to enter the application.



NOTE: Each Operator will only have access to the enabled options which depend on the permissions configuration. For more information consult Section 10.1.

7.2.1.3 Initial Checks Done by the Operator

Go to the analyzer **Status > Containers** screen (Figure 8, no. 3) and check:

- The status of the system solution and waste solution containers:
 - Sufficient level of System Solutions A & B.
 - Sufficient capacity in the waste drawers.
 - Sufficient capacity in the card waste drawers.
- Open card drawers 3 and 4 using the corresponding push buttons and remove the cards that are completely used (Figure 5, no. 3) in the case that the Service Racks are configured in this way.



NOTE: If you need the intervention of an Operator during any of these processes, proceed as follows Section 7.5.

7.3 Analyzer Programming

The Erytra analyzer allows the constant load of samples, reagents, diluents, cards and system solutions, by using the analyzer push-buttons directly or by using the software, pressing on the



7.3.1 Entering Samples

7.3.1.1 Specifications

The Erytra analyzer allows the use of the following types of tubes, without lids, for sample processing:

- **Tube Type:** Plastic or glass (siliconized).
- **Tube Diameter:** $9 \leq \varnothing \leq 16$ mm (samples holder) or $9 \leq \varnothing < 17$ mm (samples holder > 16 mm).
- **Tube Height:** ≤ 100 mm.



NOTE: Before using pediatric sample holders, a configuration file must be edited. For more information, contact your local Grifols service representative.



CAUTION: The use of non-specified sample tubes can cause the deterioration of the instrument.

If you wish to use a non-specified tube, please contact your local Grifols service representative.



CAUTION: For the proper functioning of the instrument it is important that the barcode labels:

- They are well stuck to the tube.
 - They are not hand written or defective.
 - They are correctly positioned facing the reading window.
 - They comply with the specifications described in Section 3.2.
-



CAUTION: To guarantee maximum security during sample identification, it is recommended to use barcodes with a control digit (checksum).



CAUTION: If a sample is manually identified using a barcode with checksum, this control digit should not be introduced.



NOTE: If Erytra is configured to identify samples with ISBT 128 barcodes of 16 digits, a check of the manual control digit introduced by using the keyboard is performed during the manual identification.



CAUTION: The Erytra analyzer configuration to work with the ISBT 128 barcode of 13 digits or the non introduction of the control digit during the manual identification disable the checking of the correct identification of the sample.

7.3.1.2 Nature of the Samples

The reliability of a laboratory result and its interpretation depends greatly on the quality on the samples. For this reason, their proper management from when they are obtained until they are processed is fundamentally important. Please, keep in mind the following warnings:



CAUTION: In order to obtain reliable results, you must enter tempered samples (serum, plasma or red blood cells).



CAUTION: The presence of foam, bubbles or drops on the container walls may induce alterations in the dispensation of the sample, causing a possible error in the results.



CAUTION: The following samples can cause erroneous results or generate problems during the execution of the analysis:

- Grossly haemolysed, lipemics or icterics.
- Clouds.
- Presence of clots, fibrin or particles.
- Frozen samples without tempering and/or unsettled.
- Old samples (collection time > 7 days).
- Insufficient volume.
- Incorrect proportion of plasma and cells.
- Samples with serum separator.



CAUTION: Check that the level of serum or plasma above the surface of the cell layer or the gel separator in the tube is correct. A tube with an incorrect level of plasma should be discarded as an incorrect collection and should not be introduced into the instrument for processing.

7.3.1.3 Loading Samples

Instructions for Holder Use

The analyzer allows the continuous loading and unloading of samples through the 4 sample drawers available (Figure 4, no. 4). Every drawer has two removable holders and each holder has 12 locations destined to hold the sample tubes.

The sample tubes can be placed directly in the holders without the need for adaptors, regardless of their size. These holders can be configured to use only test tubes of a specific size. For more information, please contact your local Grifols service representative.

All of the holders inserted into the Erytra analyzer must be identified with a barcode on one side.

The holders received with the instrument are already identified with the grey label . These holders will be used for the tubes mainly employed in the laboratory.

In the event of using different types of tubes, contact your local Grifols service representative, since depending on the type of tube, it will be necessary to use a holder with a different barcode

 (red or yellow labels) , as well as configure the height and diameter of the tubes.



NOTE: The holders can be removed from the analyzer to aid the sample loading.

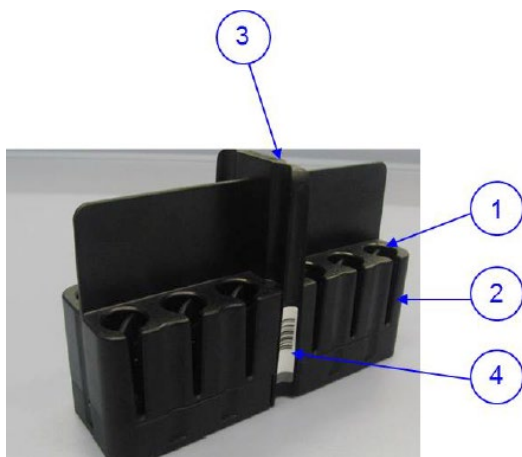


Figure 13. Sample Holder

- (1) Locations for sample tubes.
- (2) Identification window.
- (3) Handle.
- (4) Holder Barcode.



CAUTION: To avoid accidental falls: The holders are sensitive to coming apart and/or breaking if they are hit and fall. In the case of a fall they should not be put back together or reused. Do not use the holders if they are not intact.



CAUTION: Do not use any non-specified holder in the instrument.



CAUTION: Do not keep the samples inside the analyzer for a long period of time as the conditions are not the optimum for their preservation.



CAUTION: For the execution of titration techniques in the Erytra, a tube containing only plasma/serum should be loaded in the analyzer. If centrifuged whole blood samples are used for these purposes, pipetting errors might arise and possible RBCs excess error might be seen on results.



NOTE: Erytra has been validated to ensure that carry-over between samples or reagents is reduced to an extent that will not affect the results of the immunohematology tests. However, under certain situations (e.g., samples with very high titers > 1:1024) the contaminating agent concentration may not be reduced to levels low enough to completely rule out any potential carry-over issue.

- When the sample identification is automatic, place the tubes directly in its corresponding holder, facing the barcode towards the identification window (Figure 13, no. 2).
- If the sample identification is manual place the tube in the holder in the same manner and identify the tube. An identification error will appear that will allow you to enter the tube identity using the keyboard, as described in "Manual Sample Identification", in Section 7.3.1.4.

These sample holders can be configured to use test tubes other than the normal type. To do this, the holder should be configured to the size of the type of test tube, using the barcode associated with the sample holder (Figure 13, no. 4). For more information, please contact your local Grifols service representative.

In case of using sample tubes with a diameter slightly larger than 16 mm, these shall be loaded in different holders whose cavities are a bit bigger. For more information, contact your local Grifols service representative.

Holders for Pediatric Samples

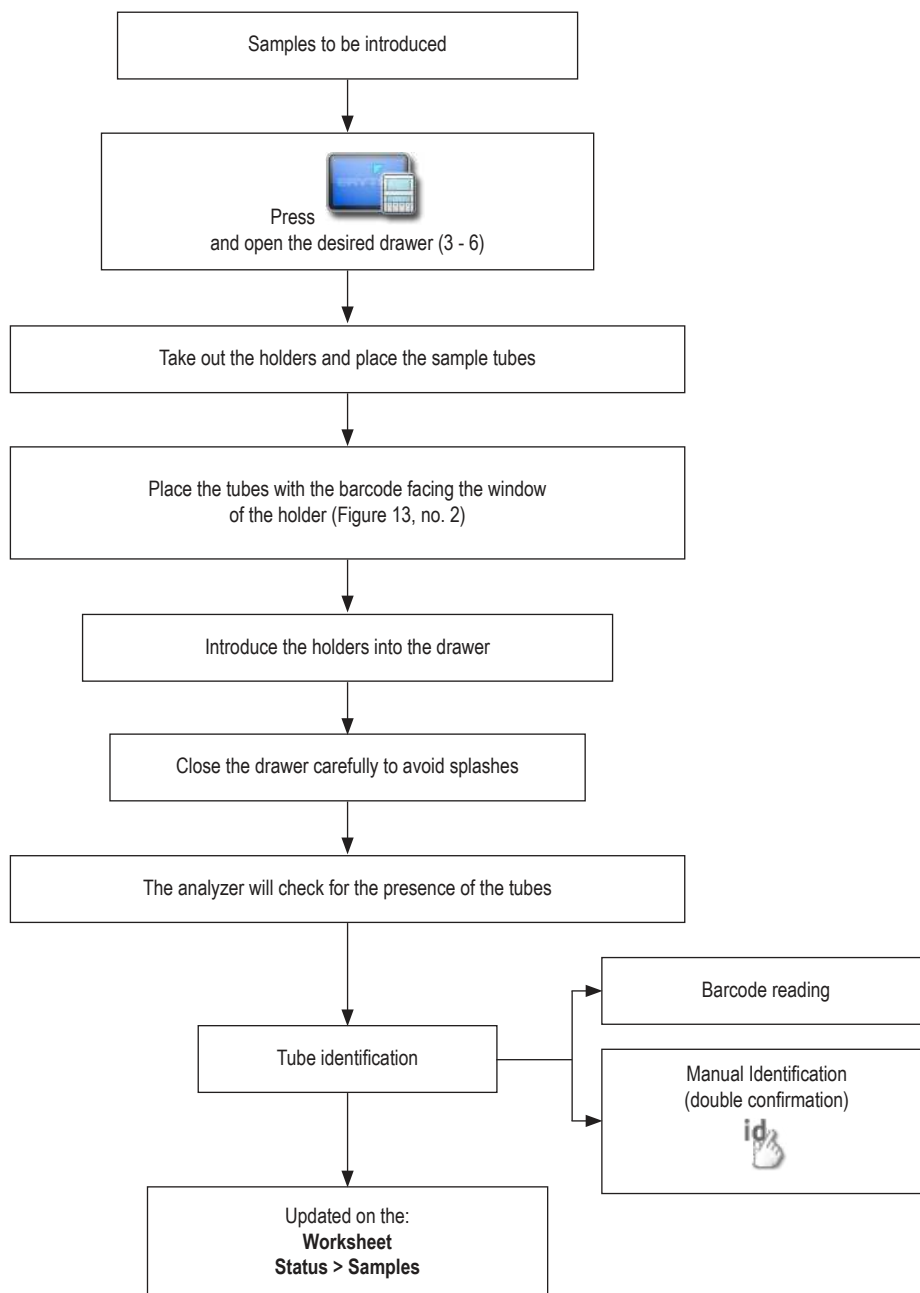
In the event of using pediatric tubes, these must be processed in the Erytra analyzer in the special holders for pediatric samples.



Figure 14. Specific Holder for Pediatric Samples

Before using these pediatric samples holders, it is necessary to edit a configuration file. For any queries, please contact your local Grifols service representative.

7.3.1.4 Procedure



To obtain more detailed information about the samples, press **Status > Samples** (Figure 8, no. 3). The Erytra analyzer software will show a detailed diagram of all of the samples present within the analyzer (Figure 15).

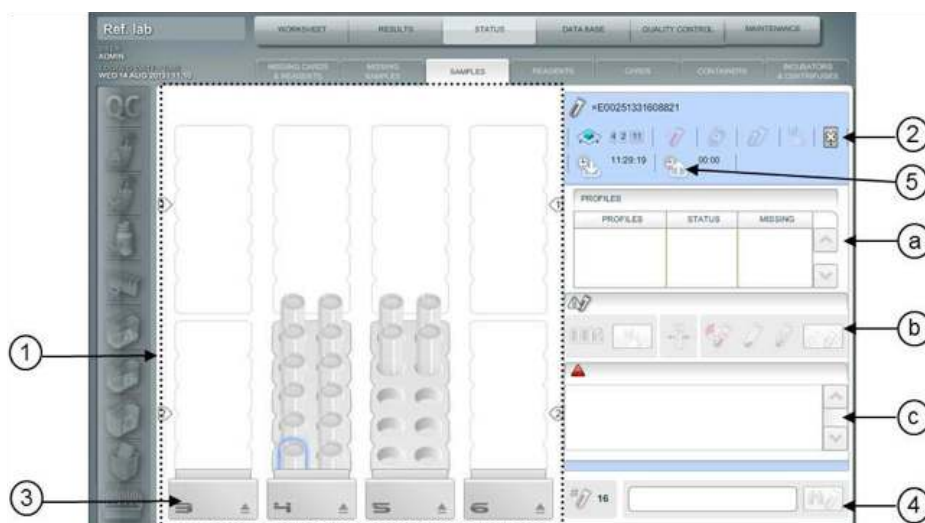


Figure 15. Screen Overview Status > Samples

- (1) Diagram of the samples within the Erytra analyzer.
- (2) Detailed information about each sample with information about the status of the profiles in execution (a), incidents and results (c) and incidents associated with the sample (b).
- (3) Push-buttons for opening the drawer.
- (4) The **Search** button  to search for samples  by barcode.
- (5) Expected time for obtaining the profile result.

The following table lists the status icons that may appear in connection with a sample, including icons that indicate that sample incidents that must be resolved before the sample can be processed.



Shows “old” for being introduced into the analyzer more than 8 hours ago



Insufficient volume



Presence of clot



New sample tube



Diameter of tube not identified



Barcode not identified



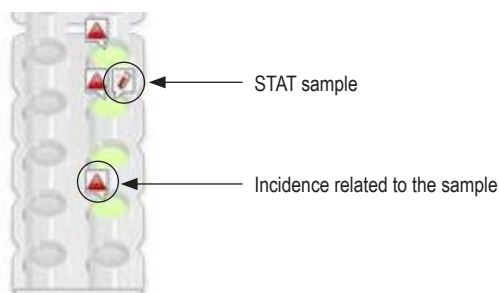
Sample was identified manually

If the sample tube appears as deactivated it means that the sample is not in use, meaning that it has no associated requests. In this case the sample tube can be removed from the analyzer.

Also, the software displays the state of the sample according to the following colour-code:

- Grey : The sample has been correctly identified and has pending requests to process.
- Green : Indicates that the sample has been correctly pipetted for all the associated requested tests, and therefore it can be removed from the analyzer.
- Yellow : Means that the sample has a request that is pending due to the requirement of some reagent, card or system solution.
- Red : Indicates that the sample has shown some kind of incident (Figure 15, no. 2). Detailed information about each sample with information about the status of the profiles in execution (a), incidents and results (c) and incidents associated with the sample (b).

Furthermore, the samples can be associated with a red warning , indicating that the affected sample has been processed and the system has encountered one of the possible incidences indicated in Section 7.6.3.



To obtain more detailed information about any of the samples present within the analyzer, select the sample. On the right-hand side of the screen (Figure 15, no. 2) the information related to this sample will appear, including the following:

- Patient identification.
- Type of tube.
- Status of the programmed profiles.
- Time that the sample was placed in the analyzer.
- Result incidents.
- Expected time to obtain the results.

Manual Sample Identification

If a problem exists during the sample identification, or if the samples do not have a barcode, an error message will appear on the screen.

- The option **Go to Samples** allows access to the **Sample Viewing** screen (Figure 15, no. 1) where the sample that was not identified during the barcode read will appear in red. To solve the problem, open the drawer that contains the unidentified sample tube using the icon  for the corresponding drawer. Replace the tube, with the barcode correctly facing the window of the holder. If the error message reappears, press  and introduce the corresponding information using the keyboard which will appear automatically after pressing on the text box. This type of manual identification requires confirmation, meaning that you must introduce the sample identification twice.

Once the drawer is closed carefully to avoid splashes, the Erytra analyzer automatically re-identifies all of the sample tubes introduced into the holder which are present in the drawer and it will update the onscreen sample information **Samples**.

- The option **Continue** will ignore all of the unidentified sample tubes and the analyzer can continue working with the rest of the introduced samples that have been correctly identified.



CAUTION: The manual entry of sample identification deactivates the positive identification function and users are responsible for assuring that the identified samples are placed in the correct location.



CAUTION: The manual identification of samples increases the risk of mix ups.

In the event of opening a drawer containing samples that have previously been manually identified, a message will appear on screen to confirm whether they are the same tubes or if they have to be identified again and also to confirm that the samples holder has been introduced in the same direction as when tubes were manually identified.

7.3.2 Programming the Worksheet

The analyzer can receive the requests through the Laboratory Information System (LIS) or from the requests generated by the user by means of the touch screen.

- **For Instruments Connected to the Laboratory Information System (LIS):** The sample requests are programmed by downloading the workload from the LIS to the **Worksheet**. The analyzer automatically executes the tests once the samples are loaded according to the sample identification numbers read by the barcode reader.
- **For Instruments Not Connected to an LIS:** The sample requests must be manually programmed from the **Worksheet** by selecting the corresponding fields for the desired profiles for each sample or by assigning a test profile to a group of samples. It can also be used to add additional requests to the workload sent by the LIS.

Requests that are not sent from LIS are shown in the **Status > Samples** screen with the

icon .

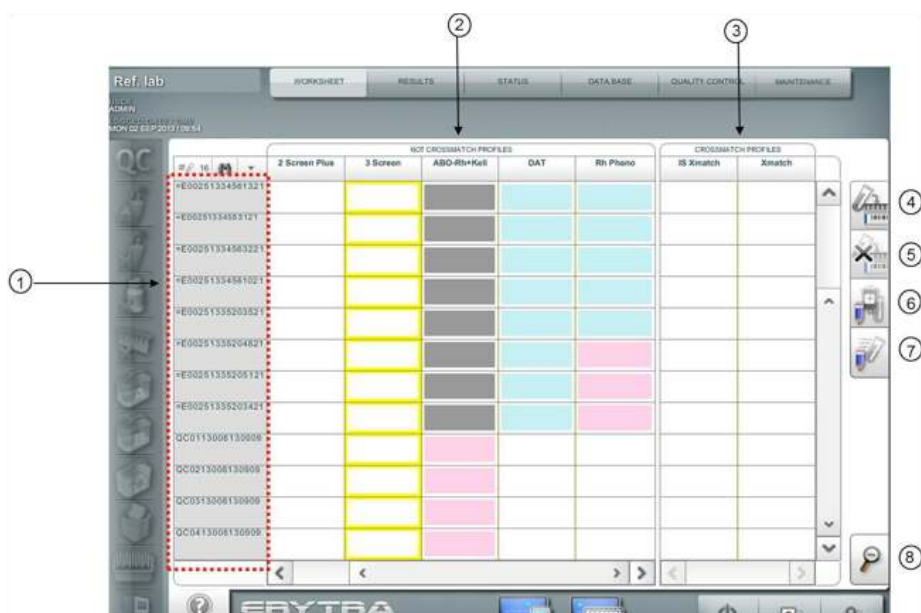


Figure 16. Overview of the Worksheet

- (1) Identification of the samples that are in the analyzer. Informs you if the sample is STAT (🚑), has been manually identified (👉) or if the sample has been programmed as a donor (👤).
- (2) Available analyzer profiles.
- (3) Crossmatching profiles available in the analyzer.
- (4) Requests programming.
- (5) Requests cancellation.
- (6) Donor assignment.
- (7) STAT sample assignment.
- (8) Zoom of the **Worksheet** (visualization 24 samples).

Pressing on sample identification from the **Worksheet** and also on **Status > Samples** screen, it is possible to obtain the following information, as well as the barcode:



Indicates that is a STAT sample



Shows the time the sample was inserted into the analyzer



Shows the expected time to obtain the result



Indicates the location of the sample in the analyzer (Drawer, holder, location)



Allows searching the sample selected in the **Status > Samples** screen



Shows if that sample ID is a double tube (RBCs and plasma separate)



Indicates that the test request does not come from the LIS



Indicates that the sample has been assigned as a donor for a crossmatch test

Viewing Reagent Information

To view information about the reagents, press **Status > Reagents**. The Erytra analyzer software displays a detailed diagram of all of the reagents present within the analyzer.

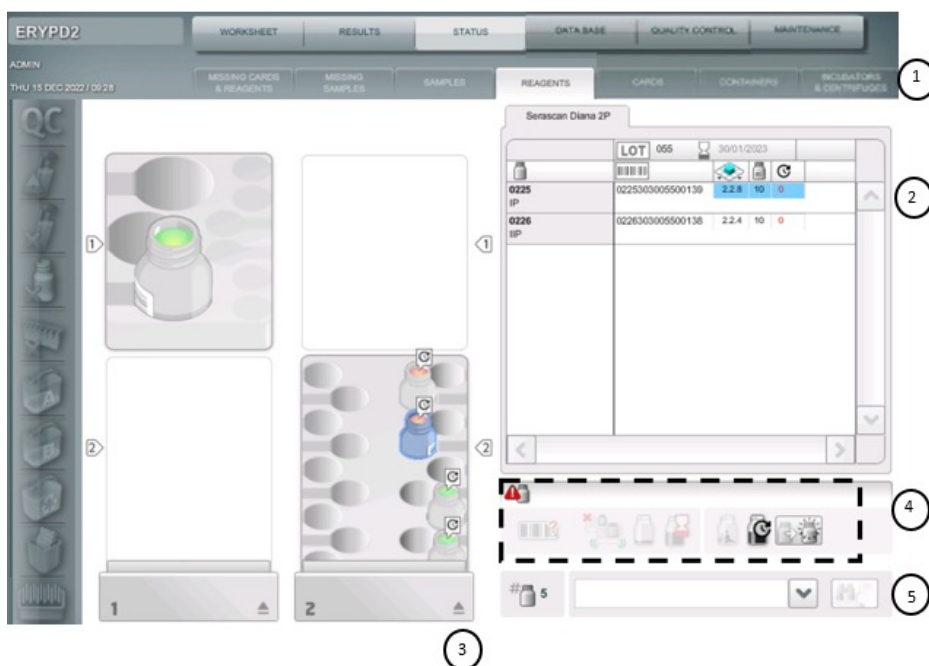


Figure 17. Screen Overview Status > Reagents

- (1) Diagram of the reagents within the Erytra analyzer.
- (2) Detailed information about each reagent including the barcode, commercial name, position, volume and the remaining on-board stability time (in hours).
- (3) Push-buttons for opening the racks.
- (4) Incidents associated to the selected reagent.
- (5) The **Search** button  to search for reagents within the analyzer by barcode.

During the identification of reagents, as well as during execution of the workload, the following symbols may be activated in the **Status > Reagents** screen:



Problem in identifying the reagent barcode. Open the drawer and make sure that the barcode is facing toward the identification window.



Reagents placed in the wrong reagent holder. Open the reagents drawer and move any reagent that require re-suspension into the appropriate holder.



Low reagent volume. Open the drawer and replace the reagent vial or bottle with a reagent vial or bottle with adequate volume and press **Update Reagent Status** button or load a new vial or bottle in a new holder position.



Reagent expired. Unload the expired reagent and replace with a reagent that has not expired.



A clot has been detected in the reagent. Open the drawer and check the reagent for clots. If clots are detected, load a new reagent and press the **Update Reagent Status** button.



Reagent on-board stability expired. Open the rack and replace the reagent vial with a new reagent vial with adequate on-board stability or press **Update Reagent Status** button to ignore the on-board stability.



Update Reagent Status button. Use this button to load a new reagent that substitutes the previous one in its exact same position in the holder or to continuous using the reagent with expired on-board stability.

Erytra is able to track the stability per each vial of red blood cells reagents with unique identification while powered on and inside.

The on-board stability tracking countdown starts when the analyzer identifies the red blood cells vial, stops when the vial is unloaded, or the instrument is switched off.

Elapsed 5 days in continuous use the on-board stability time of red blood cells expires.

Once the vial is loaded again or the instrument is switched on, the tracking countdown will be resumed.

The remaining time, in hours, is displayed in **Status > Reagents** if this feature is enabled otherwise, a hyphen will be displayed instead. When the remaining on-board stability time is less than one hour it would be displayed "< 1" and the vial content remains in green flagged with a



tooltip (Figure 18, no. 1). On the other hand, if the remaining on-board stability time expires it would be displayed a red "0" and the content of the vial turns to red (Figure 17).



CAUTION: The tracking of the on-board stability is considered for 5 days of continuous use scenarios. Otherwise, the user is considered to act under his own responsibility.



CAUTION: If the tracking of the on-board stability is ignored, the user will be able to use the expired red blood cells and is considered to act under his own responsibility.



CAUTION: The tracking of the on-board stability stops when switching off the instrument.



CAUTION: The system does not control or track the time required for the reagent to reach room temperature and homogenize before being placed on the instrument.

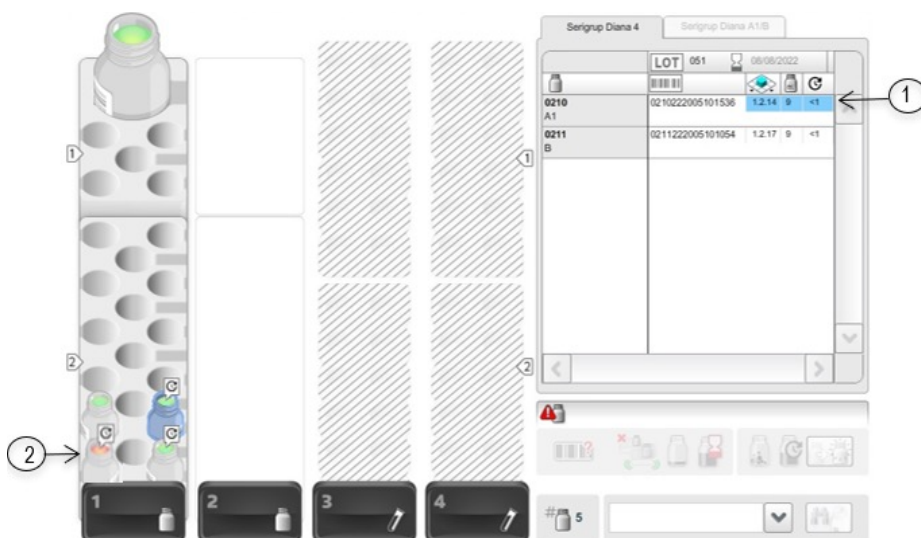


Figure 18. Remaining On-board Stability Time for each Vial

- (1) 1 hour of remaining on-board stability time.
- (2) Expired on-board stability time.



NOTE: Grifols reagent red blood cells can be placed on the instrument for up to 5 days when operating continuously on-board, without unloading the vials.

Other scenarios that vary from the recommended storage conditions or the conditions described above should be validated by the user.

For further details about Grifols reagent red blood cells use and storage refer to their Instructions for Use.



NOTE: Please, contact your local Grifols service representative to know which reagent red blood cells can be tracked by Erytra.



NOTE: To enable the tracking of the on-board stability for red blood cells, please contact your local Grifols service representative.

7.3.2.1 Programming Samples

Manually Programming the Worksheet

Manual programming of the **Worksheet** can be done in 2 ways:

- **By Sample:** Selecting the field for the desired profile in the samples row.
- **By Profile:** Assigning a group of samples to the same profile.

Manual Programming By Sample

To assign tests to samples **By Sample**:

1. From the **Worksheet**, press the field in the samples row for the profile to be assigned to highlight the field.

Multiple fields can be highlighted by pressing and dragging down through those fields to select them.

The fields that were selected are highlighted in yellow to indicate that they have tests pending.

2. After assigning all of the desired profiles, press the **Start Test Execution** button  to begin testing.

Manual Programming By Profile

To assign tests to samples **By Profile**:

1. From the **Worksheet**, press the profile name at the top of the **Worksheet** for the profile to be assigned.

The **Sample Selection** window opens:

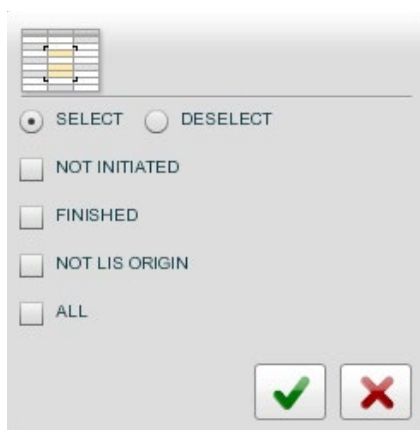


Figure 19. Samples Selection

2. Press the check box for the desired group of samples to assign the profile from the following choices:
 - Not Initiated: Selects all of the samples that do not have a running profile request (In Progress, Queued or Pending).
 - Finished: Selects all of the samples that have a finished test request (Pending Validation, Validated or Exported).
 - Not LIS origin: Selects all of the samples that do not have an LIS request for that profile.
 - All: Selects all of the samples.

Press the **Confirm** button to accept.

3. The samples that were selected are highlighted in yellow to indicate that they have pending tests.

4. After assigning all of the desired profiles, press the **Start Test Execution** button , to start test execution.

Automatically Programming the Worksheet

If the analyzer is connected to an LIS, the **Worksheet** can be programmed by downloading sample test requests after they have been entered in the LIS:

1. Load the appropriate samples into the analyzer, if this hasn't been done already.
2. From the upper right side of the **Status > Missing Samples** screen, press the **Download from LIS** button (Figure 20).

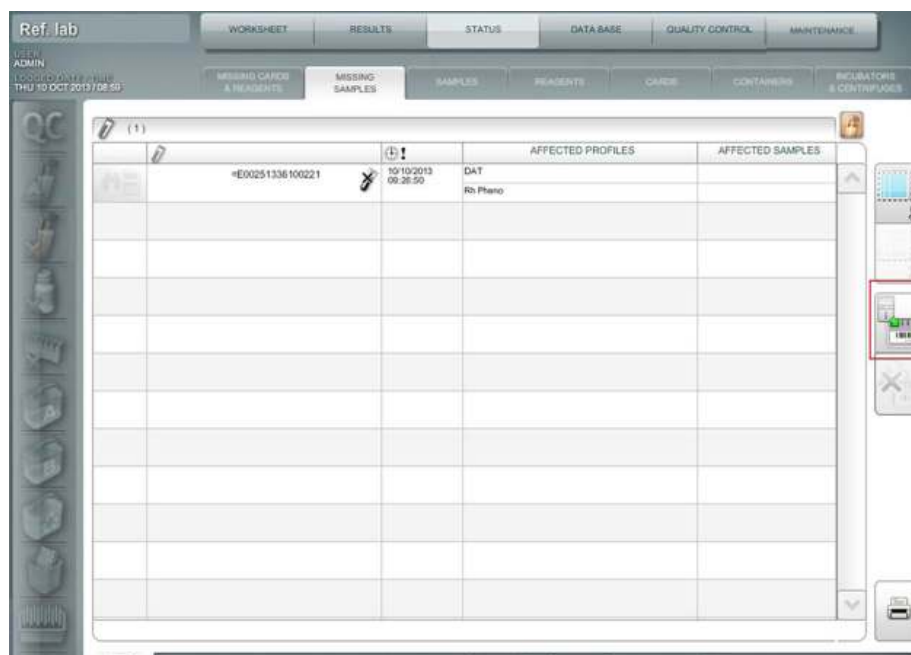


Figure 20. Status > Missing Samples Screen Showing the Download from LIS Button

3. If any samples appear on the **Status > Missing Samples** screen, load those samples into the analyzer.

The **Worksheet** is populated with the profiles and test assignments downloaded from the LIS and test processing begins automatically.

There is the possibility to configure the analyzer in a mode where all those requests related with samples that are not inside the instrument are returned to the LIS as cancelled. For more information, contact your local Grifols service representative.

It is also possible to configure an automatic query to every time a samples drawer (containing either routine or STAT samples) is closed by the user, in order to automatically download the worklist. For more information about this option, contact your local Grifols service representative.

Likewise, receivment of demographic data from the LIS in the instrument it is possible under configuration. For more information about LIS connectivity mode configuration, contact with your local Grifols service representative.



NOTE: LIS downloading connection with demographic data option enabled is the only possible entrance of Personal Data in the analyzer. The demographic data the instrument can manage is limited to Name, Surname, Gender, Birthdate and Maiden name when available. Any extra information received from the LIS will be ignored. The laboratory based on their own policies can include Personal Data in each test request.



WARNING: When the sample information is received from the LIS, it also includes the Patient ID. This Patient ID should meet the definition of “pseudonymisation” given in the corresponding regulations, as it will not be managed as Personal Data by the instrument.



WARNING: The Personal Data transmitted by the LIS to the instrument is stored in the log files for traceability and data integrity purposes. If instrument is configured to display demographic data, this data is also stored in **Database**.

7.3.2.2 STAT Programming

When a sample needs to be loaded into the analyzer and processed before the samples previously programmed on the **Worksheet**, the sample must be loaded as **STAT** (). To load STAT samples, proceed as follows:

1. Press the **STAT**  button on the front of the analyzer. A sample drawer opens.



NOTE: If there are no free locations in any of the sample holders, drawer 6 will open. Remove the necessary number of samples from the holder. These samples will appear on the **Status > Missing Samples** screen.

2. Load the STAT sample into the holder.
3. Close the drawer carefully to avoid splashes. The Erytra software identifies the recently loaded samples as STAT samples.
4. Send requests for the STAT samples from the LIS.



NOTE: If an LIS is not used, program the desired profile(s) to the STAT samples from the **Worksheet** and press the **Start Test Execution** button (see Section 7.3.2).



The analyzer displays the STAT samples on the **Worksheet** with the **STAT** icon  and begins processing the STAT sample as soon as possible.

Designating Samples Already in the Analyzer as STAT

To designate as STAT samples that are already inside the analyzer:



1. From the **Worksheet**, press the **STAT Assignment** button (Figure 16, no. 7).

The **STAT** screen opens and displays two lists. On the left, is a list of all of the samples that are currently inside the analyzer and on the right is a list of the samples to be programmed as STAT.

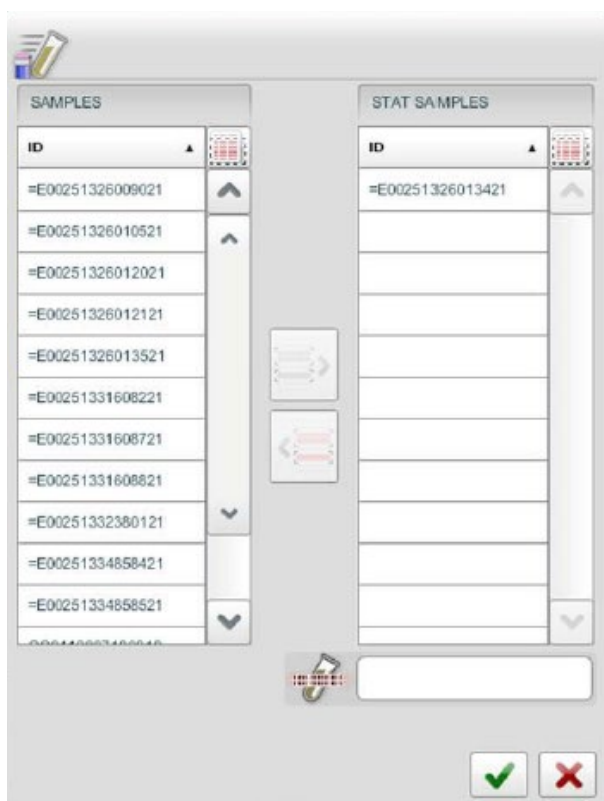


Figure 21. STAT Screen

2. Select the sample(s) to mark as STAT from the list on the left.

3. Press the **Move Sample** button .

The samples assigned as urgent will appear on the right-hand side of the screen.

4. Press the **Confirm** button .

The STAT samples will appear on the **Worksheet** with the barcode identification and the **STAT**

icon  and they will be executed as soon as possible. From the **Status > Incubators and Centrifuges** screen, the device that contains the STAT sample(s) will also display the **STAT** icon on it.

7.3.2.3 Working with a Double Tube

To work with double tubes on Erytra, a parameter must first be configured in the software. Contact your local Grifols service representative to configure the double tube parameter.

When using double tubes, the 2 tubes with the same barcode should be placed in consecutive positions in the same samples holder. The Erytra assumes that the tube closest to the front of the instrument is the plasma/serum tube and that the other tube is the RBC tube. Once the tubes are identified, only one of them will appear in the **Worksheet** with the **Double Tube**



icon

In the **Status > Samples** screen, the double tubes appear flagged either with a red circle for an RBC tube or a yellow circle for a plasma/serum tube:

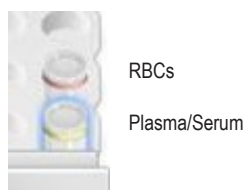


Figure 22. Double Tube in the Status > Samples Screen



WARNING: It is the responsibility of the user to load double tube samples in the positions expected by the Erytra. Placing them in the incorrect position can lead to incorrect results.



WARNING: In case of using sample tubes containing RBC and Serum, the instrument is not able to detect that RBC are properly sedimented and that there is enough serum over them for the test.

If 2 tubes with the same barcode are introduced in the analyzer in non-consecutive positions in the same holder:

- The tubes will not appear in the **Worksheet**.
- The warning related to sample incidence will be activated.
- An error message will appear on screen.
- The tubes will appear in red in the **Status > Samples** screen.

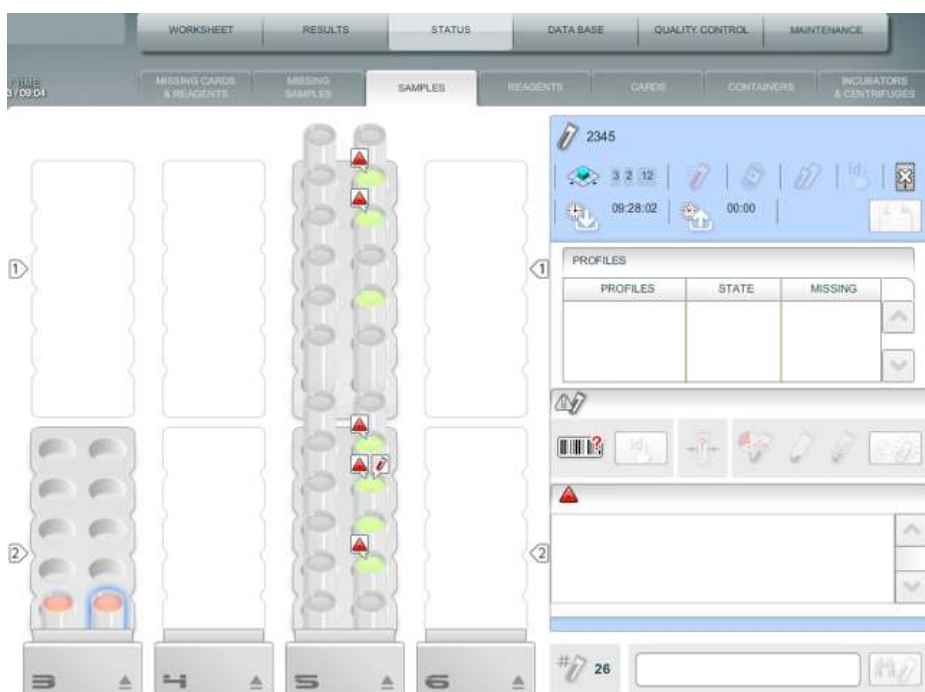


Figure 23. Image of Status > Samples with 2 Tubes with the Same Barcode Incorrectly Positioned

7.3.2.4 Programming a Crossmatching Profile

A crossmatch profile can be programmed in the followings ways, depending on whether or not the instrument is connected to a Laboratory Information System (LIS):

- For instruments connected to an LIS: The LIS automatically sends the assignment of the donors and matches them with their corresponding recipients when the workload is downloaded from the LIS to the **Worksheet**.
- For instruments not connected to an LIS: The donor and its respective recipients will be assigned manually from the **Worksheet** by following the procedure below:

Manual Crossmatch Programming

To add a new Recipient-Donor profile assignment manually:

1. Load the samples into the analyzer by following the instructions in Section 7.3.1.3.
2. Press the **Donor Assignment** button  that appears on the **Worksheet** (Figure 16, no. 6).
 - The **Crossmatching** window opens (Figure 24, no. 1) you can scan the barcode of the patient sample by using an external barcode scanner. Make sure that the barcode you have just scanned appears highlighted in blue from the display shown before going to the next step.
 - This window displays information about all of the current Donor-Recipient assignments and the programmed profiles.

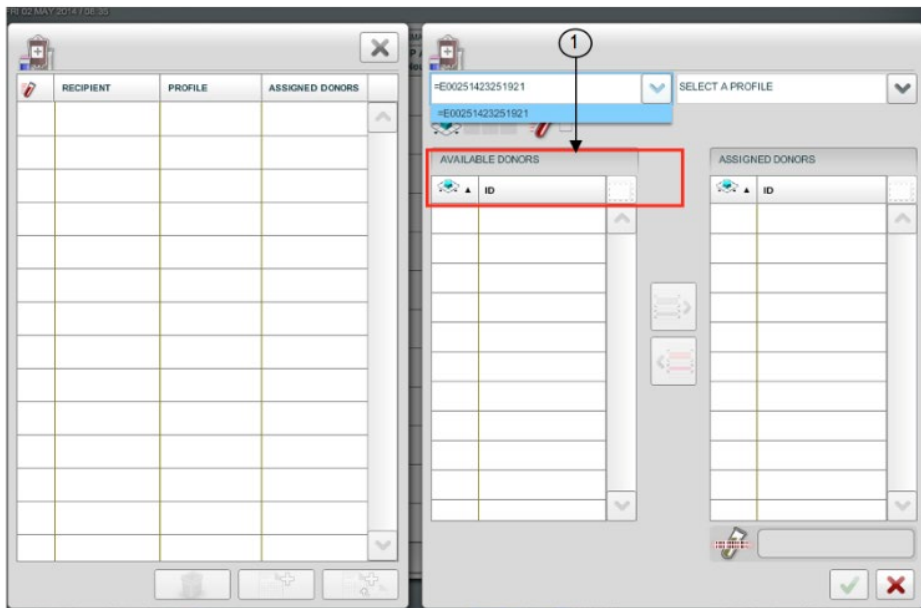


Figure 24. Crossmatching Programming Window for Assigning Recipients (Example)

- (1) Window for scanning the barcode of the patient sample by using an external barcode scanner.

The assignments can be modified by pressing the following buttons:



Delete all of the Recipient-Donor profile assignments selected in the table above.



Add new Recipient-Donor profile assignments using the **Donor Assignments** window (Figure 24).



Duplicate the Recipient-Donor profile assignments selected in the table above to assign profile requests where only the recipient varies.

3. Select the crossmatch profile from the list of available crossmatch profiles (Figure 25, no. 1).

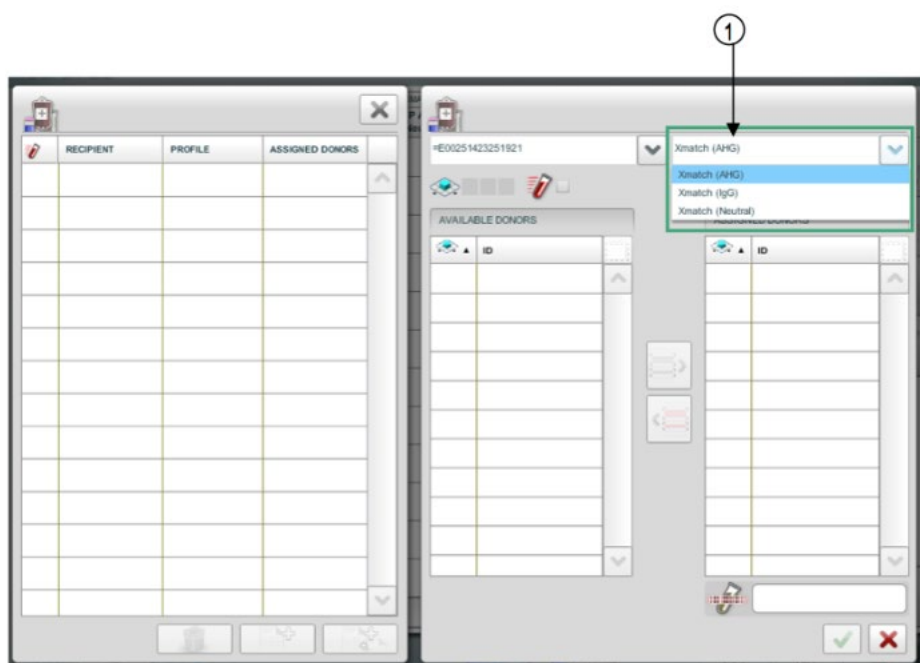


Figure 25. Crossmatch Programming Window for Assigning the Crossmatch Profile

- (1) Window for selecting the crossmatch profile.

4. Press the **Donor ID External** entry window  (Figure 26, no. 1). Using an external barcode scanner, scan the donor unit ID barcodes.

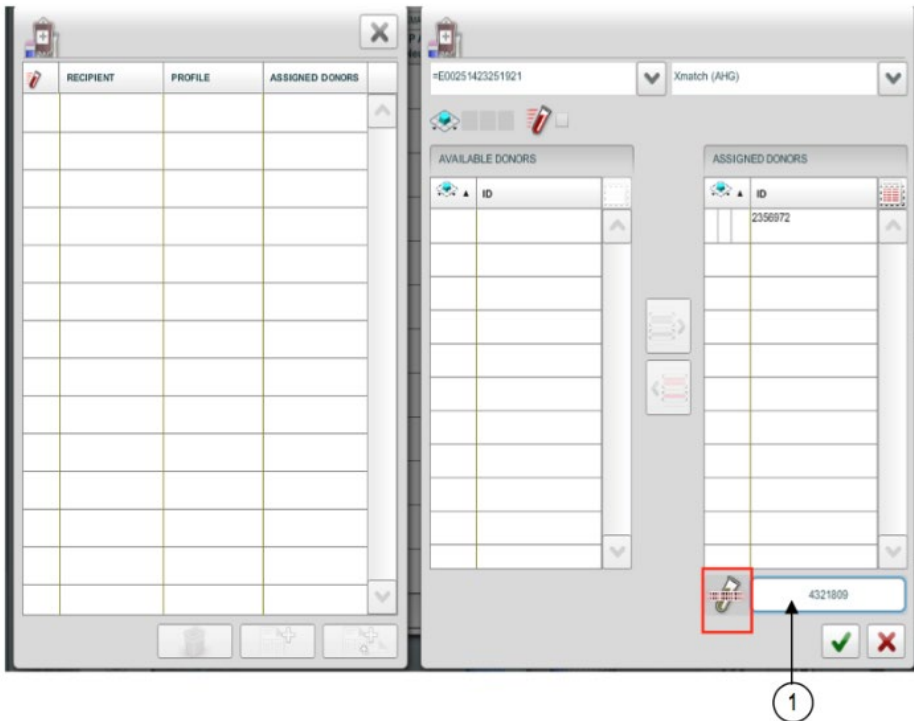


Figure 26. Crossmatch Programming Window for Assigning Donors with an External Barcode Reader (1 as Example)

5. Press the **Confirm** button .
6. Once finished editing the crossmatch list, close the **Donor Assignment** window by pressing



7. Samples that are not on board on Erytra will appear on the Missing Samples list, as shown in Figure 27.

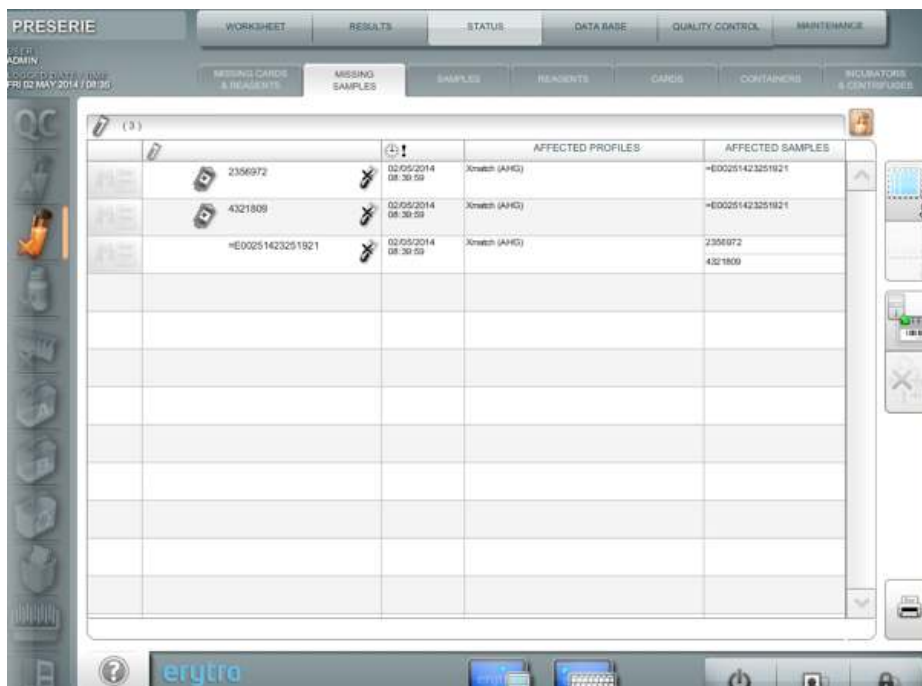


Figure 27. Missing Samples After the Crossmatch Programming

In case that any of the missing samples is not going to be processed, select the desired one



and cancel its execution with the **Cancel** button

8. Once the missing samples are added, text execution will start automatically.

7.3.2.5 Ordering Samples on the Worksheet

The samples appear by default ordered according to the following criteria:

- STAT samples.
- Samples with incidents.
- Position.

If you wish to change the order of the samples on the **Worksheet**:

1. Press the **Sort** button  that appears on the top left-hand side of the screen.
2. Choose from the following criteria:
 - **Barcode:** The samples are sorted alphanumerically, according to their barcode.
 - **Position:** The samples are sorted by location in the four available drawers, from drawer 3 to 6.
 - **Incident:** Any sample with an incident associated with their results will appear first on the **Worksheet**.
 - **STAT:** Any sample with STAT status will appear first on the **Worksheet**.
3. Press the **Confirm** button  to sort by the selected criteria. To close the window without changing the sort order, press the **Cancel** button.

QC Samples do not follow the above conditions and are also displayed at the end of the list.

7.4 Executing the Workload

If the Erytra is connected to an LIS, the execution of the workload begins automatically once the test requests are sent if the samples have been loaded into the analyzer.

To manually execute the programmed workload, press the **Start Tests Execution**  button on the **Worksheet**:

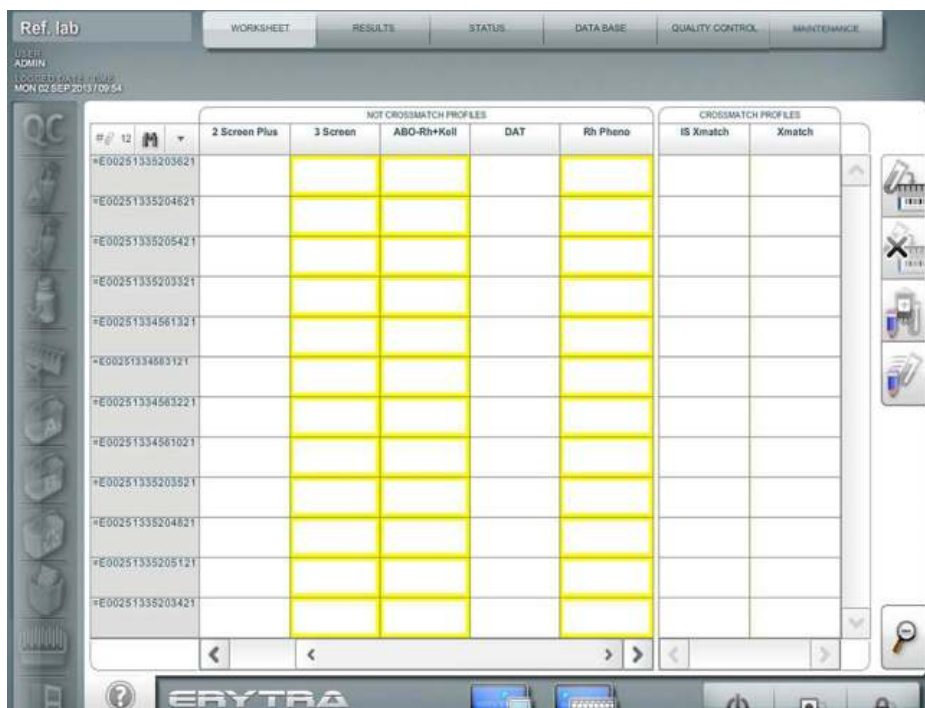


Figure 28. Workload

The Erytra software indicates the status of requests by outlining each field with an assigned profile in color, according to the color code shown in Figure 29:

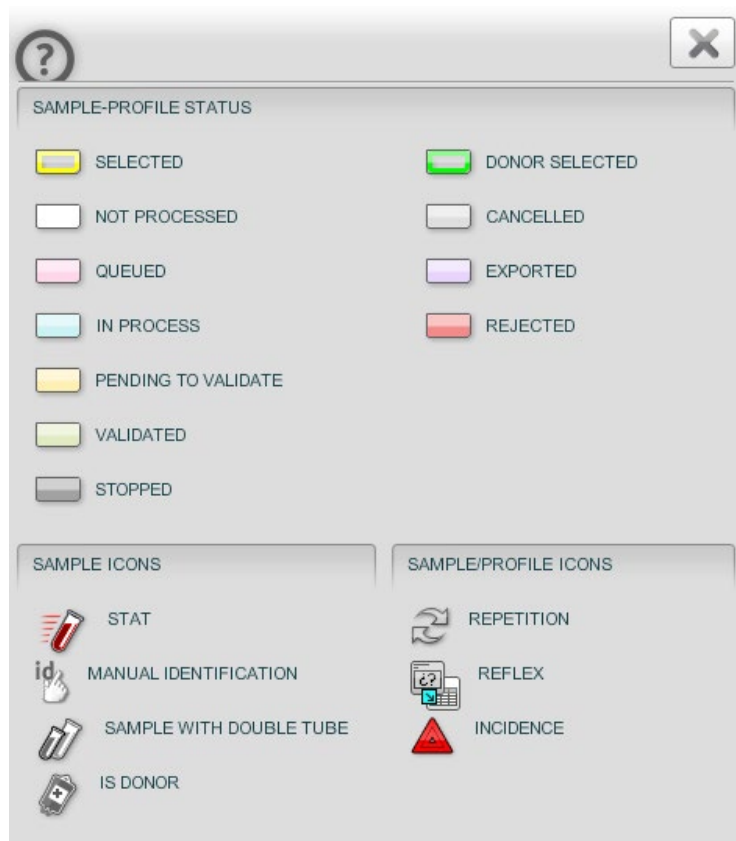


Figure 29. Color Code of the Worksheet

Once test execution begins, the Erytra performs a series of tasks as indicated by the flowchart in Figure 30:

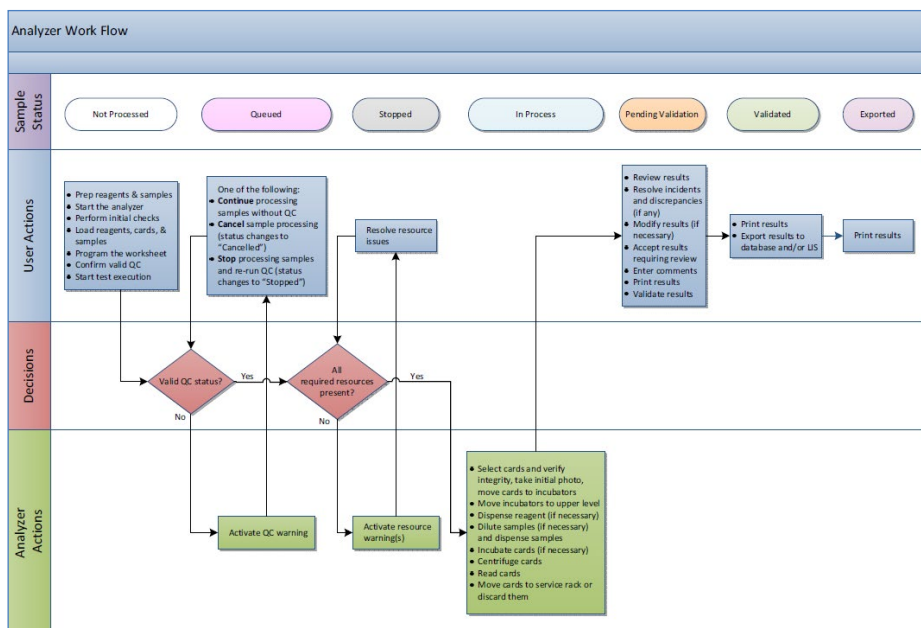


Figure 30. Erytra Flowchart

To cancel an unfinished request, those which have a status of either "In process", "Queued" or

"Pending", select them on the **Worksheet** and press the **Cancel Test Execution** button (Figure 16, no. 5).



7.5 Loading Missing Items and Resolving Resources Issues

Before executing the programmed workload, the Erytra analyzer automatically performs an assessment of the resources needed (reagents, diluent, cleaner solution, Grifols gel cards, washing solutions and waste capacity) to execute the set of programmed profiles. Also, during the execution of profiles, the Erytra analyzer software monitors in real time:

- The status and the distribution of samples in the analyzer.
- The status and the distribution of reagents and diluents in the analyzer.
- The status of washing solutions and the capacity of the cards waste container and the system solution waste containers.
- The status and the distribution of Grifols gel cards in the analyzer.

When a system component is missing, or if there are not enough resources to execute all of the programmed profiles, the icon for the corresponding resource will light up in the **Warning Area** (Figure 8, no. 7) and the color of the traffic light (Figure 1, no. 8) will change to yellow, the execution of the related tests will be stopped and the status of the requests in **Worksheet** will be updated to **"Stopped"** status.

Press the illuminated **Warning** icon, to open a screen with information about what should be done to resolve the incident.



CAUTION: If the tracking of the on-board stability is ignored, the user will be able to use the expired red blood cells and is considered to act under his own responsibility.



NOTE: The lack of resources does not interrupt the execution of other requests which have sufficient resources, those being part of this or of other profiles. The analyzer will continue with the execution until it has no more pending requests with available resources.



NOTE: Requested resources may be loaded at any time. However, if the lack of resources affects STAT requests, the analyzer continues with the requests that can be executed but it does an acoustic signal and shows an error screen listing the required resources issues that must be resolved immediately. To obtain further information, the window gives access to:

- **Missing Samples** in order to obtain information about the nature of the requirement.
- **Continue** to close the screen. The STAT samples go to a *Pending* status and will not be processed.

7.5.1 Missing Reagents, Diluents or Cleaner Solution

If a reagent, diluent or cleaner solution is missing, the on-board stability tracking is enabled and expired or if their volume is not sufficient in order to execute the set of programmed profiles, the



Missing Reagents, Diluent and/or Cleaner Solution Warning icon will appear in the **Warning Area**.

In order to solve this requirement issue, proceed as follows:



1. Press . The **Missing Reagents and Cards** screen displays information about the reagent, diluent and/or cleaner solution that must be loaded to execute the programmed workload.
2. Open a reagents drawer with free locations using the corresponding button.
3. Remove the empty vial or bottle (if necessary) and place a new vial or bottle in the holder, with the barcode facing the identification window of the corresponding holder.
4. Close the drawer carefully to avoid splashes. The Erytra performs the following actions:
 - Checks for the presence of the vials and bottles.
 - Identifies the reagents and/or diluent loaded.
 - Verifies that the reagents and/or diluent have not expired and still have remaining on-board stability, if configured.
 - Checks that RBC reagents are not loaded into the holders for reagents that do not require re-suspension.

The **Warning** icon is deactivated and the **Status > Reagents** screen is updated.

The pending workload is executed.

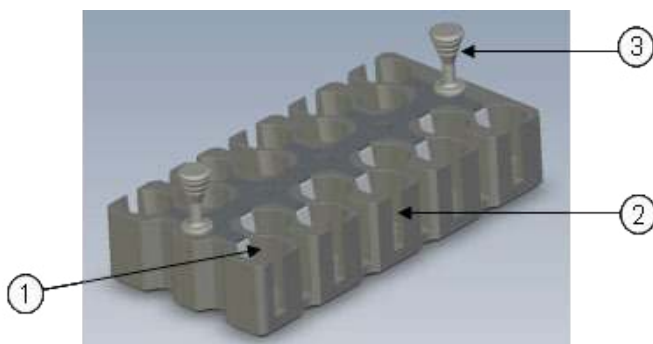


Figure 31. Holder for Reagents which Require Re-suspension

- (1) Locations for vials of reagents which require re-suspension.
- (2) Identification window.
- (3) Support handles.

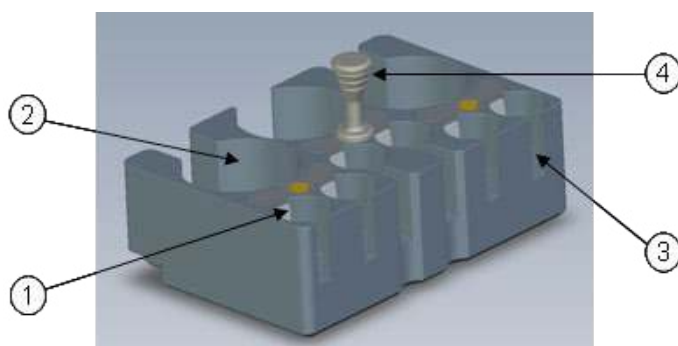


Figure 32. Holder for Diluents and Reagents which do not Require Re-suspension

- (1) Locations for vials of reagents which do not require re-suspension.
- (2) Locations for bottles of Grifols diluent. The DG Clean Adaptor has to be placed in any of these diluent positions.
- (3) Identification window.
- (4) Support handle.

Both reagent holders must be placed in the correct direction within each reagents drawer of the Erytra analyzer. Before closing the drawer, ensure that the holders fit perfectly in their corresponding positions. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

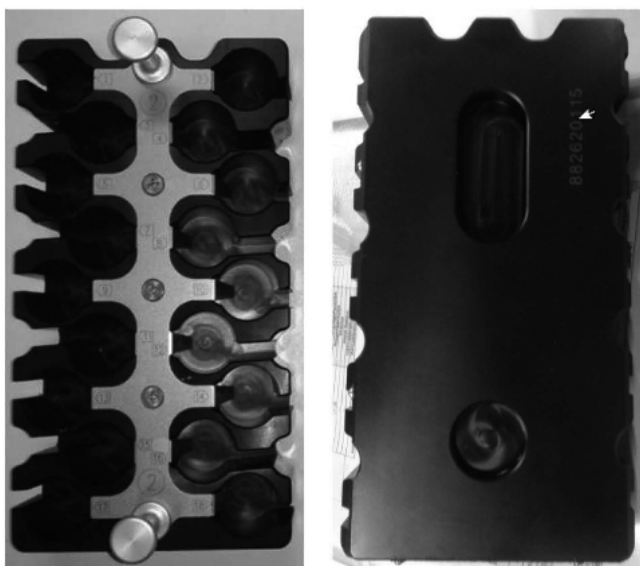


Figure 33. View of the Positions of the Shaken Reagents Holder

The reagents and diluents can be placed in any of the locations on the reagent test tube holder which have been designated to them, as the system will locate and identify every one of the reagents and diluents present in the Erytra analyzer by reading their barcodes.



CAUTION: In order to correctly execute the test, one must introduce the tempered and re-suspended reagents, the vials must be completely dry and with the barcode label in good condition.



WARNING: Remove the caps from the vials before introducing them into the analyzer.



WARNING: The exchanging of caps or the incorrect placement of the system solution containers, could be a source of crossed contamination between the reagents and between the samples and could lead to incorrect results.



WARNING: For the laboratory's own reagents, do not use reagent vials different to those specified in Section 3.2.

If you desire further information, please contact your local Grifols service representative.



CAUTION: Do not keep the reagents inside the analyzer for a long period of time as the conditions may not be of an optimum nature. Please consult the reagent user instructions for further details.



WARNING: In order to avoid splashes inside the analyzer, do not use vials which are too full.



CAUTION: The presence of foam, bubbles, or drops on the reagent vial walls may affect the dispensing of the reagent, causing a possible error in the results.



WARNING: The cleaner solution should be loaded inside the analyzer when required by the concerned test executed.



CAUTION: Expiration date of reagents is only contemplated and handled by the instrument during 5 years.



NOTE: Grifols reagent red blood cells can be placed on the instrument for up to 5 days when operating continuously on-board, without unloading the vials.

Other scenarios that vary from the recommended storage conditions or the conditions described above should be validated by the user.

For further details about Grifols reagent red blood cells use and storage refer to their Instructions for Use.

7.5.2 Missing Grifols Gel Cards

If there are not enough Grifols gel cards to execute the programmed profiles, the **Missing**



Grifols Gel Cards Warning icon appears in the **Warning Area**.

In order to solve this requirement issue, proceed as follows:



1. Press . The **Missing Reagents and Cards** screen displays information about the Grifols gel cards that must be loaded to execute the programmed workload.
2. Open a cards drawer using the corresponding button.
3. Remove the empty card racks (if this is the case) and replace them with a rack of new

Grifols gel cards with the barcode facing as described by the drawing on the base



4. Close the drawer. The Erytra automatically identifies the Grifols gel cards loaded. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

The Grifols gel cards can be placed in any of the locations designated to be occupied by cards, except for the locations which are configured as Service Racks. The system will locate and identify the type of cards present in every card location.

To speed up the identification process, the instrument reads the first Grifols gel card in the rack and then assumes that all the rest of the Grifols gel cards in the rack are of the same type. During test processing, each Grifols gel cards will be identified individually. If the instrument can not read any of the first three Grifols gel cards, those cards in that rack appear in red on the **Status > Cards** screen and can not be used for processing until they are correctly identified by the instrument.



WARNING: The Grifols gel cards should be loaded into the Erytra analyzer directly in the original packaging racks.



CAUTION: Do not use the locations designated as Service Racks to load the Grifols gel cards. The Service Racks are used by the analyzer only to hold only those cards which must be reviewed or reused by the Operator.



CAUTION: Do not keep the Grifols gel cards inside the analyzer during a long period of time as the conditions may not be of an optimum nature. Please consult the Instructions for Use of the card for further details.



CAUTION: In the event of artefacts or drops found, follow the DG Gel card Instructions for Use.

7.5.3 Missing Samples

If any of the samples which are required to execute the programmed profiles are missing, the



Missing Samples Warning icon appears in the **Warning Area**.

To load missing samples, proceed as follows:



1. Press the **Missing Samples Warning** icon. The **Missing > Samples** screen displays information about the samples that must be loaded to execute the programmed workload.

2. Open a samples drawer using the corresponding button.
3. Load the required samples or resolve the incident by pressing the **Set New** button

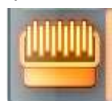


4. Close the drawer carefully to avoid splashes. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

The Erytra automatically identifies the samples loaded. The **Warning** icon is deactivated and the **Status > Samples** screen is updated. The pending workload is then executed.

7.5.4 Emptying the Service Rack

If the Service Racks are full or almost full and need to be emptied before the Erytra can execute



the programmed profiles, the **Service Racks Warning** icon appears in the **Warning Area**.

The racks in position 4 of card drawers 3 and 4 (Figure 5, no. 3) are designated as the Service Racks by default.



CAUTION: Review the cards discarded in the Service Rack. These cards should be reviewed by the Operator, as they may have a result with an incident or a special result associated with them (see Section 7.6).

The Service Racks may also hold Grifols gel cards with free microtubes which can be reused.

To empty the Service Rack, proceed as follows:



1. Press . The **Status > Cards** screen displays information about the distribution of Grifols gel cards inside the analyzer.
2. Open the Service Rack drawers through the **Status > Cards** screen or by using the corresponding push-buttons.
3. Remove all fully used Grifols gel cards in the Service Racks that have results to be reviewed and leave all partially used Grifols gel cards in the Service Racks to be reused by the instrument.

4. Close the drawer.

The Erytra checks for the presence of Grifols gel cards in the Service Racks and identifies them.

The **Warning** icon is deactivated and the **Status > Cards** screen is updated.

The pending workload is executed.

7.5.5 Missing System Solutions (System Solution A or System Solution B)

If either System Solution container (A or B) is empty or almost empty, or there is not sufficient volume to execute the set of programmed profiles in the Erytra analyzer, a **System Solution**

 or  will appear in the **Warning Area**.

To refill the system solution containers:

-  or  (depending on the case).
1. Press

The **Status > Containers** screen displays information about the volume of System Solutions (A and B) available in the analyzer.

2. Dilute the contents of a bottle of System Solution A or B with purified water until it reaches a final volume of 2 liters.
3. Open the appropriate system solution container drawer from the **Status > Containers** screen.
4. Remove the lid and refill the appropriate system solution container.
5. Repeat the process in steps 2 and 4 to add the required amount of system solution to the container (total capacity 8 liters).
6. Close the drawer. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

The **Warning** icon is deactivated and the **Status > Containers** screen is updated.

The pending workload is executed.



NOTE: Although the system solutions can be directly refilled in the analyzer, the containers can be removed from the analyzer for convenience of the Operator and also for maintenance tasks.



CAUTION: The use of different system solutions or solutions at different concentration to that specified, can be the cause of crossed contamination by insufficient washing and can cause incorrect results.



CAUTION: If any kind of deformity of the tubing, connectors or the system solution containers themselves is observed, discard them and replace them with new ones, as they could cause significant damage to the analyzer.



CAUTION: Do not use system solutions after their expiry date.

The use of degraded system solutions can be the cause of crossed contamination by insufficient washing and can cause incorrect results.



NOTE: It is recommended to record the system solution lot information in the System Solution Registry. For more information, see Section 11.2.4.

7.5.6 Emptying the Waste Solution Containers (W1 and W2)



The **Waste Solution Container Warning** indicates that the waste solution container must be emptied.

This signal will not activate on those instruments that have the waste liquid configured to drain to the laboratory sink.

In order to empty the solution waste drawer, proceed as follows:



1. Press . The **Status > Containers** screen displays information about the capacity available in the waste solution containers.
2. Open the appropriate waste solutions drawers from the **Status > Containers** screen.
3. Disconnect the tubing from the container and remove it from the analyzer.
4. Remove the lid from the container, empty the container and replace the lid.
5. Put the empty container back in the analyzer.
6. Reconnect the tubing to the container.
7. Close the drawer. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

The warning is deactivated and the **Status > Containers** screen is updated. The pending workload is executed.



WARNING: The waste solution containers should be emptied regularly. Handle, label and dispose of all liquid waste into special containers designated to the disposal of such waste, which must be eliminated according to the local regulations and practices.

7.5.7 Emptying the Card Waste Container



The **Card Waste Container Warning** indicates that the card waste container is full and that must be emptied.

The card waste container can use disposable, sealable bags (see Section 4.3) to prevent contact between the Operator and the discarded Grifols gel cards.

In order to empty the used cards container, proceed as follows:



1. Press . The **Status > Containers** screen displays information about the capacity available in the card waste container.
2. Open the card waste container drawer from the **Status > Containers** screen.
3. Remove the protective strip from the adhesive bag on the flap and close it, sticking the strip of the flap onto the opposite side of the bag.
4. Lift the bag and remove it from the container.
5. Place the bag into a laboratory rigid container for biohazardous waste.
6. Place a new empty bag into the card waste drawer, conforming it as much as possible to the shape of the containers, overlapping the sides of the container.
7. Close the card waste drawer. If a drawer remains open for more than 1 minute, an acoustic signal sounds to alert that it needs to be closed.

The **Warning** icon is deactivated and the **Status > Containers** screen is updated. The pending workload is executed.



WARNING: The contents of the waste drawers must be emptied regularly. Handle, label and dispose such waste in accordance with site procedures and local regulations and practices.

7.6 Displaying and Validating Results

7.6.1 Displaying and Generally Validating Results



CAUTION: The results obtained by the Erytra analyzer must be reviewed and validated by qualified personnel with the appropriate rights.



NOTE: The Erytra software has been designed based on web technology. This permits the authorized users to access the **Results Lists** not only from the Erytra analyzer but also from any remote desktop using http protocol (Remote Client). The remote access maintains the user privileges which are configured locally.

For further information about the activation of this service, contact your local Grifols service representative.



NOTE: Regarding to titration techniques and their results, each laboratory should establish the antibody titer threshold using the titration procedure, taking into account clinical findings and laboratory data to ensure meaningful interpretation based on its own titration values.

Results Overview

The Erytra provides a detailed analysis of test results and makes it possible to view the images of each test, the reaction grade and the interpretations.

In addition, the software provides constant information on the status of the **Quality Control** configured for the tests in the analyzer and the traceability of the reagents used.

All results must be validated before they are exported to the internal **Database** and the LIS (if configured). Validation of the results confirms acceptance and can be done individually, by sample, by profile, or in multiple ways, as long as this does not produce discrepant results, special results or results with incidents.

Erytra results fall into 4 categories:

- **Routine Results:** Results that the Erytra software is able to interpret do not to be accepted before being validated. See Section 7.6.1.
- **Special Results:** When the Erytra software is not able to provide an interpretation because some microtubes of the test have a reaction grade profile that cannot be classified, the result is considered a special result. See Section 7.6.3. The Erytra software can also be configured to list a particular type of result as a special result. Contact your local Grifols service representative for more information.

- **Results with Incidents:** During the execution of the test, incidents can occur which can affect the reaction grade. The software marks the microtube which is the cause of the incident with the **Microtube Incident** icon . See Section 7.6.3 for more information.
- **Results with Inconsistencies:** An inconsistency is considered a discrepancy between 2 or more results for samples that have the same patient identification number or sample barcode. If this option has been configured, the Erytra software checks the **Database** for any inconsistencies between results that are pending validation and any other result with the same patient ID or sample barcode. See Section 7.6.7 for more information.

When accepting a result during the validation step, take into account, in addition to the result itself, any incident deriving from the condition of the sample, the analyzer, the result itself, the **Quality Control** results, and any additional clinical information. Such incidents require acceptance prior to the validation of the result.

Managing and Printing Results

The Erytra is equipped with a **Database** (independent from the **Database** in any connected LIS) where all results are stored, including the images. All of these results and images are accessible to the Operator. See Section 7.6.9 for more information on working with the **Database**.

The Erytra provides different reports, which can be viewed and printed by the Operator at any stage after results have been generated. For more information, see Section 7.6.10.



Results without incidents and due to this they can be validated directly, without the need of a detailed review by the Operator.



Results with incident requiring review, acceptance and/or modification of the incident for its validation.

- (2) Automatic sample identification: When dealing with a crossmatch, the reagent ID appears in black while the donor ID appears in green and between quotation marks. The symbol



shows that the sample has been processed as an Urgent Sample.

- (3) **Profile:** Set of test(s) executed on the samples.
- (4) Individual **Test** which are part of the profile.
- (5) **Results:** Results with clinical significance which are obtained by the combining of the reaction grade of the microtubes which form a part of the realized test.
- (6) **Profile Status** (Pending validation, Validated).
- (7) **QC Status.**
- (8) **Date:** Date and time of the reading of the profile test.
- (9) Action buttons: Used to validate results and listed in the table below:



Filter: Search and view the results for that have not yet been exported and that match the determined search criteria (date, profile, state, barcode, incident, sample, etc.).



Delete Filter: Clears the current filter of the results being viewed and displays all of the results on the **Results List**.



Select All: Selects all of the active samples on the **Results List** screen. Once selected, the software allows you to carry out the combined action of **Validate**, **Cancel Validation**, **Export** and/or **Print**.



Deselect All: Deselects the previous selection.



Validate: The validation implies the acceptance by the user (with the necessary privileges) of ALL of the results of all of the selected profiles and is performed after the confirmation of the message that appears on the screen.

Only the profiles which show the icon  will be validated. Sample profiles which appear with the icon  should be individually reviewed from the **Profile Information** screen and the incidents or special results accepted by pressing the

Accept Results button  (see Section 7.6.3).



Results associated with the Delta Check , indicate that the system has encountered an inconsistency with a previous result and the result should be individually reviewed (see Section 7.6.7).



Cancel Validation: Once the results are validated, they can be changed back to the status **Pending Validation** by selecting them and pressing this button.



Reject: If a result is not accepted by a user with the necessary privileges, it can be rejected by pressing this button, before it has been exported. The requests and its results are deleted from the **Results List** and the results are saved in the **Database** of the Erytra software. In the case of having the LIS connection configured, the status of the result will remain as "Cancelled".



Export: Once a result is validated, it can be exported to the LIS and/or to the analyzer's own **Database**. The exportation is done after confirmation of the message that appears on the screen by a user with the necessary privileges. Once a profile's results have been exported, they disappear from the **Results** screen and the results are saved automatically in the **Database** of the Erytra. For more information see Section 8.



Print: Prints a report with the results of the selected sample profiles or results. For more information see Section 7.6.10.

Validating Routine Results

Results that do not have an associated incident, or are not special results, do not need to be accepted before being validated. To validate routine results:

1. From the **Results List** screen, select the sample profiles or results to be validated.



2. Press the **Validation** button . A window appears to confirm validation.



3. Press the **Confirm** button .



NOTE: If the users' policy is activated (see Section 10.1) the software will request the identification of a user with the necessary privileges (**User** and **Password**) during the **Validation** and **Cancel of Validation** processes.

Validated results will now appear in green in the **Results List** and show a status of "Validated". After validation, results can be exported.

Exporting Results

Export results send them to the Erytra internal **Database** and the LIS (if LIS connection is configured). Results must be validated before they can be exported.

To export results:

1. From the **Results** screen, select the results to export by pressing on each result or by



pressing the **Select All** button  to select all of the results on the **Results List**.



2. Press the **Export** button .



3. Press the **Confirm** button  to confirm.

The selected results are exported to the internal **Database** and to the LIS (if configured).

It is possible to configure that results without incidence are automatically validated and exported. Likewise, it is also possible to configure to validate and export automatically only specific profiles. To have more information about these options, please contact your local Grifols service representative.

7.6.2 Detailed Visualization and Acceptance of Profile Results

Press on  or  (Figure 34, no. 1) to obtain detailed information about the results. The software will show the **Profile Information** screen with all of the relative information related to the profile tests, including the images of the microtubes associated with every test, the interpretation and also the possible errors and associated incidents (Figure 35).

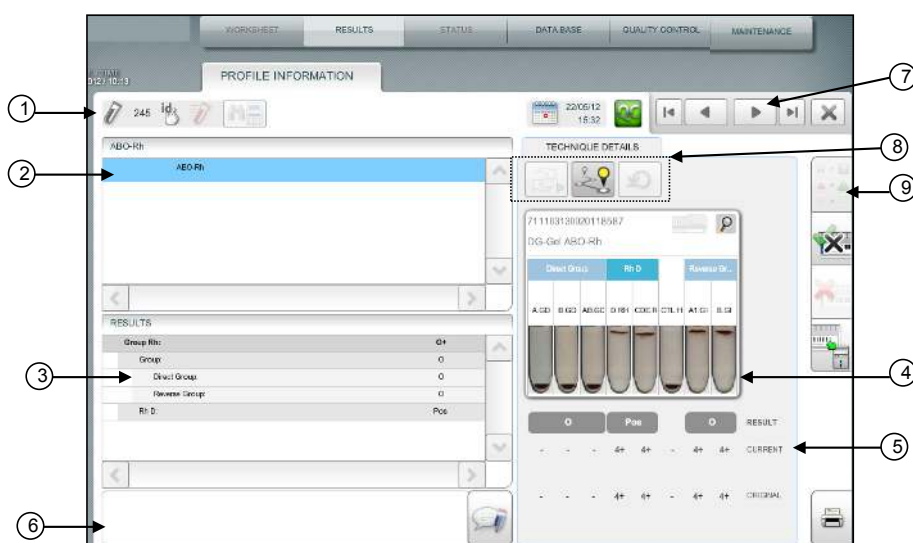


Figure 35. Screen Profile Information

- (1) General information of the sample.
- (2) Profile information and the test involved.
- (3) Results of the Tests.
- (4) Images and interpretation of the microtubes of the Grifols gel card associated with the test selected in (no. 2). In order to facilitate the review of results and of reaction grade, you can enlarge the image by pressing on the **Zoom** button .
- (5) Reaction grade information.
- (6) Comments.
- (7) Navigation buttons: Access the **Profile Information** of the results of the profiles which appear on the active **Results List** (Figure 35).
- (8) Special buttons: Described in the table below.



- (9) The **Accept Results** button  accepts all of the incidents or special results in the profile, including the results of the reaction grade of the microtubes. Once results have been accepted, the **Accept Results** button allows you to **Undo the Acceptance**.

Table with the description for special buttons of the **Profile Details** screen:



Displays information regarding the traceability of the sample profile within the analyzer.

Underneath the photograph of every microtube, there is a button with the reaction grade calculated by the Erytra software (Figure 35, no. 5).

The following table lists the reaction grades, symbols, and their descriptions:

Table 4. Reactions Grade

RESULT	REACTION GRADE	DESCRIPTION
Negative	-	Well-defined pellet of non-agglutinated red blood cells at the bottom of the gel column and no visible agglutinated cells in the rest of the gel column.
Doubtful	?	The reading algorithm is not able to decide between “-” and “+/-”. This is considered a Special Result (see below) and must be reviewed and manually assigned as a valid result. This may be caused by very few free red blood cells in the gel column or by small artefacts such as particles or bubbles in the gel.
	+/-	Barely visible small-sized clumps of agglutinated cells in the lower part of the gel column and a pellet of unagglutinated cells at the bottom. This is considered a Special Result (see below) and must be reviewed and manually assigned as a valid result.
Positive	1+	Some small-sized clumps of agglutinated cells most frequently in the lower half of the gel column. A small pellet may also be observed at the bottom of the gel column.
	2+	Small or medium-sized clumps of agglutinated cells throughout the gel column. A few unagglutinated cells may be visible at the bottom of the gel column.
	3+	Medium-sized clumps of agglutinated cells in the upper half of the gel column.
	4+	A well-defined band of agglutinated red blood cells in the top part of the gel column. A few agglutinated cells may be visible below the band.
Double Population	DP	A band of red blood cells at the top part of the gel or dispersed through the gel column, and a pellet in the bottom as a negative result.

A picture of an example of reaction grades is shown below:



NOTE: In case that the software is not able to assign any of the previous results to the microtube, a Not Determined Result (NR) will be provided.

Once the reaction grade in the different microtubes which form part of the test is classified, the Erytra software applies a mathematical algorithm to obtain the result and make an interpretation.

Reviewing and Accepting Results to Special Results

There are some situations where the Erytra software is not able to provide an interpretation because some microtubes of the test have an agglutination profile that can not be classified. These types of result are considered special results and are marked with the **Special Microtube**

Grade icon  or the **Special Interpretation** icon .

The Erytra software can also be configured to list a particular type of result as a special result. To configure a particular type of result as a special result, contact your local Grifols service representative.

A list of the types of results that are considered special results and their category is shown in the table below:

Table 5. Type of Results

ICON	RESULT	DESCRIPTION
	Special Microtube Grade	• Double Population (DP). • Doubtful reaction grade (?). • Doubtful positive reaction grade (+/-). • No Results Determined (NR). • Any result or reaction grades previously configured as a special result in the Erytra software.
	Special Interpretations	• Not Interpretable Results (NI). • Positive Autocontrol. • Positive Direct Antiglobulin Test.
No icon	Discrepant Interpretations	• Discrepancies (Dis): When the result of combining the subgroup (<i>i.e.</i> forward and reverse) of the possible phenotyping interpretations of the test does not allow the software to obtain a clinical result because it is incongruous.

A special result must be reviewed and individually accepted before it can be validated.

To review and accept a result to special result:

1. Press  next to the sample ID in the **Results List** to open the **Profile Details** screen.
2. In the Profile Information area, it is displayed the **Special Microtube Grade** icon  or the **Special Interpretation** icon  with the details about the special result (see Section 7.6.2).

3. Retrieve the processed Grifols gel card with the special result that has been stored in the Service Rack (if it has been configured to) and perform the following actions, as appropriate:



NOTE: To aid the review process, the analyzer can be configured to store any gel cards with incidences in the Service Rack so that they will be available for review.

Cards with special results and special interpretation can be configured to leave the affected cards in the Service Rack for further review of the user.

To configure the Service Rack, contact your local Grifols service representative.



NOTE: It is possible to use the external barcode reader provided with the instrument for scanning the barcode of the Grifols gel cards left in the Service Rack and review their results by introducing the scanned barcode in the proper field of the filter. For configuring the external barcode reader, contact your local Grifols service representative.

- Consult the patient's clinical information.
 - Read the Grifols gel card.
 - If appropriate, modify the reaction grade for the microtube that had a special result (see Section 7.6.8).
4. Repeat steps 1 - 3 for any other microtube of the sample profile that have special results.



5. From the **Profile Details** screen, press the **Accept Results** button to accept the results for that sample profile.



NOTE: To reject the results, press .

The request and its results are deleted from the **Results List** and saved in the **Database** of the Erytra software.

6. Press the **Confirm** button to confirm acceptance.
7. If requested, enter **Username** and **Password**, and press  if it has been configured to request it. Any **Special Microtube Grade** icons or **Special Interpretation** icons will change color from red ( or ) to green ( or ).
8. Add a comment, if desired, by pressing the **Comment** button .
9. To validate the entire profile, press the **Validate** button .



NOTE: Press  to return to the **Results List** screen, where this profile can be validated individually or along with other results.

7.6.3 Reviewing and Accepting Incidents

During the execution of the test, incidents can occur which can affect the reaction grade. The software marks the microtube which is the cause of the incident with the **Microtube Incident**

icon .

The possible incidents that can appear in the results are described as follows:

7.6.3.1 Incidents During Card Reading

The table below describes the different abnormal events that can be detected by the analyzer during the reading step, the results provided and the corresponding warning in the microtubes affected. A description of the most common actions that can be performed by the Operator is also included:

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
No RBC	NR	The analyzer has detected that the quantity of red blood cells in the microtube are below the expected range. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Confirm the absence or low quantity of RBCs. 3. If the incidence is confirmed: • Check that there are enough volume of reagent and its integrity. • Check that there are enough volume sample and its integrity. 4. Then reject the result by using the Reject button  and repeat the test. If it is a frequent incidence, contact your local Grifols service representative.

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
RBC excess	NR	The analyzer has detected that the quantity of RBCs in the microtube is higher than expected. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Confirm the excess of RBCs. 3. Identify the sample tube. 4. Check the integrity of the sample. 5. This incidence does not invalidate the test: • If the quantity of RBCs is acceptable according to the visual reading, therefore proceed to modify the result as described in Section 7.6.8 and accept the test result by pressing Accept Results button  . Then validate and export the result. • If the quantity of RBCs is not acceptable, reject the  result by using the Reject button and repeat the test. If it is a frequent incidence, contact your local Grifols service representative.

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
Fibrin haemolysis	NR	The analyzer has detected the presence of fibrin in the top of the microtube. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Confirm the presence of fibrin. 3. Identify the sample tube. 4. Inspect the sample tube and check for the presence of fibrin. 5. Only in those cases that the fibrin strand is clearly visible in the microtube image and can be distinguished from the reaction pattern, modify the NR result to a valid result as described in Section 7.6.8 and accept the test result by pressing Accept Results button . Then validate and export the result.
Haemolysis	Original result	The analyzer has detected haemolysis in the incubation chamber of the microtubes or the highest area of the gel column. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Confirm the presence of haemolysis. 3. Identify the sample tube.

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
		4. Inspect the sample tube and check if there is or not presence of haemolysis in the tube. - If haemolysis is present: - Check the historical data of the patient to confirm or discard the presence of potential autoantibodies. - Confirm that the blood samples has been collected following standard blood sampling guidelines, it has been stored in appropriate conditions and the extraction date. - Check the storage and centrifugation conditions of the sample. Then, if appropriate, accept the microtube result by using Accept Results button , validate and export. - If haemolysis is not present, reject the result by using the Reject button  and repeat the test.
Not classifiable	NR	The agglutination pattern presented by the microtube does not match any of the known agglutination patterns used by the analyzer. Proceed as follow: - Retrieve the card gel from the Service Rack and inspect it visually. - Check the result provided by the analyzer and modify the result according the visual reading. - Accept the microtube result by using Accept Results button . Then, validate and export.

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
Not detected volume	Original result	The analyzer has failed to detect the dispensed volume in the incubation chamber of the microtube. Proceed as follow: - Retrieve the card gel from the Service Rack and inspect it visually. To review it, check in the Instructions for Use of the product to identify the volume needed for the performed test. Then, compare the processed card volume with the expected volume for this test. An example of the approximated expected total volumes after centrifugation for each test are: 75 μL  100 μL  - If the incidence is confirmed, check that there are enough volume of reagent and sample, reject the result by using the Reject button  and repeat the test. - On the contrary, confirm the result provided by the analyzer or modify the result according the visual reading. Then, accept the microtube result by using Accept Results button .

WARNING	RESULT PROVIDED	DESCRIPTION AND INCIDENCE RESOLVING
Reading error	NR	An anomaly during the during the image digitalization process of the card makes impossible to obtain the results. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Modify the result according the visual reading of the gel card. 3. Accept the microtube result by using Accept Results  button If the incidence is recurrent, contact your local Grifols service representative.

7.6.3.2 Incidents During Test Execution

The table below describes the different incidences that could take place during execution and the corresponding warning in the microtubes affected. A description of the most common actions that can be performed by the Operator is also included:

DESCRIPTION	RESULT PROVIDED	TROUBLESHOOTING
Over-incubation	Original result	The incubation time of the gel card has exceeded the tolerable time (45 minutes). Proceed as follow:  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
Over-centrifugation	Original result	The centrifugation time of the gel card has exceeded the tolerable time. Proceed as follow:  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
Loss sensitivity	NR	The time between reading and the end of centrifugation exceeded the tolerable time. Proceed as follow:  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.

DESCRIPTION	RESULT PROVIDED	TROUBLESHOOTING
Missing validation step	NR	The gel card integrity check has not been performed.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
RBC to serum timeout	Original result	The time between RBC and serum dispensations has exceeded the tolerable time.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
Missing incubation step	NR	The incubation step has not been completed or executed.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
Missing pipetting step	NR	The pipetting step has not been completed or executed.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.

DESCRIPTION	RESULT PROVIDED	TROUBLESHOOTING
Missing centrifugation step	NR	The centrifugation step of the gel card has not been completed or executed.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.
Missing reading step	NR	The reading step of the gel card has not been completed or executed.  1. Reject the result by using the Reject button 2. Repeat the test. If the incidence is recurrent, contact your local Grifols service representative.

DESCRIPTION	RESULT PROVIDED	TROUBLESHOOTING
Incorrect volume	Original result	The total volume dispensed in the microtube does not coincide with the expected volume in test where serum/ plasma. Proceed as follow: - Retrieve the card gel from the Service Rack and inspect it visually. To review it, check in the Instructions for Use of the product to identify the volume needed for the performed test. Then, compare the processed card volume with the expected volume for this test. An example of the approximated expected total volume after centrifugation for each test are: 75 μL  100 μL  - If the incidence is confirmed, reject the result by using the Reject button  and repeat the test. - On the contrary, confirm the result provide by the analyzer or modify the result according the visual reading. Then, accept the microtube result by using Accept Results button .

To review and accept incidences, press the **Microtube Incident** icon  from the **Profile Information** screen near the image of the microtube to obtain more information about the incidence.

7.6.4 Reviewing and Accepting Special Microtube Grade Results

The table below describes the different special microcolumn grade agglutination which can be provided by the analyzer and the description of the most common actions that can be performed by the Operator:

DESCRIPTION	TROUBLESHOOTING
DP (Double Population)	The analyzer has detected a band of red blood cells at the top part of the gel or dispersed throughout the gel column, and a pellet in the bottom as a negative result. Precaution should be taken in the interpretation of Double Population events. - Retrieve the card gel from the Service Rack and inspect it visually. To review it, check in the Instructions for Use of the product to identify the volume needed for the performed test. Then, compare the processed card volume with the expected DP pattern for this test. An example of the approximated expected pattern for DP after centrifugation for each test are: 75 μL  100 μL  - Check for the absence of fibrin following the instructions described in Section 7.6.3. - If fibrin is discarded and the DP pattern is confirmed: - Read carefully the Instructions for Use of the gel card. - Check additional information on patient history about subgroups, transfusion or transplant events. - Follow the corresponding guidelines. - Perform additional testing, if necessary.

DESCRIPTION	TROUBLESHOOTING
	4. With this information: - Accept the microtube result by using Accept Results button . If DP result is accepted as a result, Not Interpretable result (NI) is reported by the analyzer in the result interpretation level, after combining the corresponding microtubes that form part of that interpretation. The result is ready to be validated and exported. - Modify the result according the visual reading. See more information in Section 7.6.8. Then, validate and export the result.
Doubtful reaction grade (?)	The analyzer has detected a very weak reaction pattern “?” in that microtube that is lower than “+/-” agglutination grade. Given that this reaction can be in the limit of the sensitivity of the reader, this result must be confirmed by the user. Proceed as follow: - Retrieve the card gel from the Service Rack and inspect it visually: - Accept the microtube result by using Accept Results button . If “?” result is accepted as a result, Not Interpretable result (NI) is reported by the analyzer in the result interpretation level after combining the corresponding microtubes that form part of that interpretation. The result is ready to be validated and exported. - Modify the result according the visual reading by assigning a valid result as indicated in Section 7.6.8. Then, validate and export the result.

DESCRIPTION	TROUBLESHOOTING
Doubtful reaction grade (+/-)	The analyzer has detected a weak reaction pattern in that microtube corresponding to a “+/-” agglutination grade. Given that this reaction can be in the limit of the sensitivity of the reader, this result must be confirmed by the user. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Proceed according of the guidelines of the lab: • Accept the microtube result by using Accept Results button . If “+/-” result is accepted as a result, Not Interpretable result (NI) is reported by the analyzer in the result interpretation level, after combining the corresponding microtubes that form part of that interpretation. The result is ready to be validated and exported. • Modify the result according the visual by assigning a valid result as indicated in Section 7.6.8. Then, validate and export the result.
Any result previously configured as a special result in the Erytra software	The analyzer has detected a reaction pattern that agrees to an agglutination that must be reviewed under customer request (see more information in the introduction of this section). Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Confirm the result provide by the analyzer. 3. Proceed according of the guidelines of the lab: • Accept the microtube result by using Accept Results button . If a regular agglutination (from - to 4+) is accepted, the analyzer combines the result of the defined microtubes to provide the result interpretation. The result is ready to be validated and exported. • Modify the result, if appropriate (see more information in Section 7.6.8). Then, validate and export the result.

DESCRIPTION	TROUBLESHOOTING
NR (No Result Determined)	The reaction shows a pattern that is not recognized by the reader. Proceed as follow: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Modify the results according the visual reading. See more information in Section 7.6.8. If a regular agglutination (from - to 4+) has been introduced, the analyzer combines the result of the defined microtubes to provide the result interpretation. The result is ready to be validated.

7.6.5 Reviewing and Accepting Discrepancies Results

Discrepancies between forward and reverse groups may be observed. Proceed as follow:

1. Retrieve the card gel from the Service Rack and inspect it visually.
2. If the visual reading confirms the results obtained by the user:
 - Read carefully the Instructions for Use of the used gel card.
 - Check additional information on patient history. Discrepancies between forward and reverse groups may be observed in patients with low or non-existent levels of isoagglutinins: Newborns up to the age of 4 - 6 months, elderly persons, patients with immunodeficiency or with much diluted antibodies due to plasma exchange procedures.
 - Follow the corresponding guidelines to resolve.
 - Perform additional testing if necessary.



- Accept the microtube result (if appropriate) by using **Accept Results** button . If the discrepant result is accepted as a result, **Not Interpretable** result (NI) is reported by the analyzer in the result interpretation level, after combining the corresponding microtubes that form part of that interpretation.

The result is ready to be validated.

3. On the contrary, modify the results according the visual reading. See more information in Section 7.6.8.

If a regular agglutination (from - to 4+) has been introduced, the analyzer combines the result of the defined microtubes to provide the result interpretation. The result is ready to be validated.

7.6.6 Reviewing and Accepting Specials Results

The table below describes the different special results which can be provided by the analyzer and the description of the most common actions that can be performed by the Operator:

Table 6. Specials Results

DESCRIPTION	TROUBLESHOOTING
NI (Not Interpretable)	Erytra software fails to obtain an expected interpretation when it combines the results of each microtube that form part of a phenotype interpretation (forward group, reverse group, group, phenotype, etc.). Proceed as follow to solve: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. Identify the cause of the NI: • If it due to a processing incidence, follow the procedure described in Section 7.6.3. • If it is due to a special microtube grade, follow the procedure described in Section 7.6.4. • If it is due to a discrepant result, follow the procedure described in Section 7.6.5. • If it is due to a Positive control microtube, follow the Instructions for Use of the corresponding gel card. Repeat the test, if necessary.
AUTOCONTROL NOT NEG (Autocontrol Not Negative)	Erytra software detects that a positive autocontrol. Proceed as follow to solve: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. If the visual reading confirms the results obtained by the user: • Read carefully the Instructions for Use of the used gel card. • Check additional information on patient history. • Follow the corresponding guidelines to resolve. • Perform additional testing if necessary. 3. Accept the results (if it is appropriate) or repeat the test.

DESCRIPTION	TROUBLESHOOTING
DAT NOT NEG (Direct Antiglobulin Test Not Negative)	Erytra software detects that positive direct antiglobuline test. Proceed as follow to solve: 1. Retrieve the card gel from the Service Rack and inspect it visually. 2. If the visual reading confirms the results obtained by the user: • Read carefully the Instructions for Use of the used gel card. • Check additional information on patient history. • Follow the corresponding guidelines to resolve. • Perform additional testing if necessary. 3. Accept the results (if it is appropriate) or repeat the test.

7.6.7 Reviewing Results with Inconsistencies

An inconsistency is considered a discrepancy between two results for samples that have the same patient identification number or sample barcode.

The Erytra analyzer can be configured to search for inconsistencies once a result has been obtained. Contact your local Grifols service representative to activate these inconsistencies check and set the time period from when inconsistencies should start to be reviewed.

Once configured, the system checks for inconsistencies between results that are pending validation and any other result with the same patient ID or sample barcode (depending on what has been previously configured) that is one of the following status:

- Pending validation.
- Validated.
- Exported.



If an inconsistency is encountered, the system will show the **Delta Check** icon on the **Results List** screen (Figure 36).

WORKSHEET RESULTS STATUS DATA BASE QUALITY CONTROL MAINTENANCE									
RESULTS LIST									
SAMPLE TYPE		DRAWER		PATIENT		STATE		NONE	
CARD BARCODE		REAGENT BARCODE		STATE		INCIDENCE		SPECIAL RESULT	
PROFILE		FROM		TO		INCONSISTENCIES			
7/2/13		07/02/2013		07/02/2013					
▲	SAMPLES (16/163)	PROFILE	TEST	RESULTS	STATUS	QC	DATE		
	#E00251318478521	ABO-Rh+Kell	Ag Kell	NRD	Pending validation		7/2/13		
			Group Rh	NRD					
	#E00251320070921	ABO-Rh+Kell	Ag Kell	Neg	Pending validation		7/2/13		
			Group Rh	A+					
	#E00251320071121	ABO-Rh+Kell	Ag Kell	Pos	Pending validation		7/2/13		
			Group Rh	A+					
	#E00251320071321	ABO-Rh+Kell	Ag Kell	NRD	Pending validation		7/2/13		
			Group Rh	NRD					
	#E00251321301021	ABO-Rh+Kell	Ag Kell	Neg	Pending validation		7/2/13		
			Group Rh	AB+					
	#E00251321301121	ABO-Rh+Kell	Ag Kell	Neg	Pending validation		7/2/13		
			Group Rh	A+					

Figure 36. Results Showing Inconsistencies

To review the inconsistency and validate the result:

1. Go to the **Profile Information** window by pressing .

2. Press the **Delta Check** button  to view more information about the inconsistency.

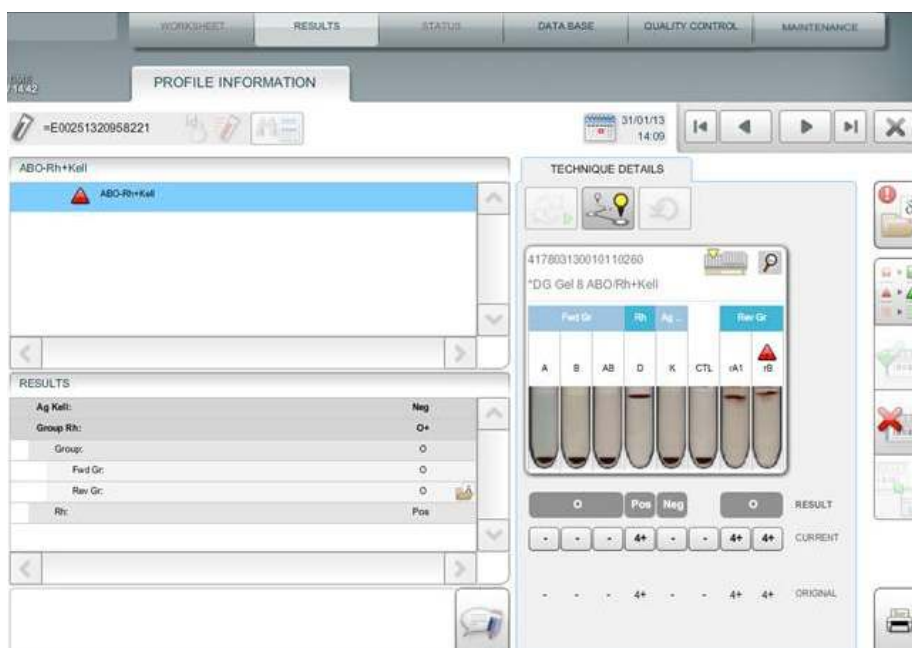


Figure 37. Detailed Result with Inconsistency



Once the inconsistency is accepted, the symbol of the **Delta Check** icon remains associated to that result if the result has a status of “Pending validation”, “Validated” or “Exported” (Figure 39). Therefore, the inconsistency, despite being accepted, can always be reviewed and the acceptance can be reversed. If the result is exported, the acceptance can not be reversed.

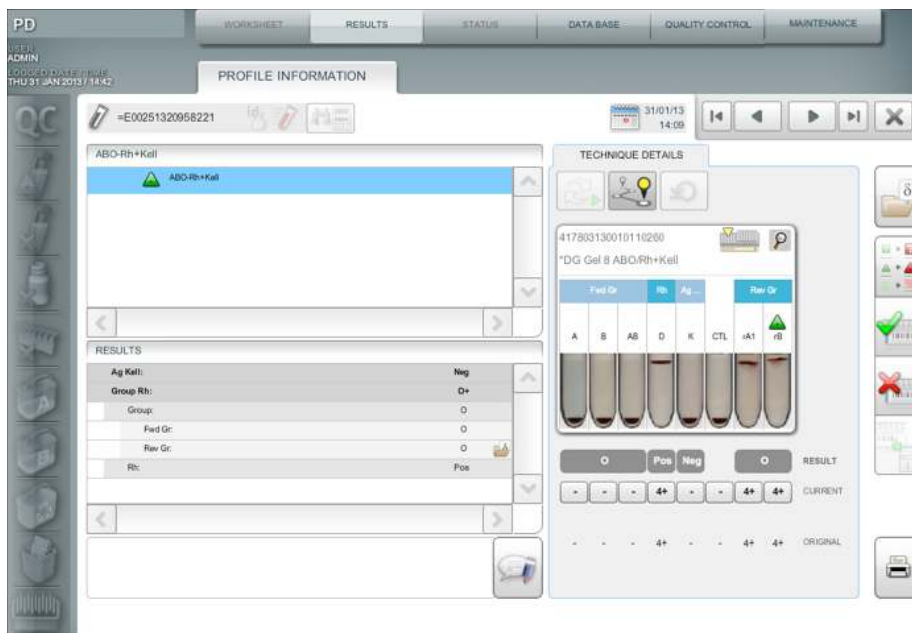


Figure 39. Reviewed Inconsistency

7.6.8 Modifying the Reaction Grade

It is possible to modify the reaction grade obtained by the analyzer by pressing on the reaction score (Figure 35, no. 5). The modifications will be updated on the line that corresponds to **Agglutination** or **Reaction Grade** and the Erytra software will automatically interpret the result using the new reaction grade. On the line Erytra **Agglutination** the software displays the result provided by the analyzer.

To modify the reaction grade:

1. Press on the reaction score to display the reaction grade drop-down list.
2. Select the appropriate reaction grade. A confirmation message displays.
3. Press the **Accept** button to accept.



NOTE: Press



to cancel the action.

4. Enter **Username** and **Password**, and press  if it has been configured to request it.
5. Add a comment, if desired, by pressing the **Comment** button .

The icon  is displayed next to any result that has had the original reaction grade modified.



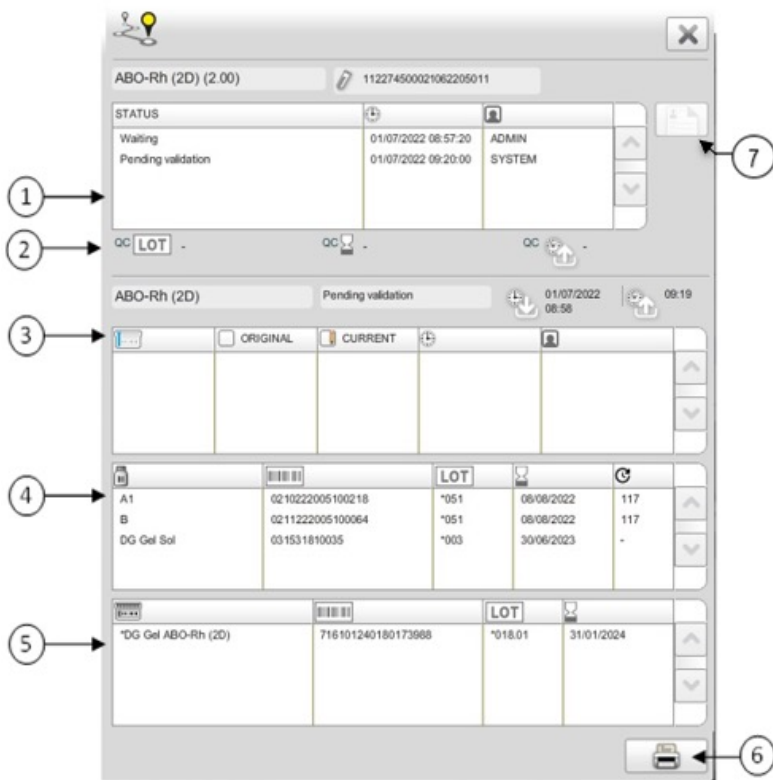
NOTE: Any modification made to the reaction grade is recorded by the Erytra software.

The modification of a reaction grade can modify the interpretation of a profile.

7.6.9 Viewing Traceability Information

It is possible to obtain all the information regarding the traceability of the test and the sample

executed. To do this, press on the **Traceability** button  on the **Profile Information** screen. A window opens (Figure 40) showing all the information related to the traceability of the execution:



ABO-Rh (2D) (2.00) 112274500021062205011

STATUS	Time	User
Waiting	01/07/2022 08:57:20	ADMIN
Pending validation	01/07/2022 09:20:00	SYSTEM

QC LOT - QC - QC

ABO-Rh (2D) Pending validation 01/07/2022 08:58 09:19

ORIGINAL	CURRENT	Time	User

LOT	Time	User
A1	0210222005100218	*051 06/06/2022 117
B	0211222005100064	*051 06/06/2022 117
DG Gel Sol	031531810035	*003 30/06/2023 -

LOT	Time	User
*DG Gel ABO-Rh (2D)	716101240180173988	*018.01 31/01/2024

Printer icon

Figure 40. Sample Traceability Information

- (1) **Test execution** information: This section contains the following information: Test name, Sample ID, timing status and User involved in the execution.
- (2) **Quality Control** traceability: This section contains the following information about the **Quality Control**: Lot number, expiry date, Finish QC time and QC protocol result. To see more information about **Quality Control** see Section 9.
- (3) **Results** traceability: This section contains information about the microtubes which results has been modified. Traceability of the instrument result, time of the modification and the user responsible of this modification is also included.

- (4) **Reagents** traceability: This section contains information about the reagents used in the test execution, including barcode, lot number, remaining on-board stability time and expiry date.
- (5) **DG Gel cards** traceability: This section contains information about the DG Gel card used in the test execution, including barcode, lot number and expiry date.
- (6) **Print** button. Pressing this button, the traceability information is printed directly by the printing.
- (7) **Sample Related Data** button. Pressing this icon, a window will open showing related information of the sample and test executed (Figure 41).



NOTE: Demographic data is displayed in **Sample Related Data** window only if LIS connection is enabled (workload is received from LIS) and the instrument is configured to display demographic data. Instead, the Patient ID is always displayed in **Sample Related Data** window.

SAMPLE	HOSPITAL
357	ID297
NAME	
JULIET WEISINGER	
GENDER	DATE OF BIRTH
WOMAN	31/12/1987

Figure 41. Sample Related Data Window

7.6.10 Printing Results

To print the results which appear on the **Results List** screen:

1. Select the range of samples that you wish to include in the report.



2. Press the **Print** button.
3. Select one of the three types of report available:



- Press  to obtain the **Reading Report** with all of the results of the sample profiles selected, with the interpretation of the results of each profile test, along with the reaction grade associated with each microtube.



- Press  to obtain the **Results List Report** with the results of the sample profiles selected, with only the final interpretation of the results for each profile test.



- Press  to obtain the **Traceability Report**, which includes the results of the sample profiles selected, with the image of the processed card microtubes, and their reaction grade.



NOTE: In order to print more detailed information in relation to a specific profile applied to a specific sample, enter the screen **Profile Information** and print it from there, as described in Section 7.6.10.



NOTE: Demographic data is displayed in **Reading** and **Traceability Reports** only if LIS connection is enabled (workload is received from LIS) and the instrument is configured to display demographic data. Instead, the Patient ID is always displayed in **Reading** and **Traceability Reports**.

The report is integrated within the interface and it is necessary to close it in order to continue working with the Erytra software.

To add the logo of the laboratory on the printed reports, contact your local Grifols service representative.



4. Press  to obtain a printed copy of the report.

It is possible to configure that each time that the user selects the printing icon, the pdf report is automatically printed through the printer connected to the Erytra. For more information, please contact your local Grifols service representative.

7.6.11 Saving Report Data



1. Press the **Save** button  of the **Report Tool Bar** (Figure 42).
2. Enter the file name and the directory to save the report to.
3. Press the **Save** button.

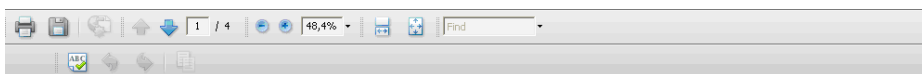


Figure 42. Report Tool Bar



WARNING: The reports stored in the instrument will be automatically deleted after a certain time.

7.7 Unloading and Shutting Down the Analyzer



WARNING: Cap the reagent vials and save or discard them according to their Instructions for Use.



WARNING: To avoid injury, do not open the analyzer while tests are being processed.

The Erytra should be unloaded at the end of every workday. To do this, proceed as follows:

1. Open the upper door.
2. Remove any reagents and samples.
Reagents and samples may also be removed by means of the drawers.
3. Close the upper door.

4. Open the card drawers.

On the **Cards** screen, identify those which should be kept in the fridge and that are marked with a (*) before their name.

5. Remove any empty Grifols gel card racks.
6. Remove the reviewed Grifols gel cards that have been placed in the Service Rack and leave any partially used cards in the Service Racks.
Be sure to review any Grifols gel cards that have associated special results.
7. Close the card drawers.
8. Open the waste solutions drawers.
9. Disconnect the tubes from the waste solution containers, remove them from the analyzer, and empty them.
10. Put the empty waste solution containers back in the analyzer and reconnect the tubes.
11. Close the waste solution drawers.



12. Press the **Shutdown** icon. A message displays to confirm shutdown.



CAUTION: The tracking of the on-board stability stops when switching off the instrument.



NOTE: When shutting down the instrument be sure to remove all reagents, diluent and Grifols gel cards that require storage conditions between 2 - 8 °C from the instrument.



13. Click the **Confirm** button to close the Erytra software.

The analyzer rinses and empties the entire fluidic system automatically, stops the on-board stability tracking and shuts down when finished.

8 Database Searches



NOTE: The Erytra software has been designed with web technology. This permits the authorized users to access the **Database** not only from the Erytra analyzer, but also from any remote desktop using http protocol (Remote Client). During remote access, the same user privileges apply which have been configured locally.

For more information about activating this service, contact your local Grifols service representative.

8.1 Searching and Viewing Results in the Database

To search and view results that have been stored in the **Database** and that have been previously exported from the **Results** screen:

1. Press the **Database** tab (Figure 8, no. 4). The **Search Filter** screen opens (Figure 43).

Figure 43. Search Results Filter Screen

2. Enter search criteria in one of the following ways:

- By reagent barcode.
- By sample barcode.
- By Grifols gel card barcode.
- By date: To facilitate the introduction of the date, there is a small calendar accessible

by clicking .

- By Profile.
- By special result.
- By incident.
- By modified result.



3. Click the **Confirm** button to apply the selected filters.

The information appears on the screen in table form (Figure 44) with the following information:

- Sample identification.
- Profile.
- Test.
- Results.
- Status.
- Read date.
- **Quality Control** status.

At the bottom, the number of pages found matching this search criteria and the total number of results are listed. The page navigation buttons allow movement between the different pages.

2 DATA BASE HITS

AMPLE BARCODE: PATIENT: NONE
ard Barcode: REAGENT BARCODE: INCIDENCE
PROFILE: FROM 01/02/2012 TO 16/02/2012 SPECIAL RESULT

	SAMPLES	PROFILE	TEST	RESULTS	STATUS	QC	DATE
	11271329275383984	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:18
	11261329275383921	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11251329275383875	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11241329275383812	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11231329275383765	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11221329275383718	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11211329275383656	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	11201329275383609	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:17
	111111329275381843	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12 5:16
	111101329275381750	2 Screen (IgG)	AbScr IgG	Pos	Exported		15/2/12

158 # 15706

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GRIFOLS

Figure 44. Pagination of the Results Found

- Press , to access the **Profile Information** screen to see more detailed information about the tests contained in a specific profile. For more information consult, see Section 7.6.2.
- To perform a new search, delete the filter by pressing the **Delete Filter** button  and enter the new search criteria in the filter.

8.2 Printing Searched Results



To print the search results, select the desired results range and press the **Print** button. The software allows you to choose from different report formats: For further information consult Section 7.6.10.



The **Select All** button and **Deselect All** button allow the selection and de-selection of all of the results obtained by the search.

8.3 Saving Searched Results

To save search results in an electronic format, see Section 7.6.11.



To perform a new search, delete the filter by pressing the **Delete Filter** button and enter the new search criteria in the filter.

8.4 Resending Results to LIS

If results are not successfully sent to LIS due to a connection problem, resend the results by



pressing the **Export** button from the **Database** screen.

9 Quality Control

The Erytra software ensures accurate test results through a module for **Quality Control** samples. The **Quality Control** module can be used to access:

- The **Results** screen, where the processed and validated results are displayed.
- The **Database** screen, where exported results can be recovered.

The Erytra analyzer has an internal **Database** with all the **Quality Control** kits available. Once the **Quality Control** samples have been loaded, the analyzer automatically recognises and assigns the expected values for each test. Additionally, this barcode enables the batch and expiry date of each **Quality Control** kit to be traced.

Quality Control samples are displayed on the **Status > Sample** screen with the **Quality Control**

icon .

9.1 Quality Control Policy

The **Quality Control** module can be configured with the following parameters:

- Enable or disable the **Quality Control** policy. Once enabled, a prospective quality policy is followed.
- Activate the **Quality Control** kit you want to use. Once a kit is activated, the QC results automatically affect all the tests whose segments are defined in this unit. Once QC is activated, the system automatically assigns QC to all tests on the **Worksheet** which contain these elements.
- Frequency, defined in hours, after **Quality Control** samples are run, after which the **Quality Control** expires.
- Enable autovalidation of QC PASS, so that any QC results that PASS without needing review are automatically validated.

It is also possible to configure the QC policy in a way that if the QC result is PASS, its results are automatically validated and exported. Furthermore, it is possible to leave them as pending validation.

Quality Control module settings are initially configured during installation. To change the configuration settings of the **Quality Control** module, please contact your local Grifols service representative.

9.2 Programming Quality Control Samples

After the **Quality Control** sample tubes belonging to an activated kit have been loaded into the analyzer, they appear on the **Worksheet**, as shown in Figure 45 below:

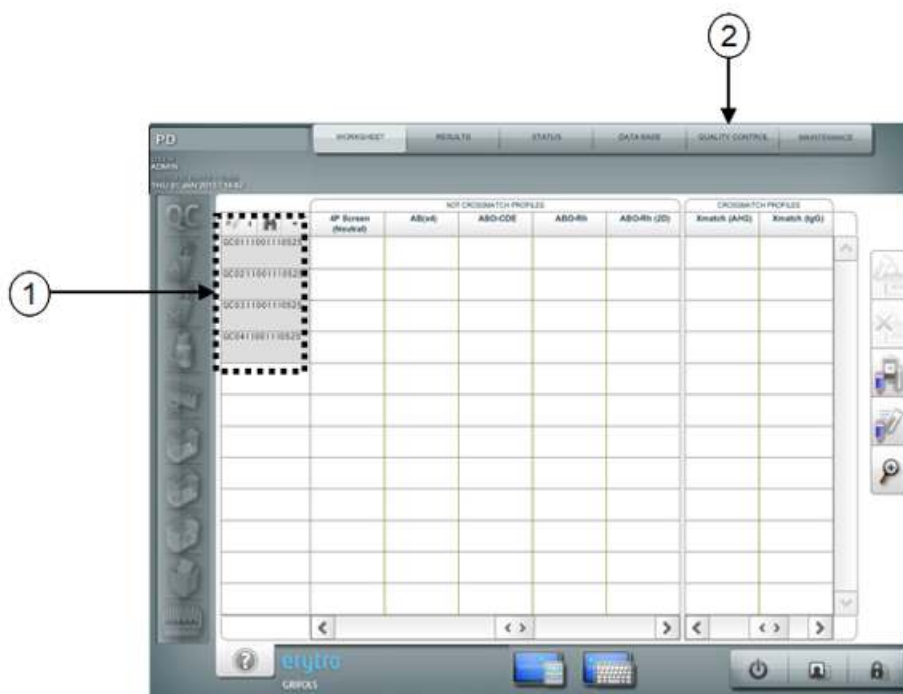


Figure 45. Overview of the Worksheet

- (1) QC samples identified in the **Worksheet**.
- (2) **Quality Control** Module.



WARNING: Should it be necessary to manually identify the tubes in the **Quality Control**, this identification should use the same numbering as the original barcode and should be entered twice.

If routine samples have been loaded into the analyzer, the QC samples will appear on the **Worksheet** below the routine samples.

Quality Control tubes will be flagged with the **Old Sample** icon  96 hours after being loaded and identified by Erytra (when the instrument has not been switch off during this period).

9.2.1 Automatic Quality Control Programming

When programming routine sample tests with segments controlled by quality policy, the Erytra software processes samples in the following manner:

- The QC samples are processed immediately when the **Quality Control** of any of these segments is pending (*i.e.* when it has expired or they have not been previously processed and the tubes are in the analyzer). The routine samples will stay as “Stopped” until the QC result has been obtained.
- The routine samples are processed immediately when the segments controlled by QC involved in the tests are valid and the result is PASS. If they are valid but the result is FAIL, choose between the different options offered in a screen message notifying this event.
- If the QC tubes are not on board but the frequency of some of these segments has expired, the **QC Warning** icon  is activated in the top left of the **Worksheet**, as well as the QC tube requirement in the **Missing Samples Warning** screen. The traffic light changes from green to yellow and an acoustic alarm sound. The routine samples will remain “Stopped” (as shown in Figure 46).

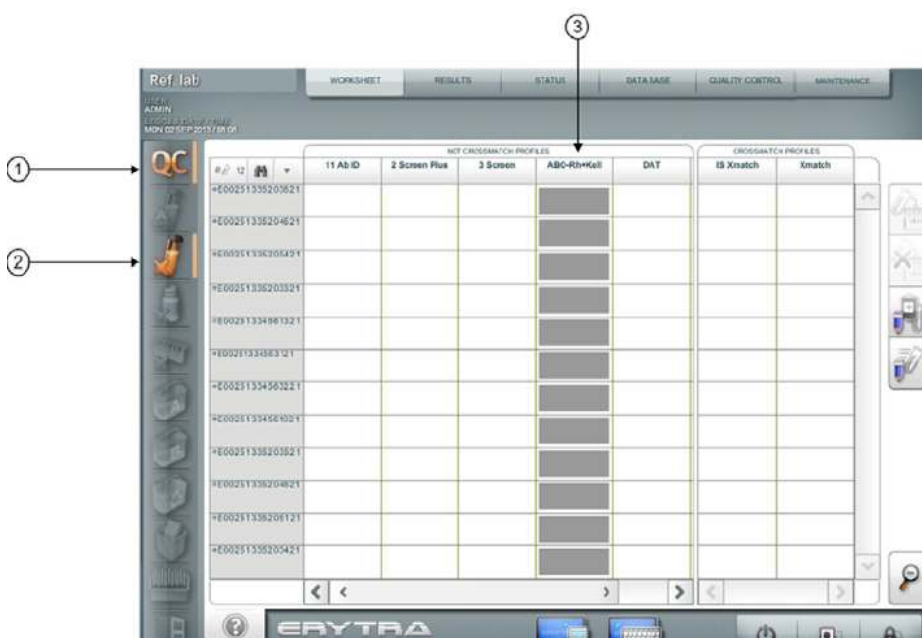
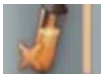


Figure 46. Activation of the Quality Control Warning

- (1) The **Quality Control Warning** icon 
- (2) The **Missing Samples Warning** icon.
- (3) Samples stopped due to lack of QC tubes.

Clicking the **Missing Sample Warning** icon  opens the screen indicating the QC tubes that have to be loaded into the analyzer, as well as the profile of the routine sample which is affected by this requirement for the QC tubes (as shown in Figure 47):

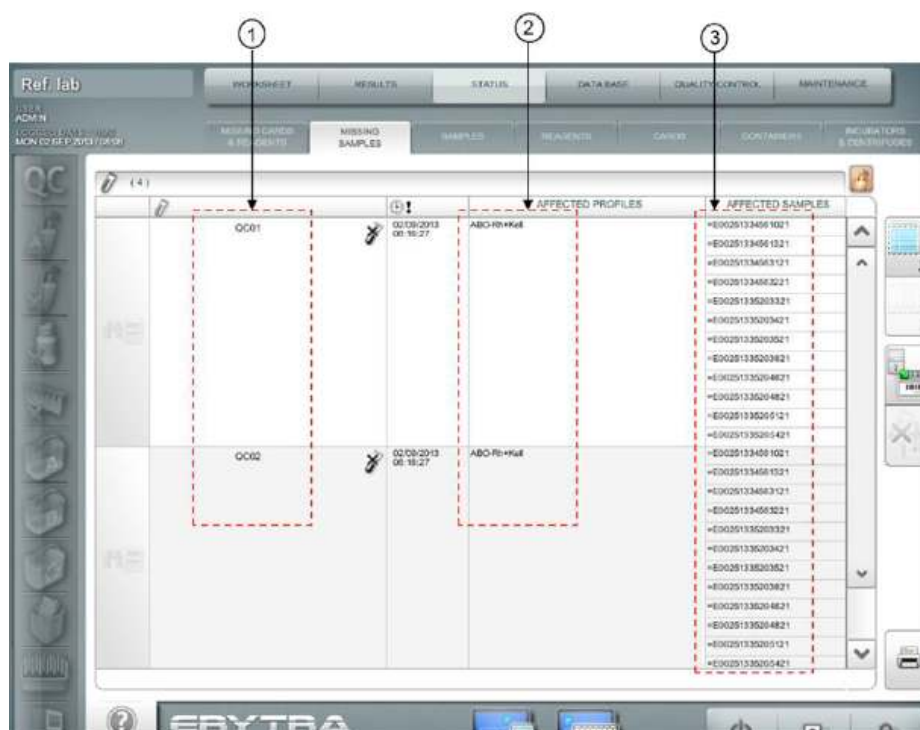


Figure 47. Quality Control Sample Requirements

- (1) QC tubes required by the system.
- (2) Profiles affected by lack of QC tubes.
- (3) Samples affected by lack of QC tubes.

9.2.2 Manual Quality Control Programming

Any QC can be created manually before it is required by the frequency assigned according to the laboratory **Quality Control** procedures. To do this, load the **Quality Control** sample tubes and manually assign the test to be performed from the **Worksheet**.

It is also possible to execute any of the tests available on the **Worksheet** to tubes that are part of the QC. The tubes will be processed but will not have the expected results for those segments that are not controlled by the Erytra analyzer. The results will be stored in the **Results List** in the **Results** tab, not in the **Quality Control** tab.

9.3 Processing Quality Control Samples

The **Quality Control** samples are processed in an independent incubator separate from the rest of the routine samples.

If there is an associated requirement (reagents, Grifols gel card, sample, system solution) to process the **Quality Control**, the traffic light will change from green to yellow and the corresponding warning will be activated in **Warning Area**, which will be accompanied by an acoustic alarm. The QC samples will be "Stopped" until the requirement is met.

If the QC samples are cancelled in the **Status > Samples** screen, an error message will appear asking the user if the routine samples should be cancelled, remain pending or still be processed.

If an expired batch of a **Quality Control** kit is identified, the Erytra analyzer will give the choice of one of the following options:

- Opening the drawer to replace the expired kit for a valid one.
- Continue executing with the expired QC kit. Once the results for the expired kit have been obtained, they will show the **Expired Batch** icon  (as shown in Figure 48):

SAMPLES (A/B)	PROFILE	TEST	RESULTS	STATUS	QC	DATE
QC11387130812	ABO-RhVital	Ag Kall	Pos	Pending validation	Pass	2/9/13
		Group Rh	An (+)		Pass	12/17
QC321387130812	ABO-RhVital	Ag Kall	Neg	Pending validation	Fail	2/9/13
		Group Rh	Or (+)		Fail	12/25
QC331387130812	ABO-RhVital	Ag Kall	Neg	Pending validation	Fail	2/9/13
		Group Rh	An (+)		Fail	12/25
QC341387130812	ABO-RhVital	Ag Kall	Neg	Pending validation	Pass	2/9/13
		Group Rh	AB (+B)		Pass	12/25

Figure 48. Example of Results Processed with an Expired Quality Control

(1) Results of an expired QC batch.

It is configurable that when the system detects a new lot of any of the resources (Grifols gel cards, RBC, antisera and diluent) involved in the processment of a QC protocol that has been obtained as PASS, it will show a warning to the user, who will have to acknowledge it. It is optional for the user to process the QC with the new lot of the identified resources (Figure 49).



Figure 49. New Lot QC Warning (Examples)

9.4 Validating Quality Control Samples

Once the **Quality Control** samples have been processed, the results will appear in the corresponding box, in the **Quality Control > Results** screen (Figure 50).

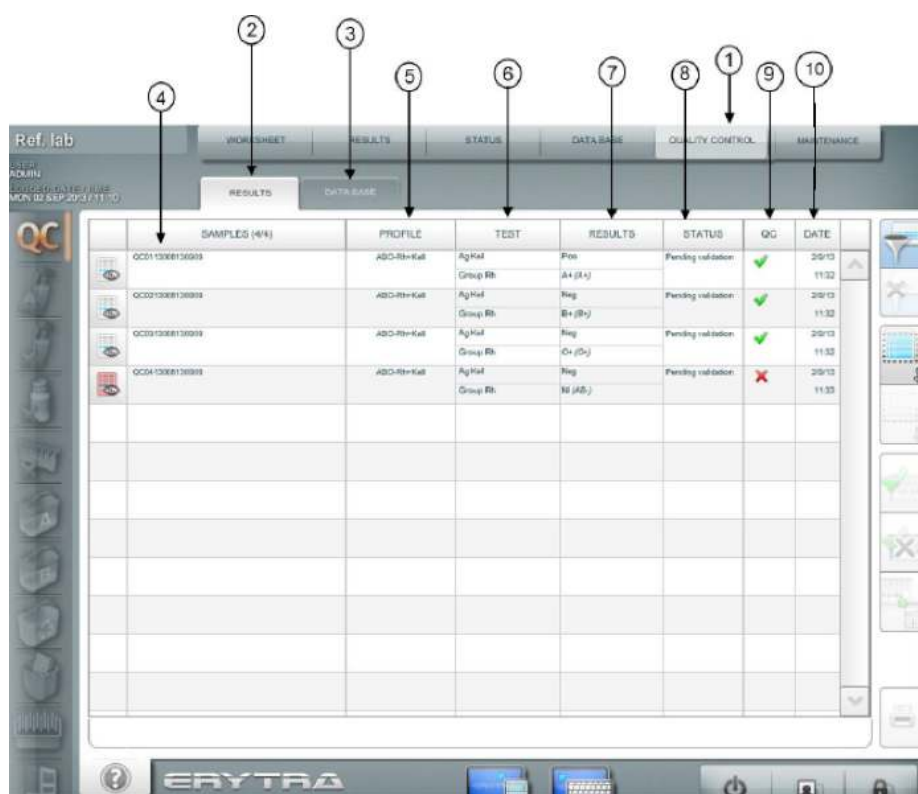


Figure 50. Quality Control Results Screen

- (1) **Quality Control** tab.
- (2) The **Results** tab of the **Quality Control** screen.
- (3) The **Database** tab of the **Quality Control** screen.
- (4) Sample identification.
- (5) Profile associated with the sample.
- (6) Test associated with the sample.
- (7) Results obtained.
- (8) Sample status.
- (9) **Quality Control** status.
- (10) Execution date.

The results expected for each tube in the **Quality Control** appear in brackets in the **Results** column, to the right of the results obtained.

The **Quality Control** status associated with the test appears in the **Quality Control > Results** screen as  or , depending on whether it is PASS or FAIL, respectively.

There are two possible QC statuses:

- **PASS**  : The QC result has been obtained in accordance with the expected result. A detailed image of the **Results** screen is given in Figure 51 below:

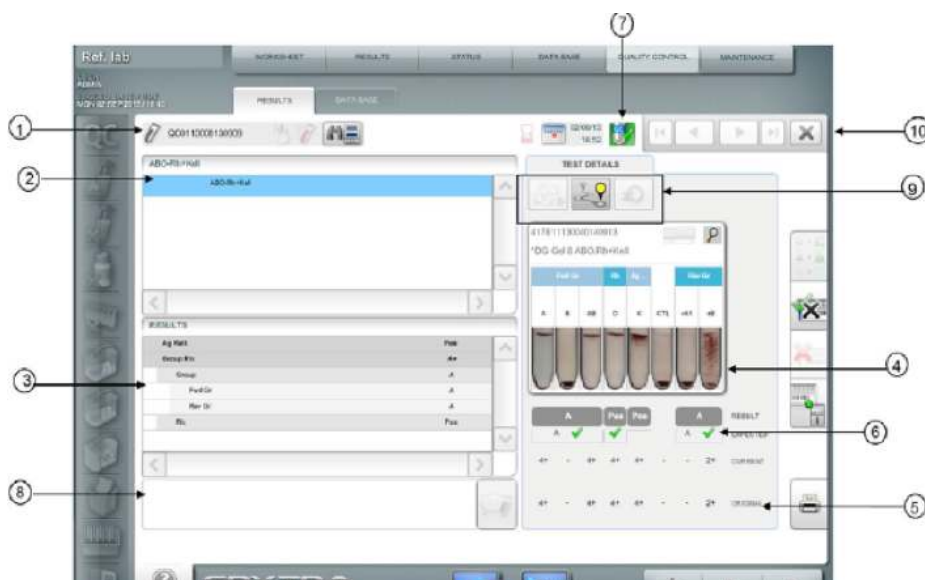


Figure 51. Detailed Screen of the PASS Result for Quality Control

- (1) General information about the **Quality Control** sample.
- (2) Profile information and the tests involved.
- (3) Results of the Tests.
- (4) Images and interpretation of the microtubes of the Grifols gel cards associated with the test selected in (no. 2). To enlarge the image, press the **Zoom** button .

-
- The screenshot shows the Erytra software interface with the following numbered callouts:
- 1: Top-left toolbar icons.
 - 2: Sample ID field (ADD-RN-vial).
 - 3: Results table.
 - 4: Test details panel (41781133040-14905).
 - 5: Bottom-right toolbar icons.
 - 6: Right sidebar icons.
 - 7: Top navigation tabs (RESULTS, STATUS, DATABASE, QUALITY CONTROL, MAINTENANCE).
 - 8: Bottom-left toolbar icons.
 - 9: Test details panel (41781133040-14905).
 - 10: Top-right toolbar icons.

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- (1) General information about the **Quality Control** sample.
- (2) Profile information and the tests involved.
- (3) Results of the Tests.
- (4) Images and interpretation of the microtubes of the Grifols gel cards associated with the test selected in (no. 2). To enlarge the image press the **Zoom** button .
- (5) Reaction grade information.
- (6) The PASS/FAIL information of the test interpretation segment.
- (7) **Quality Control** status.
- (8) Comments.
- (9) Special buttons (described in Section 7.6.2).
- (10) Navigation buttons that provide access the **Profile Information** for the results of profiles present on the active **Results List** screen.

To **Validate** the **Quality Control** result, follow the same procedure used to validate the results from routine samples (see Section 7.6).

If the **Enable Autovalidation of QC PASS** option has been enabled, the QC results that have obtained a PASS will be automatically validated by the system (see Section 9.1). Obtaining a PASS in the QC results means that all the samples in the **Quality Control** kit have been classified as PASS. To change the **Enable Autovalidation of QC PASS** option, contact your local Grifols service representative.

If any of the samples obtains a FAIL for any segment, the overall QC result for this test will be FAIL and the associated results will remain in “Pending validation”, obliging the user to review that QC sample tube and the associated Grifols gel card. Reject the result for the QC tube that failed in order to retest that tube automatically.

Once a QC test has started, all of the tubes in the QC kit must return a result of PASS. For this reason, leave the tube that have obtained the PASS without exporting them, reject the QC FAIL result of the tube that has not passed and wait until the automatic repetition of the test for the failed tube passes. Then export the results for the whole kit that has obtained a PASS for all the tubes to have a QC status of PASS.

9.5 The Quality Control Warning



The **Quality Control Warning** icon is activated when:

- The quality policy has expired, in accordance with the programmed frequency.
- The QC execution has been interrupted due to missing resources. For more information on resolving these issues, see Section 7.5.
- A FAIL result has been obtained for one of the tests executed on QC samples.

9.6 Exporting Quality Control Results

QC results must be exported before routine sample results can be validated. Follow the instructions described in Section 7.6.

The exported QC results are stored in the **Database** screen, in the **Quality Control** module, as shown in Figure 53:

SAMPLES	PROFILE	TEST	RESULTS	STATUS	QC	DATE
QC013089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	B+ (B+)			10:41
QC013089130889	ABO-Rh+val	Ap Fail	Pos	Exported	✓	29/13
		Group Rh	A+ (A+)			10:41
QC0413089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	AB+ (AB+)			10:41
QC0313089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	O+ (O+)			10:42
QC0313089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	B+ (B+)			11:32
QC0313089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	O+ (O+)			11:32
QC0113089130889	ABO-Rh+val	Ap Fail	Pos	Exported	✓	29/13
		Group Rh	A+ (A+)			11:32
QC0413089130889	ABO-Rh+val	Ap Fail	Naq	Exported	✓	29/13
		Group Rh	AB+ (AB+)			11:33
QC0113087130812	ABO-Rh+val	Ap Fail	Pos	Exported	✓	29/13
		Group Rh	A+ (A+)			12:17

Figure 53. List of Quality Control Results from the Database

- (1) **Quality Control** tab.
- (2) The **Database** tab of the **Quality Control** screen.
- (3) QC sample information.
- (4) Profile associated with the QC sample.
- (5) Test associated with the QC sample.
- (6) Results obtained.
- (7) QC sample status.
- (8) Result of the **Quality Control** in accordance with the expected result.
- (9) Execution date.

In the list of **Database** results, the detail of any processed, validated or exported **Quality Control** result can be recovered. To do this, the filter may be used, as explained in Section 8.1.

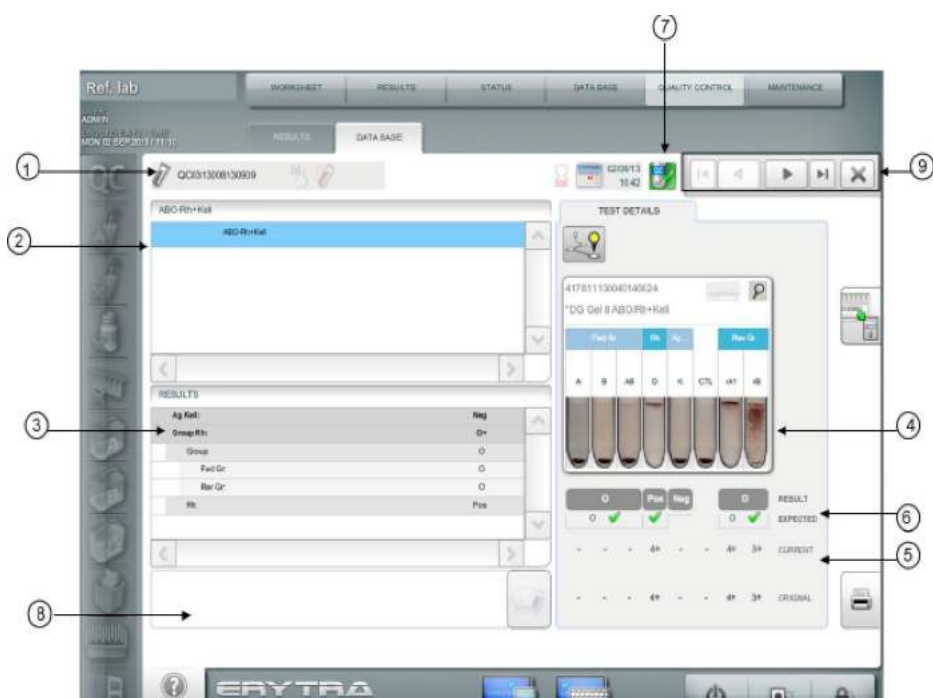


Figure 54. Detail of the Quality Control Result from the Database

- (1) General information about the **Quality Control** sample.
- (2) Profile information and the tests involved.
- (3) Results of the Tests.
- (4) Images and interpretation of the microtubes of the Grifols gel cards associated with the test selected in (no. 2). To enlarge the image press the **Zoom** button .
- (5) Reaction grade information.
- (6) The PASS/FAIL information of the test interpretation segment.
- (7) Result of the **Quality Control** in accordance with the expected result.
- (8) Comments.
- (9) Navigation buttons that provide access the Profile Information of the results of the profiles which are present on the active **Results List** screen.

FAIL results for **Quality Control** tubes cannot be exported but can be validated or rejected. Reject the FAIL results to automatically program the FAIL QC tube to repeat the QC test. After the QC tube returns a result of PASS, the PASS result can be exported.

9.7 Executing Routine Samples with Activated Quality Control Configuration

The routine samples loaded into the analyzer that are affected by a pending QC status will remain "Stopped" until the QC result is obtained.

Once the QC result is obtained, two situations can occur, depending on the QC result:

- **PASS:** The routine samples are executed automatically, the status changing from "Stopped" to "In process".
- **FAIL:** A message will appear on screen giving three possible options:
 1. Continue processing the routine samples, despite a FAIL QC result.
It is not recommended to process routine samples with a FAIL QC result as the results for these samples cannot be exported.
Continuing to process routine samples with a FAIL QC result can only be done by an Operator with the appropriate rights.
 2. Keep the routine samples "Stopped". This permits **Quality Control** to be repeated.
 3. Cancel the petitions pending.

STAT Samples will always be directly processed, without waiting to obtain the QC result. They will also have an associated QC status, corresponding to the QC status at the time of execution.

Once the routine samples have been processed, they will appear on the **Results** screen and the **QC Status** column will also indicate the result of the **Quality Control** for those tests that have an associated QC.

In the case of profiles with more than one test, the analyzer will process the routine samples when the QC result for the corresponding test has obtained a PASS result.

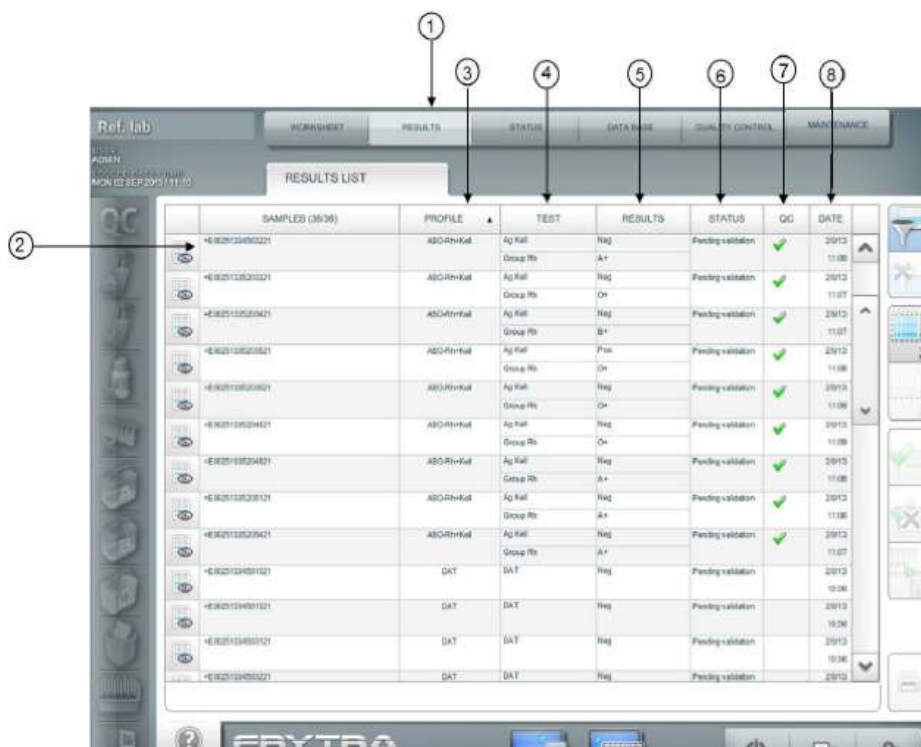


Figure 55. Detail of the Results Screen for Routine Samples

- (1) Results tab.
- (2) Sample identification.
- (3) Profile associated with the sample.
- (4) Test associated with the sample.

- (5) Results obtained.
- (6) Sample status.
- (7) Result of the **Quality Control**. The routine samples will have the associated QC status at the time of execution.
- (8) Execution date.

Once the routine samples are processed, the detailed **Results** screen will show one of the following QC statuses associated:

-  **QC PASS and Exported:** The QC results obtained are in accordance with expectations and have been exported. This status enables the results for routine samples to be validated directly.
-  **QC PASS and Not Exported:** The QC results obtained are in accordance with expectations and have not been exported. To validate the results of the routine samples, the QC results must first be exported.
-  **QC FAIL and Not Exported:** The QC results obtained are not in accordance with expectations and have not been exported. To validate the results of the routine samples, the FAIL results of the QC samples must be rejected and then retested until they obtain a PASS so that the QC samples results can be exported. Then the routine samples can be validated.
-  **QC OMITTED:** The QC protocol has not been completely executed (due to a lack of resources or the cancellation of the protocol during its execution) and the QC status was PASS before its expiration. This status enables the results for routine samples to be validated directly.

When the result of a routine sample cannot be validated due to the QC result not being exported,

the icon  will appear in the **Results** screen.

The exported routine sample results will be stored in the **Database** screen, showing the QC status in the **QC Status** column.

The routine sample reports printed from the **Results** screen and the **Database** screen have a field in the header indicating the sample QC status (Figure 56):

Reading report
Ref.:
Code: PAC002 Ver.: v3.01

UNIVERSITY HOSPITAL

Sample : =E00251330311421 Profile : Rh Pheno

Sample : =E00251330311421

First name: - Last name: -
Gender: - Date of Birth: -

Profile : Rh Pheno

Execution date: 25/11/2013 13:02:18 Status: Exported QC Status: Pass and Exported
Executed by: ADMIN Modified by: - Reviewed by: admin
Erytra (SN): 0000625

Profile interpretation
Rh Pheno:CEce

Comment

DG-Gel Rh Pheno

LOT: 1006.01 2014-02 717602140060124961

1	2	3	4	5	6	7	8
				4+	3+	4+	4+
				C.Fw	E.Fw	c.Fw	e.Fw

Incidences

Rh Pheno 7176
Rh Pheno:CEce

Figure 56. Detailed Print Report with Exported PASS and Exported QC Status



NOTE: Not all of the functions listed above are available in all software versions.

10 User Configuration

10.1 User Configuration Overview

This section provides a description of the user configuration and management of the analyzer.

The **Users** module allows the quickly and efficiently managing of user groups and their access permissions across the analyzer interface. Supervisor can configure users from the **Maintenance > Users** screen of the Erytra software. If the **Enable Passwords** option is activated, users must be configured to work with the Erytra.

10.2 User Groups and Access Permissions

There are four predefined users groups or roles with different access permissions to work with Erytra:

- Operator Basic.
- Operator.
- Operator Plus.
- Supervisor.

The following table is a summary of the permitted actions within each access level:

Table 7. Permitted Actions for each Access Level

	OPERATOR BASIC	OPERATOR	OPERATOR PLUS	SUPERVISOR
Execute tests	Access	Access	Access	Access
Print results	Access	Access	Access	Access
Validate routine and QC results (no alarms)		Access + confirmation message	Access + confirmation message	Access+ confirmation message
Export routine and QC results		Access + confirmation message	Access + confirmation message	Access + confirmation message

	OPERATOR BASIC	OPERATOR	OPERATOR PLUS	SUPERVISOR
Modify results			Access + identification required (configurable) + confirmation message	Access + identification required (configurable) + confirmation message
Cancel validated results			Access + confirmation message	Access + confirmation message
Reject results			Access + confirmation message	Access + confirmation message
Execute samples with QC pending			Access + identification required (configurable) + confirmation message	Access + identification required (configurable) + confirmation message
Validate samples with QC pending		Access (only if the previous QC status for the profile is PASS) + identification required (configurable) + confirmation message	Access + identification required (configurable) + confirmation message	Access + identification required (configurable) + confirmation message
Accept special results or results with incidents			Access + identification required (configurable) + confirmation message	Access + identification required (configurable) + confirmation message
Accept inconsistencies			Access + confirmation message	Access + confirmation message
Insert comments on the results	Access	Access	Access	Access

	OPERATOR BASIC	OPERATOR	OPERATOR PLUS	SUPERVISOR
Activate and deactivate modules		Access	Access	Access
Backup Data			Access	Access
Support Pack			Access	Access
Access User management				Access
Probes replacement			Access	Access
Diagnostic program	Access	Access	Access	Access
Record Technical Service inspection				Access



NOTE: Diagnostic program is accessed through the **Diagnostic Menu** tab from the initial loading window of the analyzer. See Section 12 for further information.

10.3 Creating New Users

To create new users in the Erytra software, go to the **Maintenance > User** screen and click the **New** button. Fill in the fields, keeping in mind the following restrictions:

- **User:** The unique numbers and/or letters that identify the user in the **Database**. It must have a length of between 4 and 10 characters in length. If you select a **User** that is already in use, the system will display an error message requesting a new **User**.
- **Name:** User first name. This field cannot be left blank.
- **Surname:** User last name. This field cannot be left blank.
- **Allow User:** If this field is not checked, the user will not be able to access the application and this will be reflected on the software login screen when the user attempts to enter.
- **Delete Password:** This option deletes the password for a specified user. This user must then enter a new password the next login.

After a user is created, the software generates a temporary password that matches the user's username. When a new user logs on to the software for the first time, he should use the username as their password. The software then requests a new password, which must **meet the requirements of length between 4 to 10 alphanumeric characters, be case sensitive and be entered twice to confirm.**

After 3 incorrect attempts to enter the password, the system automatically disables the user. To enable this user, go to **Maintenance > Users** screen, where only the Supervisor and the Administrator can access.

In case the Supervisor user is also disabled for wrongly introducing the password, it is recommended to create, at least, 2 different Supervisor user roles.

10.4 Modifying User Data

To modify **User** data in the Erytra software:

1. Go to the **Maintenance > Users** screen to view the User List.
2. Select the user whose data are to be modified. On the **User Information** screen, press the **Edit** button.
3. Modify the data to be changed. The **User** field cannot be changed.
4. After completing the changes, press  to confirm the changes (or  to cancel the changes).

10.5 Changing Passwords

Users can change their password from the Erytra software. To change a password:



1. Press the **Password Change** button on the main screen of the Erytra software (Figure 8, no. 14).
2. Fill in the field **Current Password** with the password that the user current password.
3. Fill in the **New Password** field with a new password that meets the requirements of the software (it must be a made up of letters and/or numbers and be between 4 and 10 characters of length).
4. Re-enter the same password again in the **Confirm Password** field.

If the data entered is not correct (the **New Password** does not match its confirmation, the password does not meet the system requirements, etc.) the Erytra software will notify the user on screen and will not save the changes that were made.

To configure a frequency with which the system requires users to change their passwords, as well as which specific actions require passwords to access, contact your local Grifols service representative.

Erytra

11 Maintenance

11.1 Maintenance Plan

The Maintenance Plan is a set of operations carried out at regular intervals to preserve the function of the Erytra. As a general rule and when there is no irregular occurrence in instrument operation, follow the maintenance plan listed in the table below to ensure the proper functioning of the analyzer.

Table 8. Maintenance Plan

PERIOD	ACTIVITY	DONE BY
When drops of samples or of other substances are spilled inside the analyzer	Clean the affected area (see Section 11.7)	Supervisor/Operator
After each session (when needed)	• Check for liquid leaks (see Section 11.6.1). • Check for the growth of microorganisms (see Section 11.6.2).	Supervisor/Operator
Every week	Clean the surfaces, if necessary (see Section 11.7)	Supervisor/Operator
Every month	Decontaminate the instruments and its components (see Section 11.8)	Supervisor/Operator
Every month	Decontaminate the liquid containers	Supervisor/Operator
Every year	Review of the general status of the Erytra analyzer (see Section 11.9)	Qualified Technician

If working continuously with the analyzer, it is recommended to turn it off and on every seven days for the Windows operating system to work correctly.

The maintenance operations done on the instrument must be recorded in the “Maintenance Services form” included at the end of the Instructions for Use and also in the “Technical Service Inspection Registry”.

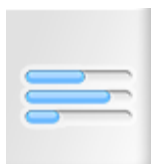
The technical assistance can be provided by your supplier or by other services authorized by Diagnostic Grifols, S.A.



CAUTION: Do not use the instrument if the Maintenance Plan has not been followed.

11.2 Checking Status and Performing Maintenance

The **Maintenance > Analyzer** screen contains the following tabs related to instrument status and maintenance:



The **Instrument Status** tab provides information about the general status of the instrument, including the expiration dates of the decontamination procedures and of Technical Service inspections, the next required restart of the system and whether any modules are disabled.



The **Decontamination** tab provides information about the expiration dates of previous decontamination procedures, as well as a button to perform a fluidic system decontamination.



The **Probes Replacement** tab provides information on when probes were replaced, as well as buttons to facilitate replacing a probe.



The **System Solution Registry** tab provides information about when the system solution containers have been refilled, as well as a button to facilitate recording the refilling of the system solution.



The **Technical Service Inspection Registry** tab provides information about when Technical Service inspections have been performed and when the current Technical Service inspection expires. This tab also contains a button to facilitate recording Technical Service inspections and a button to facilitate running the Support Pack for support purposes.



The **Activate/Deactivate Modules** tab provides information about the current status of the modules, as well as information about previous module activation and deactivation actions. This tab also contains a button to facilitate activating and deactivating modules.



The **Backup** tab provides information about the last backup performed, as well as a button to facilitate running a manual backup of certain stored data.



The above tabs are connected to the **Maintenance Warning** icon. This warning is activated under the following conditions:

- When there are one or more disabled modules.
- When there are one or more actions (decontamination procedure, Technical Service inspection, system restart, etc.) that have expired (they appear in red in the **Maintenance > Analyzer** screen).
- When there is an action that is about to expire (these appear in orange in the **Maintenance > Analyzer** screen).



The warning is fully activated when the cause that provokes this warning activation occurs for the first time. This warning appears with the background and outline highlighted.



The warning is partially activated when the user has accessed this section of the **Maintenance > Analyzer** screen before. This warning appears with only the outline highlighted.

11.2.1 The Instrument Status Tab

Use the **Instrument Status** tab (Figure 57) to check the general status of the instrument.

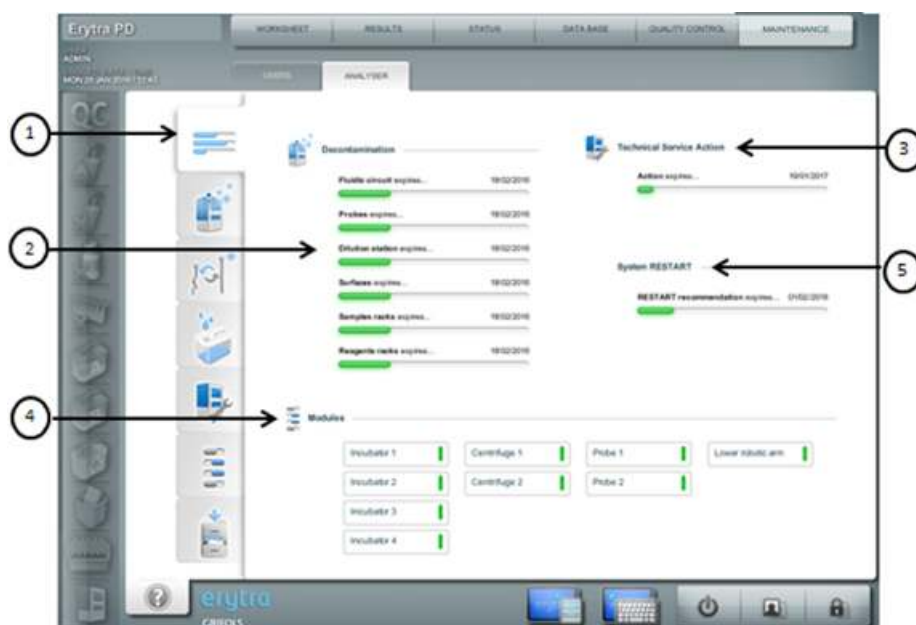


Figure 57. Instrument Status Tab

- (1) The **Instrument Status** tab icon.
- (2) **Decontamination** status: Including the expiration date of the decontamination of the following areas:
 - Fluidic system.
 - Probes.
 - Dilution Station.
 - Surfaces.
 - Sample holders.
 - Reagent holders.
- (3) **Technical Service Inspection**: Including the expiration date of the latest Technical Service inspection (based on yearly inspections) and the expiration date of the next required system restart (based on weekly system restarts).

- (4) **Modules** status: The activated modules appear in green and the deactivated modules appear in red.
- (5) **System Restart** status.

The information for the areas in (no. 2, 3 and 5) appears with the following color code:

- Green: If they are within the expiration date period.
- Orange: If they are close to the expiration date.
- Red: If they are expired.

11.2.2 The Decontamination Tab

Use the **Decontamination** tab (Figure 58) to perform the fluidic system decontamination listed in the Maintenance Plan. For the full fluidic system decontamination procedure, see Section 11.8.2.

This tab provides information about the expiration date of each of the decontamination procedures, as well as a way to register the different actions related to decontaminating any part of the instrument.

The screenshot displays the 'Decontamination Tab' interface. At the top, there is a navigation bar with tabs: WORKSHEET, RESULTS, STATUS, DATA BASE, QUALITY CONTROL, and MAINTENANCE. Below this is a sub-navigation bar with 'USERS' and 'ANALYZER'. The main area is divided into a left sidebar with icons (1-8), a central table, and a right sidebar with icons (2-7). The table lists decontamination actions with columns: Date and time, User, Type, Result, and Next date. The right sidebar shows expiration dates for various components: Fluidic system, Surfaces, Probes, Sample racks, and Dilution station, each with a green status indicator.

Date and time	User	Type	Result	Next date
27/08/2013 13:04	ADMIN	Fluidic system	Ok	26/09/2013
23/08/2013 10:54	ADMIN	Dilution station	Ok	22/09/2013
23/08/2013 10:54	ADMIN	Probes	Ok	22/09/2013
23/08/2013 10:54	ADMIN	Reagent racks	Ok	22/09/2013
23/08/2013 10:54	ADMIN	Sample racks	Ok	22/09/2013
23/08/2013 10:54	ADMIN	Surfaces	Ok	22/09/2013
22/07/2013 14:47	System	Dilution station	Ok	21/08/2013
22/07/2013 14:47	System	Probes	Ok	21/08/2013
22/07/2013 14:47	System	Reagent racks	Ok	21/08/2013
22/07/2013 14:47	System	Sample racks	Ok	21/08/2013
22/07/2013 14:47	System	Surfaces	Ok	21/08/2013
22/07/2013 14:47	System	Fluidic system	Ok	21/08/2013

Figure 58. Decontamination Tab

- (1) The **Decontamination** tab icon.
- (2) Information about the expiration date of each decontamination procedure.
- (3) The **Filter** button.
- (4) Button to remove filter criteria.
- (5) The **Decontamination Registry** button.
- (6) The **Fluidic System Decontamination** button.
- (7) Button to print the decontamination registry.
- (8) Register of previous decontamination procedures.

Press the **Filter** button (Figure 58, no. 3) to open a window (Figure 59) to filter and search for one or several specific decontamination procedures.

DATE RANGE

FROM TO

TYPE

☐ ALL

☐ Fluidic system ☐ Surfaces ☐ Other

☐ Probes ☐ Sample racks

☐ Dilution station ☐ Reagent racks

☐ ☐

Figure 59. Decontamination Filter Window

Press the **Decontamination Registry** button (Figure 58, no. 5) to open a window (Figure 60) to record the decontamination procedures performed by the user.

Figure 60. Decontamination Registry Window

Press the **Fluidic System Decontamination** button (Figure 58, no. 6) to run the fluidic system decontamination procedure. To start the procedure, click the **Confirm** button  on the message that appears (Figure 61), and then follow the instructions shown on the screen.

Figure 61. Confirmation Message for Starting the Decontamination Procedure

11.2.3 The Probes Replacement Tab

Use the **Probe Replacement** tab (Figure 62) to check information on previous probe replacements or to replace a probe. For the full probe replacement procedure, see Section 11.10.

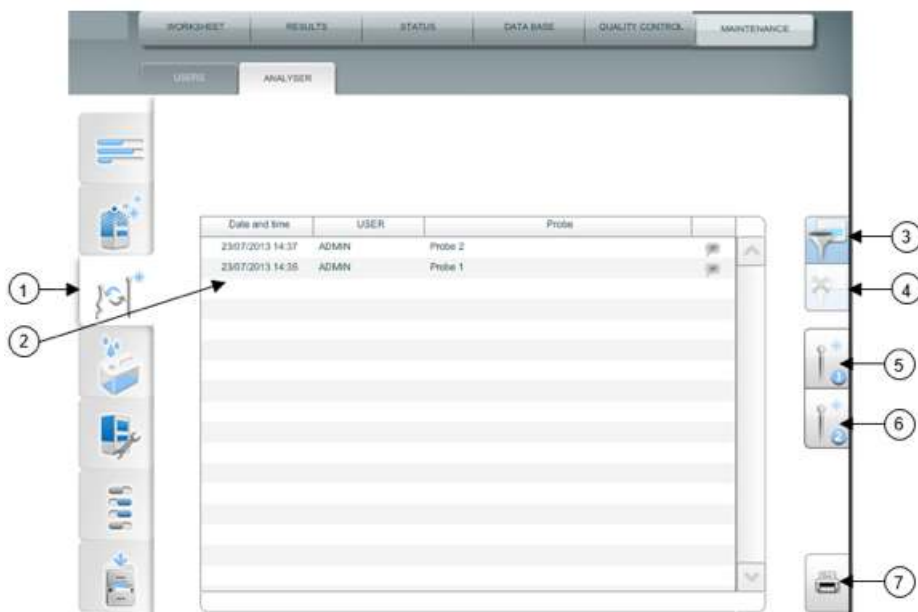


Figure 62. Probe Replacement Tab

- (1) The **Probe Replacement** tab icon.
- (2) Registry of previous probe replacements.
- (3) The **Filter** button.
- (4) Button to remove filter criteria.
- (5) Button to start replacement for probe 1.
- (6) Button to start replacement for probe 2.
- (7) Button to print the probe replacement registry.

Press the **Filter** button (Figure 62, no. 3) to filter and search previous probe replacements by date (Figure 63).

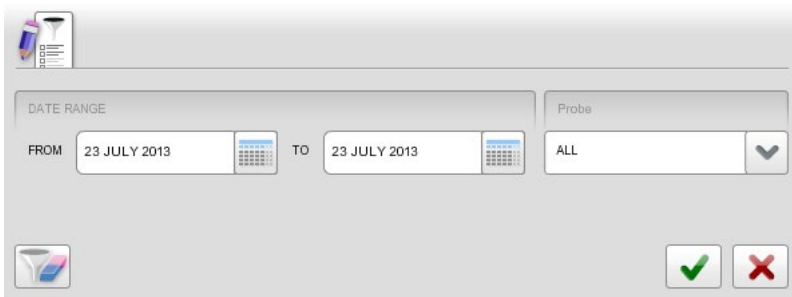
The image shows a software window titled "Probe Replacement Search Window". It features a "DATE RANGE" section with "FROM" and "TO" date pickers, both set to "23 JULY 2013". To the right is a "Probe" dropdown menu currently set to "ALL". At the bottom right, there are two buttons: a green checkmark and a red X. A small icon of a probe is visible in the top left corner of the window.

Figure 63. Probe Replacement Search Window

To replace a probe, press one of the button to start probe replacement (Figure 62, no. 5 or no. 6). After replacement, the **Probe Replacement Registry** window opens (Figure 64) to record this action. For the full probe replacement procedure, see Section 11.10.

The image shows a software window titled "Probe Replacement Registry Window". It has an information icon (i) in the top left. Below it, a table displays the following information: "Date and time" is "23/07/2013 14:35", "User" is "ADMIN", and "Probe" is "Probe 1". Below the table is a "Comments" section with a yellow text area. At the bottom right, there are two buttons: a printer icon and a green checkmark.

Figure 64. Probe Replacement Registry Window

11.2.4 The System Solution Registry Tab

Use the **System Solution Registry** tab (Figure 65) to check information on previous refillings of system solution and record the refilling of the system solution containers.

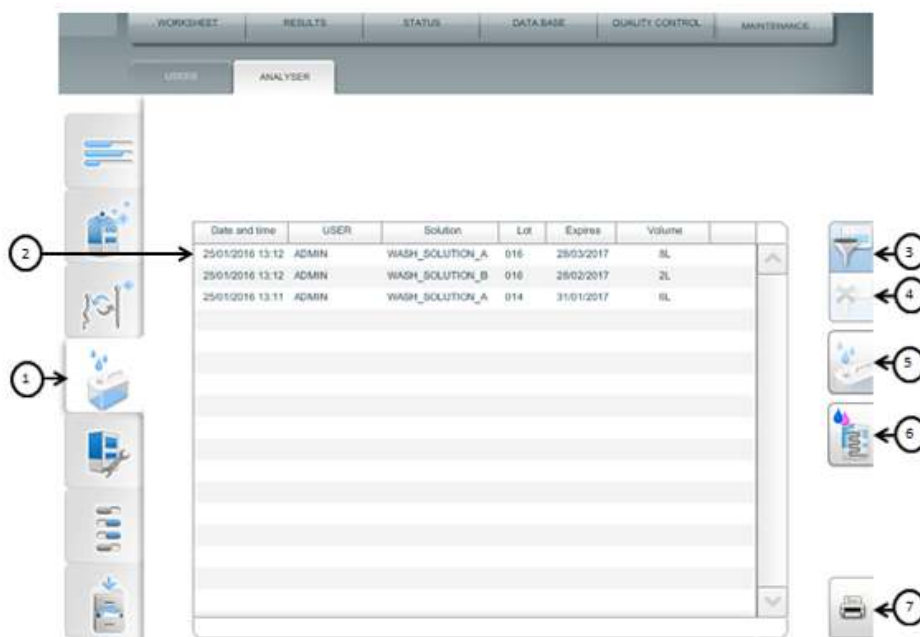


Figure 65. System Solution Registry Tab

- (1) The **System Solution Registry** tab icon.
- (2) Registry of the previous refillings of the system solution containers.
- (3) The **Filter** button.
- (4) Button to remove filter criteria.
- (5) The **System Solution Registry** button.
- (6) Option to manually prime the fluidic circuit.
- (7) Button to print the registry of the refilling of the system solution containers.

Press the **Filter** button (Figure 65, no. 3) to filter and search previous refillings of system solution containers by date (Figure 66).

DATE RANGE

FROM TO

Select solution

ALL

Figure 66. Filter for Searching Refillings of System Solution Containers

Press the **System Solution Registry** (Figure 65, no. 5) to open a window (Figure 67) to record each refilling action, where the type of system solution, its lot and expiration date, and the total volume diluted can be noted. It is also possible to add comments and print the registry.

Add fill of wash solution record

Date and time 23/07/2013 14:41

User ADMIN

Wash solution

WASH_SOLUTION_A

Lot

Expires

Volume

☒ 2L ☐ 4L ☐ 6L ☐ 8L

Comments

Touch to add a comment

Figure 67. System Solution Registry Window

It is possible to manually prime the fluidic circuit at any moment. To do it, press the **Fluidic**



Action icon that appears in the **System Solution Registry** tab (Figure 65, no. 6). If the action can not be performed at that moment, an error message will appear on the screen, informing about that (Figure 68).

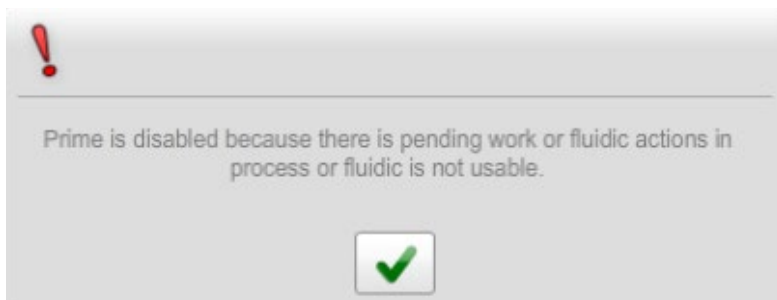


Figure 68. System Prime Error Message

It is possible to maintain the instrument always primed, so it is ready to process samples at any moment. To configure this option, please contact your local Grifols service representative.

11.2.5 The Technical Service Inspection Registry Tab

Use the **Technical Service Inspection Registry** tab (Figure 69) to check information about previous Technical Service inspections and to record Technical Service inspections.

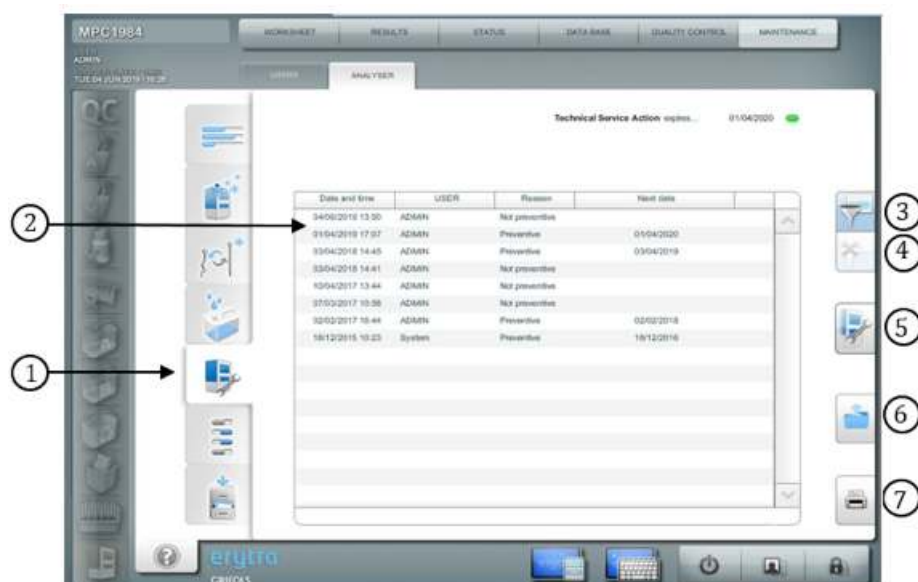


Figure 69. Technical Service Inspection Registry Tab

- (1) The **Technical Service Inspection Registry** tab icon.
- (2) Record of the previous Technical Service inspections.
- (3) The **Filter** button.
- (4) Button to remove filter criteria.
- (5) The **Technical Service Inspection Registry** button.
- (6) **Support Pack** button.
- (7) Button to print the Technical Service inspection registry.

Press the **Filter** button (Figure 69, no. 3) to open a window (Figure 70) to filter and search previous Technical Service inspections by date.

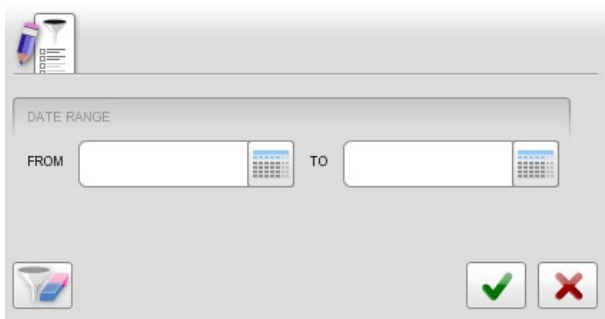
The image shows a software window titled "Technical Service Inspection Registry Search Window". It features a "DATE RANGE" section with "FROM" and "TO" labels, each followed by a text input field and a small calendar icon. At the bottom left is a printer icon, and at the bottom right are two buttons: one with a green checkmark and one with a red X.

Figure 70. Technical Service Inspection Registry Search Window

Press the **Technical Service Inspection Registry** button (Figure 69, no. 5) to open a window (Figure 71) to record each Technical Service inspection, with a check box for preventive inspections, a **Comments** field, and a button to print the registry.

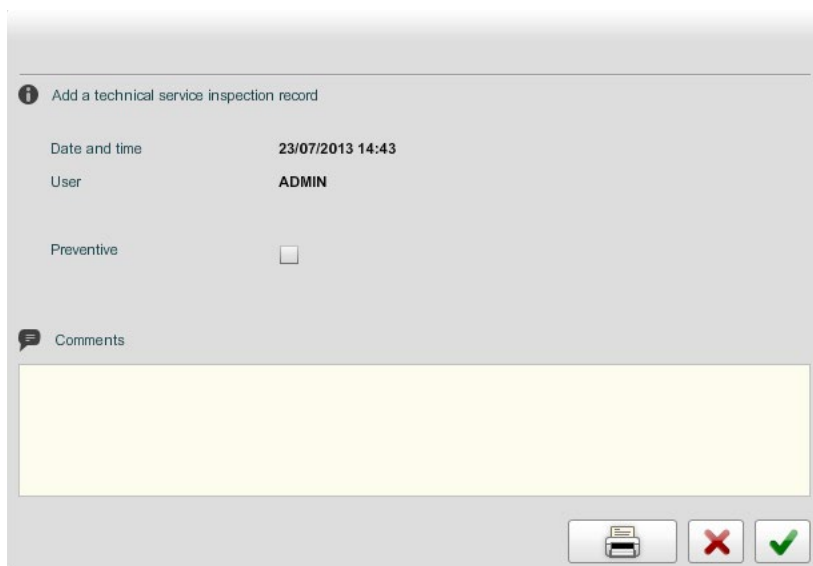
The image shows a software window titled "Technical Service Inspection Registry Window". It has a header bar with an information icon and the text "Add a technical service inspection record". Below this, there are three rows of labels and values: "Date and time" with the value "23/07/2013 14:43", "User" with the value "ADMIN", and "Preventive" with an unchecked checkbox. Below these is a "Comments" section with a speech bubble icon and a large, empty yellow rectangular text area. At the bottom right, there are three buttons: a printer icon, a red X button, and a green checkmark button.

Figure 71. Technical Service Inspection Registry Window



Press the **Support Pack** button (Figure 69, no. 6) to open a window (Figure 72) to run a manual backup which contains certain stored data for support purposes. In this window, it is possible to filter the information to be included according to dates and type of file:

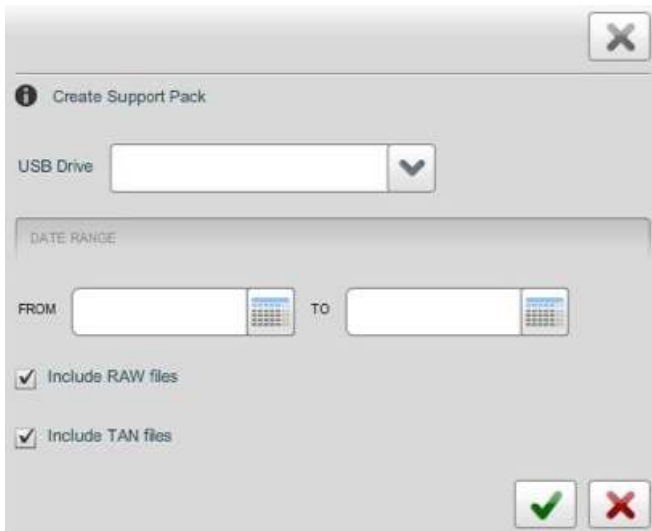


Figure 72. Support Pack Window



NOTE: The **Support Pack** does not store any Personal Data, even being the instrument connected to LIS and demographic data displaying configuration enabled.

11.2.6 The Activation/Deactivation Modules Tab

Use the **Activate/Deactivate Modules** tab (Figure 73) to check the current activation status of the modules, view information about previous module activation or deactivation, or activate or deactivate a module. For the full module activation/deactivation procedure, see Section 11.11.

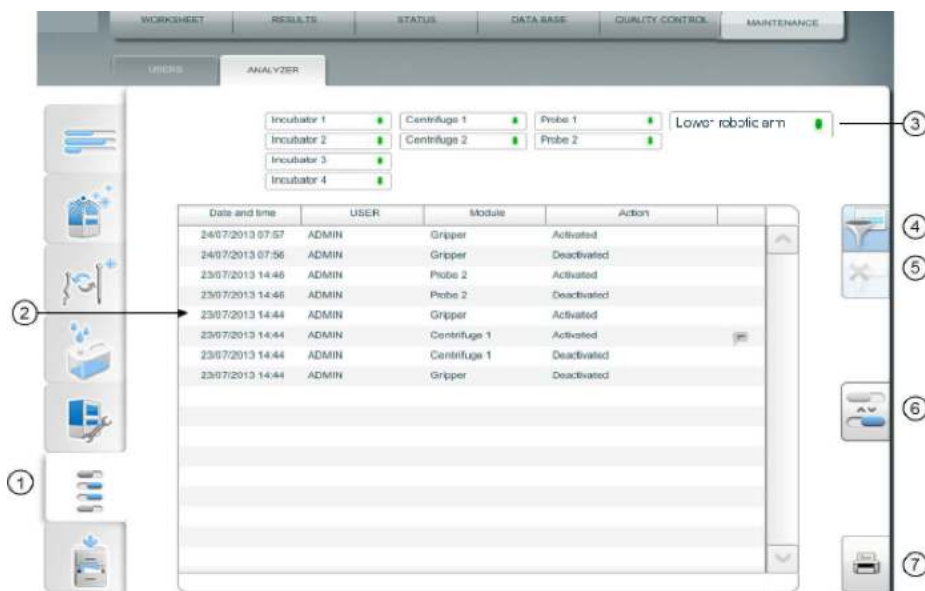


Figure 73. Activate/Deactivate Modules Tab

- (1) The **Activate/Deactivate Modules** tab icon.
- (2) Record of the previous activation and deactivation of modules.
- (3) Current status of the different modules of the instrument (Incubators, Centrifuges, Probes and Gripper). If the module is enabled, it appears in green. If it is disabled, it appears in red.
- (4) The **Filter** button to open a window (Figure 74) to filter by date range and by module.
- (5) Button to remove filter criteria.
- (6) The **Activate/Deactivate Modules** button.
- (7) Button to print the registry of modules activated/deactivated.

Press the **Filter** button (Figure 73, no. 4) to open a window (Figure 74) to filter by date range and by module.

DATE RANGE

FROM TO

MODULE

☐ ALL

☐ Incubator 1 ☐ Centrifuge 1 ☐ Probe 1 ☐ Lower robotic arm

☐ Incubator 2 ☐ Centrifuge 2 ☐ Probe 2

☐ Incubator 3

☐ Incubator 4

☐ ☐

Figure 74. Activate/Deactivate Modules Search Window

Press the **Activate/Deactivate Modules** button (Figure 73, no. 6) to open a window (Figure 75) to activate or deactivate a specific module, where the desired module(s) can be selected.

1/2

i Select one module for activate / deactivate

Incubator 1 ☐ Centrifuge 1 ☐ Probe 1 ☐ Lower robotic arm ☐

Incubator 2 ☐ Centrifuge 2 ☐ Probe 2 ☐

Incubator 3 ☐

Incubator 4 ☐

☐ ☐

Figure 75. Activate/Deactivate Modules Window

For the full procedure to activate or deactivate a module, see Section 11.11.

11.2.7 The Backup Tab

Use the **Backup** tab (Figure 76) to view information on previous backups and to perform a backup of the following information stored in the Erytra computer:

- Configuration files.
- Results Data.
- Log files.
- **Database** backups.



NOTE: If LIS connection is enabled and LIS transmits Personal Data, the **Backup** stores Personal Data (in the log files for data integrity and traceability purposes, and also in **Database** if instrument is configured to display demographic data).



NOTE: The tests requested and results obtained are stored in **Database**. If LIS connection is enabled (and workload is received from LIS) and the instrument is configured to display demographic data, the **Database** also includes Personal Data. This **Database** is password protected.

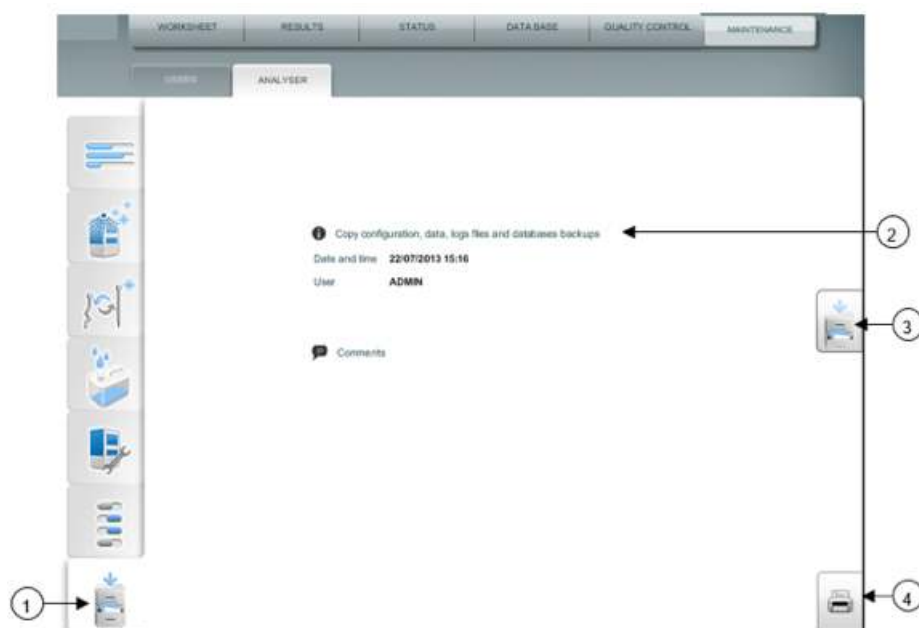


Figure 76. Backup Tab

- (1) The **Backup** tab icon.
- (2) **Record** of the last backup performed, showing information about the user, date and time and comments associated to that backup.
- (3) Button to **Start the Backup** procedure.
- (4) Button to **Print** information about the last backup performed.

For the full **Database** backup procedure, see Section 11.3.

11.3 Database Backups

To perform backups of the system data:

1. Go to the **Maintenance > Analyzer** screen and select the **Backup** tab icon (Figure 76, no. 1). For more information, see Section 11.2.7.
2. Click the **Start Backup** button (Figure 76, no. 3).

By default, the information that has been backed up will be stored in the E:\Backup folder. To configure another folder to store the backup, contact your local Grifols service representative.



WARNING: To avoid the loss of data due to a temporary corruption of the system, make a backup copy of the results stored in the Erytra analyzer **Database** on a regular basis.



WARNING: If an error occurs during the execution of a back-up, an error message will appear on the screen. If the problem persists, please contact your local Grifols service representative.



NOTE: Backing up the **Database** is only possible when the instrument is idle.



NOTE: If backup is configured in any other unit defined in the instrument configuration, as external or network drives, it should be according to the laboratory security policies.

11.4 Restoring the Database

The backup file is not accessible from the user interface, to restore the **Database**, contact your local Grifols service representative.



CAUTION: It is recommended to do a copy of the **Database** of the Erytra software before restoring it. To do this, follow the instructions described in Section 11.3.

11.5 Restoring the User Database

The backup file is not accessible from the user interface, to restore the user **Database** (containing user configuration information), contact your local Grifols service representative.

11.6 Basic Maintenance

The following maintenance tasks should be performed **when needed** (see Section 11.6 for more information).

11.6.1 Checking for Spills

Check for any liquids that have leaked or been spilled in the following areas:

- The Sample and Reagent Station.
- The Dilution Station.
- On the seal of the cards in the incubator.
- The protective lid of the incubator on the upper level of the analyzer.
- The waste solution drawers.
- The floor of the analyzer.

Clean spills or leaks carefully with a dampened cloth, as described in Section 11.7.1 and contact your local Grifols service representative if the issue occurs again.

11.6.2 Check for the Growth of Microorganisms

The growth of microorganisms must be avoided with systematic cleaning of the Erytra analyzer. It is recommended to check the following areas **periodically**:

- The system solution and waste solution containers and their lids.
- The connecting tubes of the containers and of the Washing and Dilution Station.
- The surface of the Sample, Reagent, Grifols gel Card and Dilution Station.
- The probe washers.
- In general, all of the interior surfaces of the instrument.

If you observe spills in areas that are not accessible, contact your local Grifols service representative.

11.7 Weekly Maintenance: Cleaning the Instrument Surfaces



WARNING: The cleaning process of the instrument surfaces, the Dilution Station and the probe, must always be done with the instrument unplugged from the power supply.



WARNING: Do not dismantle the instrument under any circumstances. If liquid has penetrated into its interior, unplug it from the power supply and wipe up the liquid. The cleaning operation must be done by the Operator. If liquid has spilled into an area of the analyzer that is inaccessible, contact your local Grifols service representative.



WARNING: Waste materials should be disposed of according to local regulations.



CAUTION: During the cleaning and/or decontamination procedures, wear personal protection equipment (gloves, gown, laboratory glasses, etc.).



CAUTION: Before starting the cleaning and decontamination procedures, all of the samples and reagents must be removed from the inside of the instrument.



WARNING: Before using any means of cleaning or decontamination that are different from those recommended by the manufacturer, the Supervisor must verify with the manufacturer that the proposed methods are not going to damage the instrument.

11.7.1 Cleaning the External Surfaces

The Erytra analyzer has been manufactured using materials of the best of quality. However, biological samples, saline solutions, acid or alkaline solvents can cause damage to the exterior surface of the analyzer and should be removed immediately. Periodically wipe the exterior surfaces with a cloth dampened with a mild detergent.



WARNING: Solutions containing alcohol should not be used to clean the acrylic plastic of the front doors or side panels of the Erytra.



NOTE: Before cleaning the exterior surfaces of the analyzer, unplug it from the power supply.



NOTE: The cleaning of the front and side plastic must be done with mild products that are non-abrasive, to avoid scratching the plastic.

11.7.2 Cleaning the Touch Screen

Clean regularly the touch screen with a cloth that does not leave behind lint and that has been dampened with a solution of 50% denatured alcohol (v/v) or 50% isopropyl alcohol (v/v).

Likewise, 70% isopropyl alcohol pads can also be used for wiping the touch screen. After cleaning, dry the touch screen with an absorbent material that does not leave behind lint.



WARNING: Do not scrub the touch screen directly with cleaning agents as there is a danger that they will leak inside and result in an electric shock to the Operator, as well as cause the deterioration of the touch screen.



CAUTION: It is important to avoid caustic chemical agents when cleaning the touch screen.



CAUTION: Denatured alcohol and isopropyl alcohol are highly flammable and irritating. Read the Safety Data Sheet (SDS) before use and follow the appropriate precautions.

11.7.3 Cleaning the Internal Surfaces

Clean the internal surfaces, once the holders have been removed, with a cloth which has been dampened with mild detergent and water.

Clean the upper part of the Washing Station of the probe with a swab dampened with purified water by inserting the swab into the openings in the robotic arm.



WARNING: Take care to not spill liquids into the equipment openings.

Remove the system solution and waste solution containers from the instrument to be emptied and cleaned. Wash the outside of these containers with mild detergent and water.



WARNING: The contents of the waste drawers must be dumped into special containers designated for waste disposal and disposed of in accordance with regulatory requirements.

11.8 Monthly Maintenance: Decontaminating the Instrument

To avoid health risk, decontaminate the analyzer on a month basis or in the following situations:

- After spills, leaks, etc. of potentially contaminating liquids.
- Preparation of the instrument for its transport or storage.
- Before the intervention of the Technical Service.
- Before disposing the instrument.

The fluid circuit and the surfaces in contact with potentially contaminating liquids must be decontaminated.



WARNING: The Erytra analyzer must be decontaminated before performing any preventative and/or corrective maintenance operations.



WARNING: Waste materials should be disposed of according to local regulations.



NOTE: It is recommended to record decontamination activities after they are performed. For more information, see Section 11.2.2. It is also possible to fill in the record 18.1 which is attached to these Instructions for Use before any intervention of the Technical Service or whenever necessary.

11.8.1 Decontamination Material Required

To decontaminate the Erytra analyzer, gather the following supplies:

- 2000 mL of 0.5% sodium hypochlorite solution (v/v).
- 5000 mL of purified water when the instrument is not draining the waste into an external drain (mono mode). If so, 7000 mL of purified water are needed (dual mode).
- A decontamination container.



During the use of the decontamination liquids adhere to the following recommendations:

- The corresponding Instructions for Use.
- Laboratory security rules.
- Current local risk prevention legislation.



WARNING: Some cleaning or decontamination products, such as sodium hypochlorite solution, may be corrosive, irritating for the skin or eyes, toxic if by inhaled, absorbed or ingested. Read the Safety Data Sheet (SDS) before use and follow appropriate precautions.



CAUTION: Before using any means of cleaning or decontamination that are different from those recommended by the manufacturer, the Supervisor must verify with the manufacturer that the proposed methods are not going to damage the instrument.



WARNING: During the use of the decontamination liquids, adhere to the following recommendations:

- The corresponding Instructions for Use.
 - Laboratory security rules.
 - Current local risk prevention legislation.
-

11.8.2 Decontaminating the Fluidic System

To decontaminate the fluidic system of the Erytra analyzer:



1. From the **Maintenance > Analyzer** screen press the **Decontamination** tab (Figure 57, no. 1).
2. Press the **Start Decontamination** button (Figure 58, no. 6) to run the fluidic system decontamination procedure. To start the procedure, click the **Confirm** button on the message that appears (Figure 61), and then follow the instructions shown on screen.



NOTE: During the decontamination process the screen will display key information to guide the user:

- The specific action that is being performed at the moment.
 - Next user intervention, the real time marked by clocking which the user must interact with the analyzer.
 - Expected time left, the total time required to complete the whole decontamination process.
-

3. When the analyzer has indicated, remove all the System Solution A and B containers from the analyzer.

Decontamination step:

4. Connect the decontamination solution container filled with 2000 mL of decontamination solution in the position of the System Solution A container.
5. Press **Continue**. The fluidic system starts priming.
6. When the analyzer has indicated, remove the System Solution A container and place the decontamination solution container in the position of the System Solution B container.



CAUTION: Make sure that decontamination solution container is filled with 2000 mL of decontamination solution.

7. Press **Continue**. The fluidic system starts priming.
8. When prime, rinse and soak steps are completed and when the analyzer has indicated, remove the decontamination solution container.

Water step:

9. Place the solution containers filled with 4000 mL of purified water in the position A and B, respectively.
10. Press **Continue**. The fluidic system starts priming with purified water.
11. When prime, rinse and empty steps are completed, remove the solution containers.
12. Empty the purified water from the solution containers.

Solutions step:

13. Refill the solution containers with the corresponding diluted System Solutions A and B.
14. Open any reagent drawer to automatically initiate the priming of the fluidic system.
Once the prime is finished, the analyzer will record automatically the decontamination of the fluidic system and it will be ready to perform analysis.



NOTE: The decontamination procedure takes into account the configuration of the fluidic system (*i.e.* whether or not the instrument is set up to drain to a lab drain or not).



NOTE: Before starting the decontamination of the fluidic system make sure that the Waste Solutions Container are totally empty.



NOTE: The decontamination process could be interrupted for various reasons. If the interruption happens during the water step or the solution step the process could be resumed at the beginning of this step if has not been elapsed more than one hour. Otherwise, the entire decontamination process must be restarted.



NOTE: Once the fluidic system decontamination process is finished, it is recommended to discard the remaining decontamination solution and rinse with water the inside of the Decontamination container.

If working in Dual mode, to decontaminate the fluidic system of the analyzer:

1. From the **Maintenance > Analyzer** screen (Figure 57) press the **Decontamination** tab (Figure 58, no. 2).

The **Decontamination** screen is displayed (Figure 58).

2. Press the **Fluidic System Decontamination** button  (Figure 58, no. 6) to run the fluidic system decontamination procedure.
3. Click the **Confirm** button on the message that appears on screen to start the procedure and follow the instructions shown on screen.
4. When the analyzer has indicated, remove all the System Solution A and B containers from the analyzer.

Decontamination step:

5. Connect the decontamination solution container filled with 2000 mL of decontamination solution on the first position for System Solution A container (container 1).
6. Press **Continue**. The fluidic system starts priming.
7. When the analyzer has indicated, remove the System Solution A container and place the decontamination solution container on the first position for System Solution B container (container 2).
8. Press **Continue**. The fluidic system starts priming.
9. When the analyzer has indicated, remove the decontamination solution container from position 2 and place it on the second position for System Solution A container (container 3).
10. Press **Continue**. The fluidic system starts priming.
11. When the analyzer has indicated, remove the decontamination solution container from position 3 and place it on the second position System Solution B container (container 4).
12. Press **Continue**. The fluidic system starts priming.
13. When prime, rinse and soak steps are completed and when the analyzer has indicated, remove the decontamination solution container from the analyzer.

Water step:

14. Place the solution containers filled with 4000 mL of purified water all positions, 1 to 4 respectively.
15. Press **Continue**. The fluidic system starts priming with purified water.
16. When prime, rinse and empty are completed and when the analyzer has indicated, please remove all the solution containers.
17. Empty the purified water from the containers.

Solutions step:

18. Refill the solution containers with the corresponding diluted System Solution A and B.
19. Place them again on their respectively locations.

Once the prime is finished, the analyzer will record automatically the decontamination of the fluidic system and it will be ready to perform analysis.



NOTE: Once the fluidic system decontamination process is finished, it is recommended to discard the remaining decontamination solution and rinse with water the inside of the Decontamination container.

The decontamination procedure takes into account the configuration of the fluidic system (whether or not the instrument is set up to drain to a lab drain or not).

11.8.3 Decontaminating the Probes

Decontaminate the probes periodically, and whenever they get dirty, to keep the exterior of the probes clean and to ensure that the probe level detection system performs properly.

To do this, proceed as follows:

1. Go to the **Maintenance > Analyzer** screen and press the **Probe Replacement** tab



(Figure 62, no. 1). (For more information, see Section 11.2.3).

2. Press the **Replace Probe 1** button (Figure 62, no. 5) to move probe 1 to an accessible position.
3. Clean the exterior of the probe with a cloth dampened with 0.5% sodium hypochlorite solution.
4. Wipe the exterior of the probe with a cloth dampened with purified water.
5. Decontaminate probe 2 in the same manner by repeating steps 2 - 4.

11.8.4 Decontaminating the Dilution Station

1. Open the top door (Figure 1, no. 3) and, with a pipette, add 2.5 mL of 0.5% sodium hypochlorite to the Dilution Station.
2. Close the upper door.
3. Let the decontamination solution stand for 15 minutes.
4. Remove the sodium hypochlorite solution from the Dilution Station with a pipette.
5. The previous steps are automatically performed during the monthly decontamination of the fluidic system. If the decontamination of the fluidic circuit has been performed recently, start with step 5 of the procedure to decontaminate the Dilution Station.
6. Open the upper door and wipe the interior of the Dilution Station using a cotton swab dampened with purified water.

11.8.5 Decontaminating the Instrument Surfaces

To decontaminate the instrument, proceed as follows:

1. With the instrument plugged in, open the upper door and take out the holders from the Sample and Reagent Stations.
2. Turn off the instrument and unplug it from the power supply.
3. Open manually all the drawers of the different containers on the lower level (Section 17.3).
4. In the case of a spill, wipe up the liquid with an absorbent material (for example paper towels or a bunch of gauzes or tissues).
5. Decontaminate the surfaces with a solution of 0.5% sodium hypochlorite (v/v). To do this, clean the surfaces with disposable swabs or wipes soaked in disinfectant so they are completely wet.
6. Wipe up the disinfecting solution with a disposable material.
7. Wipe the surface with a mild detergent mixed with water in order to remove the residual harmful chemical components.
8. Dry the surface.
9. Discard all of the contaminated material that was used during the decontamination process into a container for biological waste.

11.8.6 Decontaminating the Samples and/or Reagents Holders

To decontaminate the sample and reagent holders:

1. Remove the sample and reagent holders from the analyzer.
2. Soak them for half an hour in a preparation of mild detergent and an approximate 0.5% sodium hypochlorite (v/v) solution in water.
3. Rinse completely.

4. Allow the holders to dry, taking care not to force the straps or the flaps and making sure that they are completely dry before reloading them into the analyzer.

11.8.7 Decontaminating the Liquid Containers

Clean the outside of the liquid containers with a cloth dampened with mild detergent and water. The inside of the containers can be cleaned by rinsing them with an aqueous solution of 0.5% sodium hypochlorite (v/v).



WARNING: If the liquid containers are cleaned with sodium hypochlorite, intensive rinses with purified water should be performed in order to remove all traces of the sodium hypochlorite before using the container again.

11.9 Preventive Maintenance

Preventive maintenance of the Erytra analyzer must be performed on a year basis by a Qualified Technician.



CAUTION: Before the Qualified Technician performs any preventive and/or corrective maintenance operations, remove all the samples and reagents from the instrument.

This maintenance includes, among others:

- Mechanical review.
- Fluidic system review.
- Review of the status of the probe washers and the probes.
- Review of the reader and its components.
- Review of the IT system.

It is recommended to record the preventive maintenance Technical Service inspections. For more information, see Section 11.2.5.

11.10 Replacing a Probe

In the case of breakage, bending or persistent obstructions, the probes can be replaced without the need for Technical Service. The probes are named as Probe 1 (left) and Probe 2 (right) when looking at the analyzer from a frontal view. To change a probe:

1. Go to the **Maintenance > Analyzer** screen and press the **Probe Replacement** tab



(Figure 62, no. 1). (For more information, see Section 11.2.3).

2. Press the **Replace Probe 1**  button or the **Replace Probe 2**  button, depending on the case, and follow the program instructions.



NOTE: To help during the probe replacement, use the replacement tool provided with the instrument.

3. The analyzer opens one sample drawer and requests the removal of holders.
4. Once holders have been removed, close the drawer and press **Continue**. The upper robotic arm is situated in the drawer position and it lowers its head so that the probes are made more accessible.
5. Open the upper door and replace the probe in question by unscrewing the nut that holds the probe on the robotic arm, turning it clockwise until it comes loose and free.
6. Carefully remove the probe from its support, holding it by the bottom part. Discard the probe.
7. Insert the new probe in the probe holder and then in its hole of the probes washer. The upper part of the probe must fit properly inside the probe holder. Once the probe is correctly placed, tighten probe fitting to the maximum.
8. Make sure that the probe is securely fastened (a click it is heard in the safe position).
9. Close the upper door and press **OK** to continue.

The software restarts the robotic arm and leaves it prepared for test processing.

It is recommended to record probe replacement activities. For more information, see Section 11.2.3.

11.11 Activating and Deactivating Modules

To activate or deactivate a module:

1. Go to the **Maintenance > Analyzer** screen and press the **Activate/Deactivate Modules**



tab (Figure 73, no. 1). For more information, see Section 11.2.6.

2. Press the **Activate/Deactivate Modules** button (Figure 73, no. 6).

A window opens (Figure 75), with the active modules shown in green and the inactive ones in red.

3. Select the module to change and click the **Confirm** button.

Module activation is only possible when the instrument is idle.

The software records module activation and deactivation activities. For more information, see Section 11.2.6.

The following window opens (Figure 77), where comments can be entered and the registry can be printed.

Figure 77. Activate/Deactivate Modules Registry Window

4. Click the **Confirm** button  when finished to close the window.

Erytra

12 Diagnostic

The Diagnostic program performs a series of tests that check for the correct functioning of the different modules of the instrument. To run it, it is possible to access from the initial **Login** window

of the analyzer (Figure 78), selecting the **Diagnostic** icon

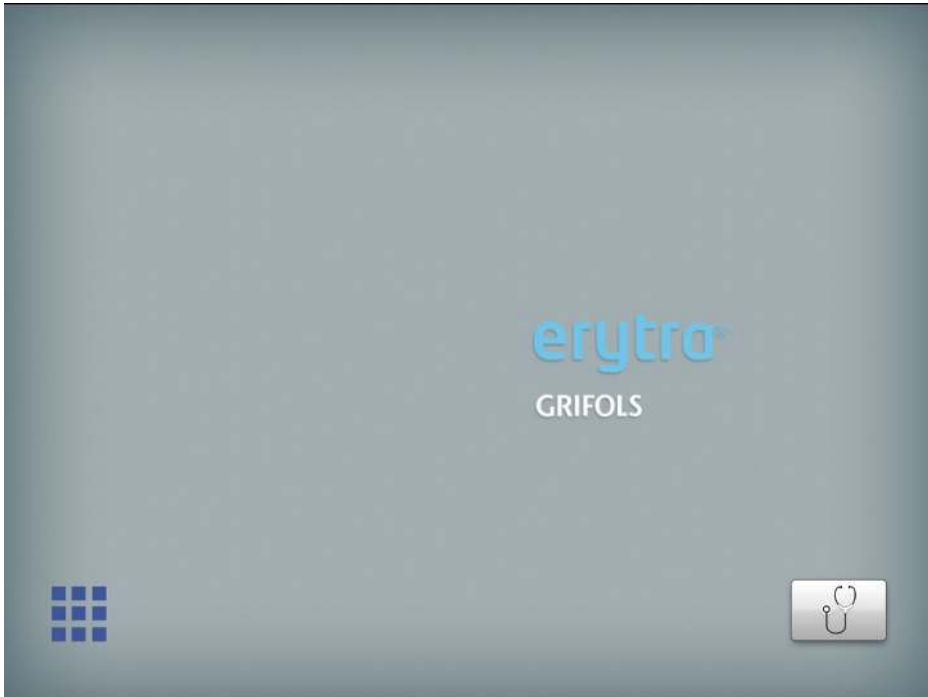


Figure 78. Initial Login Window of Erytra



NOTE: The **Diagnostic** icon is only available to press during the initial loading window.

When pressed the **Diagnostic** icon, a menu tab opens (Figure 79):

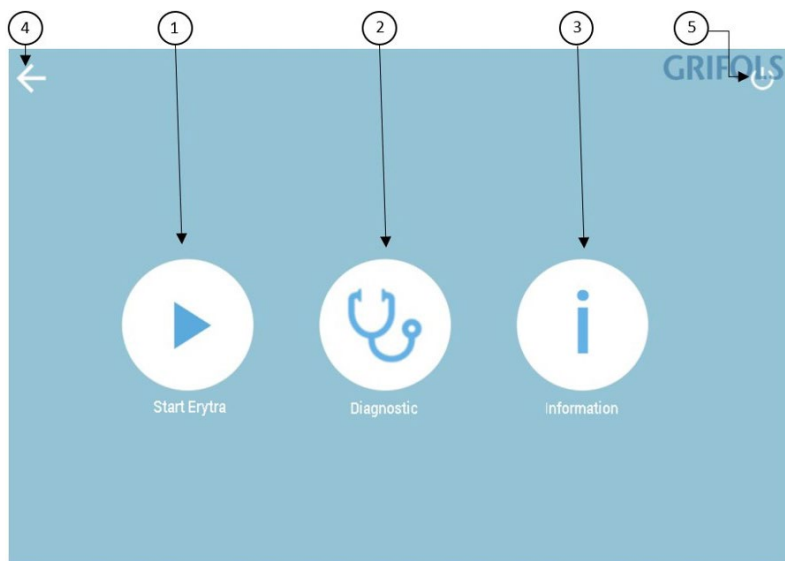


Figure 79. Diagnostic Menu Tab

- (1) **Start Erytra** button.
- (2) **Diagnostic** button. 
- (3) **Information** button.
- (4) **Backwards** button.
- (5) **Shutdown** button.



Select the option **Diagnostic** in the **Diagnostic Menu** tab (Figure 79) to run the Diagnostic program.

Once the **Diagnostic** window opens (Figure 80), select the modules to be checked and press the

Green Arrow button  to start the diagnostic.

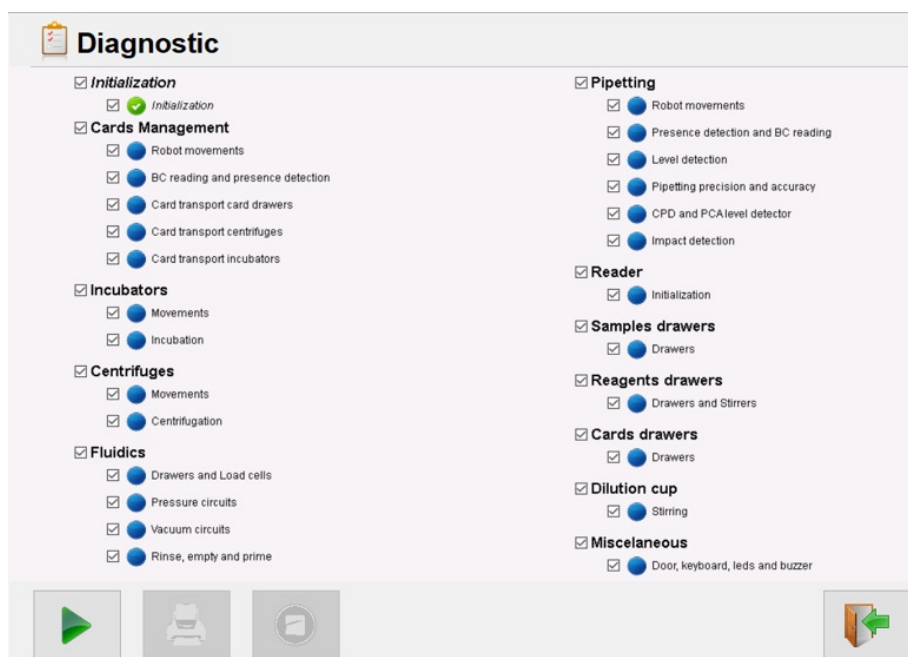


Figure 80. Diagnostic Window

To run the diagnostic on all of the analyzer modules, the following requirements must be met, as shown in the message that appears on the screen:

- The fluid system must be primed.
- A full system solution container (8 L) must be available.
- The fluid drawers must be closed.
- All of the sample drawers must be closed.
- All of the reagent drawers must be closed.
- All of the card drawers must be closed.
- The lower bottom robotic arm clamp must not have a card selected and must be in the home position.
- The incubators must be in the home position.
- The screw for the lower door must be removed.

- A 2 kg weight must be available.
- System Solution containers A and B must each contain a minimum of 4 L of the appropriate system solution.
- An empty system solution container must be available.
- A minimum capacity of 8 L in the two waste drawers must be available.
- There must be no samples in drawers 4, 5 and 6.
- Holder 2 in sample drawer 3 must contain tubes filled with 0.9% saline solution:
 - 10 mm diameter tubes in positions 1 to 4.
 - 12 mm diameter tubes in positions 5 to 8.
 - 15 mm in diameter in positions 9 to 12.
- There must be reagent vials filled with reagent erythrocytes or 0.9% saline solution in drawer 1, holder 1, positions 2 and 9, and in drawer 1, holder 2, positions 1, 6, 11 and 18.
- There must be a card holder with first 12 consecutive cards only (all in lower floor) in drawer 1, position 4.
- There must not be any Grifols gel cards in drawers 2, 3 or 4, or in the incubators or centrifuges.
- There must be empty card racks in position 1 of drawer 2, position 2 of drawer 3 and position 3 of drawer 4.

To open drawers for the requirements above, use the buttons on the front of the analyzer.

Follow the instructions that appear on the screen while the diagnostic is running.



WARNING: Do not open the lower door while the Diagnostic program is running, as the lower robotic arm may be moving.



WARNING: Diagnostic must be executed with drawers and doors closed as Diagnostic Requirements indicate.

The Diagnostic application indicates the status of the test via a specific icon:

- Non executed.
- Pass.
- Fail.
- Warning.
- Running.
- Waiting.
- Stopped.

After the Diagnostic program is complete, the print icon will become active so that an execution report containing the results of the different modules checked by the program (Figure 81) can be printed.

Diagnostic

Serial number: 0625 (ERYPD2)

Initialization	
Initialization test	pass
Cards Management	
Robot movements	not executed
Card transport centrifuges	not executed
Card transport card drawers	not executed
BC reading and presence detection	not executed
Card transport incubators	not executed
Incubators	
Movements	pass
Incubation	pass
Centrifuges	
Movements	pass
Centrifugation	pass
Fluidics	
Pressure circuits	fail
Vacuum circuits	fail

Figure 81. Execution Report

The Execution Report shows the results of the Diagnostic grouped by module, by the result of the diagnostic whether a status of "PASS" or "FAIL", and by whether the diagnostic has been executed on that module or not. These results may be saved and/or printed from the tool bar:

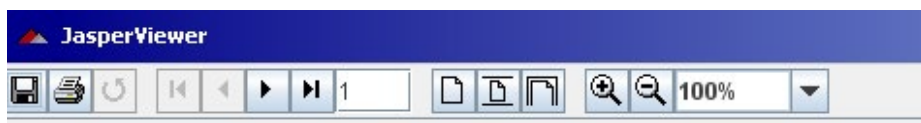


Figure 82. Diagnostic Tool Bar

To close the Diagnostic program and initialize the analyzer, press  and **Diagnostic**

Menu tab will appear again. Then press **Start Erytra** option  from the menu.

If any modules do not pass, contact your local Grifols service representative.

13 Transport and Storage

13.1 Storing the Instrument

If the Erytra analyzer must be stored for a long period of time, it is recommended that it is packaged as if it were to be transported.

The environmental conditions of storage must be those specified in Section 3.1.

The requirements for storing the instrument are:

- Size: Width 1200 mm by depth, 800 mm by height 2000 mm.
- Weight: 350 kg.



CAUTION: Only use the original packaging for the transport of the instrument.



WARNING: The instrument must be decontaminated before being transported.

13.2 Unpacking the Instrument

To unpack the instrument, remove the packaging by following the instructions:

1. Remove the wooden box keeping in mind the screws of the pallet.
2. Remove the foam covers that protect the instrument.

13.3 Placing the Instrument in its Location Place of Operation

Unpack the Erytra analyzer as close as possible to its installation location.

Here it will be unpacked as explained in the previous section and it will be placed in the chosen location.

The Erytra analyzer has 4 sets of 4 wheels on each corner. To move the instrument the levelling feet that stop the wheels from moving are unscrewed and the instrument is then pushed to the desired location. A minimum of 2 people are needed to push the instrument.



WARNING: The weight of the instrument is considerable, and therefore it is advised to strictly follow these installation instructions.



WARNING: The screw on feet must be completely removed before moving the Erytra analyzer, as they might otherwise break.

Erytra

14 Equipment Disposal

At the end of its useful life or in the case of wanting to dispose the instrument, remove all remains of samples and reagents, and then clean and decontaminate it. Upon completion, dispose of the analyzer as electronic waste, according to local and state regulations, and sent it to an authorised centre for the disposal of this type of waste.



WARNING: This instrument must only be dismantled by authorized specialists.



CAUTION: When an instrument has reached the end of its useful life, keep in mind that all current local legislation must be complied with during disposal.

The Erytra analyzer is considered to be electronic waste. Electronics can contain hazardous materials. Dispose of all electronics in accordance with local and state regulations.

If you have any question about the disposal of the analyzer, contact your local Grifols service representative.

Erytra

15 Cybersecurity and Data Protection

15.1 Cybersecurity

Technical safeguards (*i.e.*, security features) are found in the Erytra analyzer manufactured by Diagnostic Grifols, S.A. to achieve a good cyber hygiene and to reduce cybersecurity risks events to an acceptable residual risk level. With the aim to facilitate healthcare providers' efforts in maintaining effective security programs and meeting any relevant regulatory requirements and/or standards, a Manufacturer Disclosure Statement for Medical Device Security (MDS²) can be provided upon request. This document supplies healthcare providers with important information to assist them in assessing the vulnerability and risks associated with protecting private data transmitted or maintained by the Erytra instrument.

15.2 Privacy

The Erytra analyzer is designed with the premises of "Privacy by default" and "Privacy by design". The software located in the analyzer computer runs on the Microsoft Windows 10 Enterprise 2019 LTSC Operating System and it includes different applications to manage the entire flow of Personal Data in the instrument.



NOTE: LIS downloading connection with demographic data option enabled is the only possible entrance of Personal Data in the analyzer. The demographic data the instrument can manage is limited to Name, Surname, Gender, Birthdate and Maiden name when available. Any extra information received from the LIS will be ignored. The laboratory based on their own policies can include Personal Data in each test request.



WARNING: The Personal Data transmitted by the LIS to the instrument is stored in the log files for traceability and data integrity purposes. If instrument is configured to display demographic data, this data is also stored in **Database**.



WARNING: When the sample information is received from the LIS, it also includes the Patient ID. This Patient ID should meet the definition of "pseudonymisation" given in the corresponding regulations, as it will not be managed as Personal Data by the instrument.



NOTE: Demographic data is displayed in **Sample Related Data** window only if LIS connection is enabled (workload is received from LIS) and the instrument is configured to display demographic data. Instead, the Patient ID is always displayed in **Sample Related Data** window.



NOTE: Demographic data is displayed in **Reading** and **Traceability Reports** only if LIS connection is enabled (workload is received from LIS) and the instrument is configured to display demographic data. Instead, the Patient ID is always displayed in **Reading** and **Traceability Reports**.



WARNING: The reports stored in the instrument will be automatically deleted after a certain time.



NOTE: The **Support Pack** does not store any Personal Data, even being the instrument connected to LIS and demographic data displaying configuration enabled.



NOTE: If LIS connection is enabled and LIS transmits Personal Data, the **Backup** stores Personal Data (in the log files for data integrity and traceability purposes, and also in **Database** if instrument is configured to display demographic data).



NOTE: The tests requested and results obtained are stored in **Database**. If LIS connection is enabled (and workload is received from LIS) and the instrument is configured to display demographic data, the **Database** also includes Personal Data. This **Database** is password protected.



NOTE: If backup is configured in any other unit defined in the instrument configuration, as external or network drives, it should be according to the laboratory security policies.

16 Warranty

The warranty conditions are established in a separate document. For more information, contact your provider.

The following parts are excluded from the guarantee:

- RS 232 Cable and network cable.
- The tubing and connectors.
- The syringe.
- The containers.
- The probes.
- Any part that has been used improperly.

Erytra

17 Troubleshooting

17.1 Incident List

During the normal working of the Erytra analyzer some error messages and incidents may appear that can be resolved by the Operator.

Lower-level Incidents can affect the working of the analyzer without stopping test execution completely.

Intermediate Level Incidents related to the lack of a resource (reagent, Grifols gel card, sample, system solutions, etc.) appear in the **Warnings Area** of the Erytra software and do not require immediate intervention. To obtain detailed information about how to solve these types of incidents, see Section 7.5.

High-level Incidents require immediate intervention to correct the situation. High-level incidents are accompanied by an acoustic alarm.

Incidents are shown on the screen as error messages, along with a red ten-digit code associated with it, in the top right portion of the error message. These digits are used in the following way:

- The first 2 digits identify the module that has produced an error.
- The next 2 digits identify the device that has produced an error in the module.
- The next 3 digits indicate the type of error.
- The last 3 digits indicate whether the error is a fluidic error. If the last 3 digits are zeros ("000"), the error is a non-fluidic error. For fluidic errors, the last three digits identify the type of fluidic error.

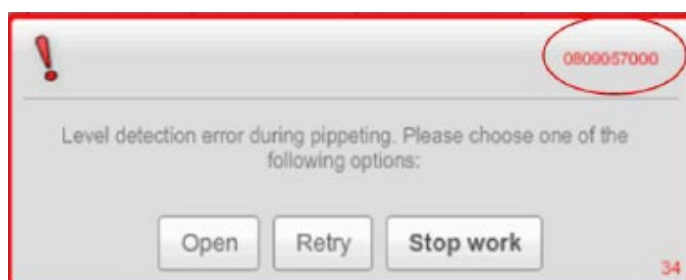


Figure 83. Incident Error Message

This information should be given to contact your local Grifols service representative if the incidence requires contacting them.

17.1.1 Low-Level Incidents

Low-level incidents appear on the Erytra software screen in the form of error messages. The following table shows a list of messages that are shown by the Erytra software, grouped by the type of error.

Table 9. Low-Level Incident Descriptions

MESSAGE	DESCRIPTION
IDENTIFICATION	
One or more tubes have not been identified. Please, go to Samples in order to repair the problem.	The identification process has found one or more tubes not detected or with a problematic barcode number. Go to the Samples screen to locate the tube that is responsible for the incident. Open the drawer, face the tube's barcode toward the identification window, and close the drawer. If the problem persists, identify the tube manually.
Identification process has found one or more vials not detected or with a wrong identifier. Please, go to Reagents in order to repair the problem.	Identification process has found one or more vials not detected or with a wrong identifier. Go to the Reagents screen to locate the vial that is responsible for the incident. Open the drawer, correctly face the vial's barcode and close the drawer.
The system has detected tubes manually identified. Please, select Yes to confirm that tubes have not been substituted by new ones and to confirm that the sample holders have been introduced in the same orientation as when those tubes were manually identified.	The process has detected tubes that have been previously identified manually. If you select Yes , the system will assume that the tubes have not been replaced with new tubes and will maintain the manual identification.
The system has detected more than one tube with the same barcode.	The system has detected more than one tube with the same barcode. Go to the Samples screen to locate the tubes that are responsible for the incident. Open the drawer and check the tubes that are responsible for the incident.
Samples holder identifier does not match with the configured ones.	The holder barcode identified by the system does not match with any of the configured ones.

MESSAGE	DESCRIPTION
Samples holder identifier has not been detected. Please ensure that the holder is properly labelled and try again.	The holder barcode has not been detected by the upper robotic arm. Open the drawer and check that the holder is labeled with the proper barcode.
EXECUTION	
One or more process incidents have been detected in tubes. Go to the Samples screen to resolve the problem.	One or more process incidents have been detected in tubes. Go to the Samples screen to check the type of incident as well as the affected tube and take the necessary actions.
The requested action is already in process. Please wait.	The analyzer has been requested to perform an action that cannot be executed immediately. Wait a few seconds until the analyzer executes the requested action.
There is no empty position to insert new samples. Select Open Immediately in order to replace an existing tube or Open After Works to wait until a position becomes available.	If the STAT button on the front of the instrument is used to insert a STAT sample and there are no positions available, the system immediately opens a drawer and gives the user the choice of replacing an existing tube or closing the drawer and waiting until a loaded sample has been pipetted so that it can be replaced with the STAT sample.
QUALITY CONTROL MANAGEMENT	
The identification process has found one or more expired QC tubes. Please, open drawer in order to repair the problem.	During the identification process, the system has detected that the expiry date has passed for one or more of the Quality Control samples. Please, open the drawer in order to repair the problem.

MESSAGE	DESCRIPTION
QC protocol is wrong or cancelled and some samples are stopped by QC. Please, press Continue if you want to queue with QC protocol omission, Stop if you want to maintain them stopped until QC is passed or Cancel if you want to cancel that works.	The system has registered a Quality Control result that is incorrect or cancelled and there are samples pending processing. In this case, the system offers the following options: • Continue processing them and omit the QC protocol. • Stop Work if you wish to maintain them on hold until the correct QC result is obtained. • Cancel if you wish to cancel all the jobs.
USER MANAGEMENT	
Access level insufficient for this action.	The identified user does not have permission to access the action requested, only a user with sufficient permission for this action can access it.
Error creating New User.	There has been an error creating a New User. Close the window and try again.
Error updating User.	There was an error when updating the User. Close the window and try again.
Some form required fields have not been filled in.	Some fields have not been filled. Complete the information to finish the process.
Error loading User Management screen.	There was an error on loading the User Management screen. Close the window and try again.
This is a required field!	There is still one field that has not been filled in. Complete the information to finish the process.
Username exist in Database .	A new user is being created, assigning an already existing username.

MESSAGE	DESCRIPTION
VALIDATION	
Are you sure that you want to Cancel the Validation for the selected results?	Press the Confirm button  to cancel the validation or press the Cancel button  to continue without cancelling the validation.
Are you sure that you want to Export the selected results?	Press the Confirm button  to continue exporting the selected results or press the Cancel button  to cancel the export.
Are you sure that you want to Reject the selected results?	Press the Confirm button  to reject the selected results or press the Cancel button  to cancel the rejecting of the selected results.
Samples with modified results are being validated. Are you sure you want to continue?	Press the Confirm button  to continue validating results that have been modified or press the Cancel button  to cancel the validation of results that have been modified.
Are you sure you want to modify this microtube result?	Press the Confirm button  to accept the modification made on the results or press the Cancel button  to not save the modification made on results.
There are no selected results.	There are no selected results, so no action can be performed on them.

MESSAGE	DESCRIPTION
PROBES	
Probe has been disabled.	One of the probes has been disabled. The error message will indicate if the deactivated probe is number 1 or 2. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
Probes cannot be changed whilst there is pending work.	Probes cannot be changed whilst there is pending work. Wait until the assigned workload is finished before changing the probes.
CENTRIFUGES	
Centrifuge has been disabled.	One of the centrifuges has been disabled by the system. Click Accept . The error message will indicate if the deactivated centrifuge is number 1 or 2. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
INCUBATORS	
Incubator has been disabled.	One of the incubators has been disabled. The error message will indicate if the deactivated incubator is number 1, 2, 3 or 4. If the problem still persists after restarting the Erytra software, contact your local Grifols service representative.
Temperature out of margins in incubator. Incubator has been disabled and tests (if any) cancelled.	The incubator temperature is outside the optimum range to perform the test and therefore will not be used. The error message will indicate if the deactivated incubator is number 1, 2, 3 or 4. If the problem still exists after restarting the Erytra software, contact your local Grifols service representative.

MESSAGE	DESCRIPTION
DECONTAMINATION	
Decontamination is disabled because there is pending work or fluidic actions in process.	The decontamination process cannot be started while there are pending tasks. Wait until the pending task is finished before proceeding with decontamination.
Decontamination process cannot be executed if waste capacity is smaller than 8 L.	The decontamination process has been cancelled as the total capacity of the waste drawer is less than 8 L. The container must be emptied before proceeding with the decontamination of the analyzer.

17.1.2 High-Level Incidents

High-level incidents require immediate Operator intervention. All incidents are accompanied by an error message with the option to “Retry” in the situations where this is possible. This option allows you to repeat execution of the process that generated the incident.

Some of these high-level incidents can only be resolved by rebooting the Erytra system. In these cases, you must accept the only option available that appears on the error message.

Table 10. High-Level Incident Descriptions

MESSAGE	DESCRIPTION
GENERAL	
There is an initialization problem. Please, Restart the System .	The system has detected an initialization problem. Press Restart the System .
Server is not available.	The system has detected a problem. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
Virtual keyboard is not available.	The system has detected that the virtual keyboard is not available. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
Error during Reader initialization. Please Restart the instrument.	The system has been unable to start the viewer module. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
Error setting security data.	The system has detected an error while setting security data. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.
Error opening Erytra program.	The system has detected an error while opening the software. If the problem persists after restarting the Erytra software, contact your local Grifols service representative.

MESSAGE	DESCRIPTION
HARDWARE	
Unrecoverable error. Please press Ok to Restart .	The system has detected an unreparable hardware error. Press Ok to restart the program.
The system has a significant error. Please Restart the instrument before requesting any further test.	The system has detected an irreparable error. It is necessary to restart the Erytra software.
FLUIDIC SYSTEM	
Required fluidic drawer is opened. Please, close it.	The system requires a fluidic drawer that is currently opened. To continue, close the drawer.
A lower drawer is opened. Please close it and select Retry . Select Cancel if the problem persists only.	The system has detected that a lower drawer is opened. Once closed, select Retry . If problem persists, select Cancel .
There is an opened Waste drawer. Please, close it and then press Retry fluidic operation.	The system has detected that there is an opened waste drawer. Please close it and, after it, press Retry to retry the fluidic operation.
Fast prime error.	The system has detected an error when priming or rinsing. Press Ok to Continue.
The fluidic system has been marked as not usable. Please Restart the instrument as soon as possible.	The fluidic system is not usable and the instrument should be restarted whenever possible.
There are one or more instrument fluidic incidents.	The system has detected one or more incidents related with the fluidics.
Pressure error on Solution A circuit. Please ensure: - Containers are properly connected. - Check that there is no spilling in the upper level.	The system has detected a pressure error in the Solution A circuit. Please ensure that containers are properly connected and that there is no spilling in the upper level.

MESSAGE	DESCRIPTION
Pressure error on Solution B circuit. Please check: - Containers are properly connected. - There is no spilling in the upper level.	The system has detected a pressure error in the Solution B circuit. Please ensure that containers are properly connected and that there is no spilling in the upper level.
Vacuum error. Please check: - Lab drain tube and connection. - There is no spilling in the upper level.	The system has detected a vacuum error. Please check the lab drain tube and connection and that there is no spilling in the upper level.
TRANSPORT	
Error in the lower robotic arm when taking a card. Please inspect middle level for lost card.	There has been an error in the lower robotic arm when taking a card. Please inspect the middle level to look for the lost card.
Error in the lower robotic arm when releasing a card. Please inspect middle level for lost card.	There has been an error in the lower robotic arm when releasing a card. Please inspect the middle level to look for the lost card.
Error during card drawer presence detection. Please choose one of the following options Retry or Cancel.	The system has detected an error during a card drawer presence detection performed by the lower robotic arm. Select Retry or Cancel.
An error in a barcode reader has been detected. Please, restart the system. In case the problem persists, contact your local Grifols service representative.	The system has detected an error during the reading of a barcode performed by the lower robotic arm. If the problem persists after restarting, contact your local Grifols service representative.
Card lost while reading. Select Retry or Cancel.	A card has been lost during the reading process. Select Retry or Cancel.

MESSAGE	DESCRIPTION
A card has been lost while transporting to the Service Rack. Ensure that the lost card does not affect the card transport and remove it as soon as possible. Select Retry or Cancel .	A card has been lost during the transport process. Select Retry or Cancel . Moreover, ensure that the lost card does not affect the card transport and remove it as soon as possible.
Card barcode not expected. Card will be discarded and samples cancelled. Press Ok .	The system has detected an unexpected barcode of a Grifols gel card. Press Ok to Continue discarding the sample of the associated card.
Service Rack is full. Please empty one Service Rack holder in order to resume cards reading.	Service Rack is full. It is necessary to empty one Service Rack holder to continue with the cards reading.
PIPETTING	
Obstruction detected during pipetting sample(s) X located in position(s) Y. Please Open the drawer and remove the obstruction.	The system has detected an error during the pipetting due to the detection of a clot in sample X located in position Y. Please Open the drawer, check the integrity of the sample and remove the clot.
Level detection error during pipetting. Please check there is enough volume and the absence of bubbles.	The system has detected an error during the level detection. Please check there is enough volume and the absence of bubbles.
Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options: Retry or Cancel Jobs .	The system has detected an error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options: Retry or Cancel Jobs .
Probe of upper robotic arm has found and obstacle. Please check: - Samples and reagents suitability. - Samples and reagents holders positioning.	The probe of upper robotic arm has found and obstacle. Please check: - Samples and reagents suitability. - Samples and reagents holders positioning.

MESSAGE	DESCRIPTION
PROBES	
It is not possible to process samples as there is no probe available.	None of the probes are available, so it is not possible to process samples.
The system was not able to complete the probe replacement process.	The process of replacing the probe has not been successful. Close the software and try again. If the problem persists, contact your local Grifols service representative.
INCUBATORS	
It is not possible to process samples as there is no incubator available.	None of the incubators are available, so it is not possible to process samples.
Error when trying to move an incubator up/down. Please choose one of the following options: Retry or Cancel Jobs .	The system has detected an error when trying to move an incubator up or down. Please choose one of the following options: Retry or Cancel Jobs .
CENTRIFUGES	
There is no balance card to allow centrifuging. Open a card drawer to introduce it and then press Retry .	There is no Grifols gel card available in the Erytra analyzer that can be used as a balance card for the centrifuge. To continue with the process open any drawer and insert a Grifols gel card. Press Retry .
It is not possible to process samples as there is no centrifuge available.	None of the centrifuges are available, so it is not possible to process samples.
Error during centrifuge positioning.	The system has detected an error during the movement of any of the centrifuges.
CONTAINERS	
There is an opened Waste drawer. Please, close it and then press Retry fluidic operation.	The system has detected an open waste drawer. Close the waste drawer that is open and press Retry to continue with the fluidic operation.

MESSAGE	DESCRIPTION
Required fluidic drawer is opened. Please, close it.	The system requires a fluidic drawer that is opened. Close it and select Ok to continue.
Waste container is full. Please open the drawer and empty it.	The system requires the waste containers and has detected that they are full. Open the drawer, empty the container and close it with the empty container inside. Then select Ok to continue.
DRAWERS	
Reagent drawer has been disabled. Please Restart the system as soon as possible.	The system has disabled a reagent drawer. It is necessary to restart the system as soon as possible.
The system was unable to open/close the drawer. Please, try again.	The system could not open or close the required drawer. Try again. If the problem persists, contact your local Grifols service representative.
The drawer is being accessed by probe, try it again later.	The system can not open the drawer because it is being used by the system. Wait for some seconds and try again. If the problem persists, contact your local Grifols service representative.
DECONTAMINATION	
The instrument is disabled because the decontamination process was not executed correctly. Please execute the decontamination process again in order to enable sample testing.	The decontamination process has not been done successfully. Ensure that the container placed in the System Solution B drawer is full and that the waste drawers are empty before repeating the procedure. If the problem persists, contact your local Grifols service representative.
The instrument did not complete the decontamination process. The system will not allow executing any workload until decontamination procedure is successfully completed.	The instrument has not finished the decontamination process. It is necessary to perform a successful decontamination procedure to continue executing any workload.

MESSAGE	DESCRIPTION
During decontamination process, the system has detected an opened Container Drawer . Please, close it and press retry.	The system has detected an open container during the decontamination. Please close it and press Retry .
Decontamination process cannot be executed if waste capacity is smaller than 8 L.	The system has detected that there is not enough capacity in the liquid waste container to perform the decontamination process.
Decontamination process could not be performed correctly. Please ensure that container placed in Solution B position is full and Waste bottles are empty and repeat the decontamination process.	The system has detected that there is not enough volume in the container placed in Solution B or that there is not enough capacity in the liquid waste container to perform the decontamination process. Check it and repeat the decontamination process.
DOORS	
Lower door is open. Close it to continue.	The system has detected that the lower door is open. The door must be closed in order to continue processing samples.
Upper door is open. Close it to continue.	The system has detected that the upper door is open. The door must be closed in order to continue processing samples.
ARM MOVEMENTS	
Error during the upper robotic arm movement. Please, inspect the instrument upper level Please choose one of the following options: Retry or Cancel Work .	The analyzer has detected that the upper robotic arm was not able to complete its movement. Press Retry or Cancel Work to continue.
Error during the lower robotic arm movement. Please, inspect the instrument middle level. Please choose one of the following options: Retry or Cancel Works .	There has been an error during the lower robotic arm movement. Please, inspect the instrument middle level. And choose one of the following options: Retry or Cancel Works .

MESSAGE	DESCRIPTION
STAT SAMPLES	
Missing cards affect one or more STAT samples. The workload will not be processed until these cards are introduced. Please, go to Missing Cards and Reagents to obtain more information.	The Grifols gel card(s) required by the analyzer affect one or more of the STAT samples. Go to the Missing Cards and Reagents screen of the analyzer and load the required Grifols gel cards.
Missing container affects one or more STAT samples. The workload will not be processed until this container is introduced. Please, go to Containers to obtain more information.	The container required by the analyzer affects one or more of the STAT samples. Please go to the Containers screen of the analyzer to identify the container that is needed.
Missing reagent kit affects one or more STAT samples. The workload will not be processed until these reagents are introduced. Please, go to Missing Cards and Reagents to obtain more information.	The reagent required by the analyzer affects one or more of the STAT samples. Go to the Missing Cards and Reagents screen and load the required reagent.
Missing sample affect one or more STAT samples. The petitions will not be processed until these samples are introduced. Please, go to Missing Samples to obtain more information.	The sample required by the analyzer affects one or more of the STAT samples. Go to the Missing Samples screen of the analyzer and load the sample required.
PROGRAMMING	
Due to a compatibility problem, only a subset of the request analysis will be programmed. Do you want to continue?	The system has detected compatibility issues and only some of the selected requests can be programmed. Press Yes or No to continue.
The selected combination of Recipient/Donor already exists. Press Ok if you want to edit the existing assignment or press Cancel to create a new one.	The system has detected that the Donor/Recipient combination selected already exists. Press Ok to modify this assignment or Cancel to create a new assignment.

MESSAGE	DESCRIPTION
This sample is not inside the analyzer.	The system has detected that the sample is not inside the analyzer.
LIS	
LIS error: Unable to initialize Host Database . Please contact your local Grifols service representative.	The system was not able to initialize the host Database . You should contact your local Grifols service representative.
LIS error: Unable to initialize file "Pocillos.def". Please contact your local Grifols service representative.	The system was unable to initialize the files "Pocillos.def". You should contact your local Grifols service representative.
LIS error: Unable to initialize file "Interp.def". Please contact your local Grifols service representative.	The system was unable to initialize the files "Interp.def". You should contact your local Grifols service representative.
LIS error: Unable to initialize file "Alias.def". Please contact your local Grifols service representative.	The system was unable to initialize the file "Alias.def". You should contact your local Grifols service representative.
LIS error: Unable to initialize file "Idioma_I.def". Please contact your local Grifols service representative.	The system was unable to initialize the file "Idioma_I.def". You should contact your local Grifols service representative.
LIS error: Unable to initialize file "DSC.def". Please contact your local Grifols service representative.	The system was unable to initialize the file "WadianaConnection.def". You should contact your local Grifols service representative.
LIS error: Unable to find "erytra_data" environment variable. Please contact your local Grifols service representative.	The system could not find the "erytra_data" environment variable. You should contact your local Grifols service representative.
LIS error: Timeout exceeded during a request to DianaServer. Please contact your local Grifols service representative.	The system has detected that the time limit in communicating with the DianaServer has been exceeded. You should contact your local Grifols service representative.

MESSAGE	DESCRIPTION
DianaServer request error. Please contact your local Grifols service representative.	The system has detected a communication error with the DianaServer. You should contact your local Grifols service representative.
LIS error: DLL has not been initialized. Please contact your local Grifols service representative.	The system has detected that the DLL has been not initialized. You should contact your local Grifols service representative.
LIS error: Gru not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected that the Gru has not been found in "DSC.def". You should contact your local Grifols service representative.
LIS error: Error loading template. Please contact your local Grifols service representative.	The system has detected an error while loading the template. You should contact your local Grifols service representative.
LIS error: Error loading XML. Please contact your local Grifols service representative.	The system has detected an error while loading XML. You should contact your local Grifols service representative.
LIS error: Result of segment not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected that the result of the segment is not found in "DSC.def". You should contact your local Grifols service representative.
LIS error: Microtube not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected that the microtube has not been found in "DSC.def". You should contact your local Grifols service representative.
LIS error: Result of microtube not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected that the microtube result has not been found in "DSC.def". You should contact your local Grifols service representative.
LIS error: Microtube incident not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected a microtube incident error in the "DSC.def". Contact your local Grifols service representative.

MESSAGE	DESCRIPTION
LIS error: Gru not found in "DSC.def". Please contact your local Grifols service representative.	The system has detected that the GRU has not been found in the "DSC.def". Contact your local Grifols service representative.
LIS error: Error loading template. Please contact your local Grifols service representative.	The system has detected an error when loading the template. Contact your local Grifols service representative.
LIS error: Error in "DSC_custom.def". Please contact your local Grifols service representative.	The system has detected an error in the "DSC_custom.def". Contact your local Grifols service representative.
LIS error: Error deleting batch. Please contact your local Grifols service representative.	The system has detected an error deleting the batch. Contact your local Grifols service representative.
Error when Exporting to LIS.	The system has detected an error exporting from LIS. Contact your local Grifols service representative.
Error when Importing from LIS.	The system has detected an error importing from LIS. Contact your local Grifols service representative.
LIS connection failed. LIS will not be available until the system is restarted. If problem persists, please contact your local Grifols service representative.	The system has detected a failure in the LIS connection. If problem persists after restarting the Erytra software, contact your local Grifols service representative.
LIS error: Microtube image not found. Please contact your local Grifols service representative.	The system has detected that the microtube image has not been found. Contact your local Grifols service representative.
CLOSING PROGRAM	
There are some unsaved changes. Are you sure to Cancel them?	The system has detected that there are some unsaved changed. Confirm if you want to cancel them.

MESSAGE	DESCRIPTION
There are tests in process. Are you sure you want to shutdown the Erytra? Before switching off, please ensure that all the reagents and diluents are removed from the instrument.	The system has detected that there are tests in process. Confirm if you want to shutdown the instrument. Before switching off, please ensure that all the reagents and diluents are removed from the inside of the instrument.
There are tests pending. Are you sure you want to shutdown the Erytra? Before switching off, please ensure that all the reagents and diluents are removed from the instrument.	The system has detected that there are pending tests. Confirm if you want to shutdown the instrument. Before switching off, please ensure that all the reagents and diluents are removed from the inside of the instrument.
There are tests waiting. Are you sure you want to shutdown the Erytra? Before switching off, please ensure that all the reagents and diluents are removed from the instrument.	The system has detected that there are tests waiting. Confirm if you want to shutdown the instrument. Before switching off, please ensure that all the reagents and diluents are removed from the inside of the instrument.
Are you sure you want to shutdown the Erytra? Before switching off, please ensure that all reagents and cards that require storage conditions between 2 - 8 °C are removed from the instrument.	Confirm if you want to shutdown the instrument. Before switching off, please ensure that all the reagents and diluents are removed from the inside of the instrument.
Movement error during shutdown. Due to this error, the system could not be rinsed and emptied. Please choose one of the following options. Cancel option will continue with shutdown procedures without rinsing and emptying the fluidic system.	The system has detected a movement error during shutdown. Due to this error, the system could not be rinsed and emptied. Please choose one of the following options. Cancel option will continue with shutdown procedures without rinsing and emptying the fluidic system.
BACK UP AND ARCHIVE PROCEDURES	
Error on Archive process.	The system has detected an error during the Archive process. If problem persist after restarting the Erytra software, contact your local Grifols service representative.

MESSAGE	DESCRIPTION
It is not possible to start the backup process because there are pending works.	It is not possible to start the backup process because there are pending works. Wait until the workload has finished and try to start the backup again. If problem persists, contact your local Grifols service representative.
Plug in an USB drive into any USB port and try again.	It is not possible to generate the Support Pack due to a problem with the USB drive. Try again to plug an USB drive into any of the USB ports. If problem persists, contact your local Grifols service representative.
Error generating the Support Pack. Please try again.	The system has detected an error during the Support Pack generation process. Try to start again the process. If problem persists, contact your local Grifols service representative.
REPORTS	
The report could not be shown due to other notifications. Please close all notifications and create the report again.	The report could not be shown due to other notifications. It is necessary to close all notifications and create the report again. If problem persists, contact your local Grifols service representative.
Please select a maximum of XXX entries.	The report could not be shown. It is necessary to select a lower number of fields and then create the report again. If problem persists, contact your local Grifols service representative.
The resulting Activity report will only show the first XXX entries.	The Activity report displayed will show the first XXX entries selected. For the non shown results, select them in a second time and create the report again.
The resulting Report will only show the first XXX entries.	The report displayed does not show all the results selected. It only displays the first XXX entries. Select the rest of results not shown in a second time and create the report again.

MESSAGE	DESCRIPTION
RESULTS FILTERING	
Some fields are required or wrong.	Some fields are required or wrong. Select the fields correctly in the filter and press Ok to confirm your selection.
Final date must be greater than initial.	The final date must be higher than the initial one in the calendar field of the filter. Select the range date appropriately.
MAINTENANCE	
Module activation is not allowed because there are pending works.	It is not possible to access the Module Activation tab because there are pending works. Wait until the workload has finished and then access it. If problem persists, contact your local Grifols service representative.
Probe activation is not allowed because fluidic system is disabled.	Probe activation is not allowed because fluidic system is disabled. If problem persists after restarting the system, contact your local Grifols service representative.

17.2 Opening the Lower Door

The lower door of the Erytra analyzer has a protective screw which prevents it from opening to protect users from accessing moving parts. However, it is possible to open the lower door for maintenance purposes, for cleaning or for the resolution of certain incidents. To open the lower door, stop the analyzer, take out the worktable and unscrew the protective screw.



WARNING: If the lower door needs to be opened, make sure that the analyzer is previously unplugged from the power supply.



WARNING: The screw of the lower door must always be kept tightened during the normal operation of the analyzer.

17.3 Manual Opening of the Lower Level Drawers

In the case of emergency or if the Erytra software does not respond, the analyzer has a manual opening system for the lower level drawers. To open the lower level drawers manually:

1. Locate the small latches situated on the base of the drawers.
2. Push the latch forward. The analyzer will open the drawer and allow access to the containers.

To close the drawers, push them in the normal manner.

18 Maintenance Services

Model:

Serial number:

Installation date:

MAINTENANCE SERVICES DONE			
DATE	TASK PERFORMED	PERFORMED BY AND DATE	SUPERVISOR AND REVIEW DATE

18.1 Decontamination Certificate

The following Decontamination Certificate must be filled in every time that the instrument is decontaminated. This Decontamination Certificate should be given to local Grifols service representative before any intervention takes place as required by local or site regulations.

DECONTAMINATION CERTIFICATE OF THE ERYTRA ANALYZER	
Serial Number	
Centre	
Decontamination was successfully completed according to the procedure described in Section 11.8	☐ PASS ☐ FAIL Comments:
Date	
Name and position	
Signature	
E-mail address	

19 Performance Studies

Erytra was tested in parallel at three (3) clinical sites in United States. The data obtained in the Method Comparison studies included 38307 comparison tests (39533 microtubes processed) in comparison testing.

More than 8000 individual unique specimens representing diverse population groups in broad geographic areas and composed of approximately 63% hospital patients and 37% blood donors were successfully tested with the Erytra.

	POSITIVE		NEGATIVE		TOTAL	
	N° of samples	Percent Agreement (Lower 95% CI)	N° of samples	Percent Agreement (Lower 95% CI)	N° of samples	Percent Agreement (Lower 95% CI)
Concordances Obtained for Blood Grouping Reagents						
Anti-A	1449	100.00% (99.79%)	1899	99.89% (99.67%)	3348	99.94% (99.81%)
Anti-B	602	100.00% (99.50%)	2746	100.00% (99.89%)	3348	100.00% (99.91%)
Anti-AB	1907	99.95% (99.75%)	1441	99.93% (99.67%)	3348	99.94% (99.81%)
Anti-D	2839	99.93% (99.78%)	509	99.61% (98.77%)	3348	99.88% (99.73%)
Anti-C	1169	100.00% (99.74%)	723	100.00% (99.59%)	1892	100.00% (99.84%)
Anti-E	560	100.00% (99.47%)	1332	99.85% (99.53%)	1892	99.89% (99.67%)
Anti-c	1543	100.00% (99.81%)	349	99.71% (98.65%)	1892	99.95% (99.75%)
Anti-e	1774	100.00% (99.83%)	118	100.00% (97.49%)	1892	100.00% (99.84%)
Anti-K	331	99.09% (97.67%)	3775	99.81% (99.65%)	4106	99.76% (99.59%)

	POSITIVE		NEGATIVE		TOTAL	
	N° of samples	Percent Agreement (Lower 95% CI)	N° of samples	Percent Agreement (Lower 95% CI)	N° of samples	Percent Agreement (Lower 95% CI)
Concordances Obtained for Anti-Human Globulin						
Ab. Screening	201	98.51% (96.19%)	1025	99.12% (98.47%)	1226	99.02% (98.42%)
Ab. Identification	612	99.35% (98.51%)	532	96.05% (94.37%)	1144	97.81% (96.96%)
IgG Crossmatch	185	100.00% (98.39%)	185	92.97% (89.06%)	370	96.49% (94.47%)
DAT	63	100.00% (95.36%)	251	97.61% (95.34%)	314	98.09% (96.26%)
Concordances Obtained for the Neutral Gel						
ABO Compatibility	191	98.95% (96.74%)	191	100.00% (98.44%)	382	99.48% (98.36%)
Reverse ABO Group A ₁	1859	100.00% (99.84%)	1427	99.44% (98.99%)	3286	99.76% (99.56%)
Reverse ABO Group A ₂	1773	99.27% (98.84%)	1460	99.59% (99.19%)	3233	99.41% (99.14%)
Reverse ABO Group B	2696	99.89% (99.71%)	590	99.66% (98.94%)	3286	99.85% (99.68%)

In addition, each of the three (3) Clinical Study Sites was assigned to perform reproducibility studies.

The study was done using the same Reproducibility Panel shipped to the three (3) Clinical Study sites in accordance to the following profile: 1 lot of test cards x 3 sites x 5 days (during 20 days) x 2 runs (am/pm) x 2 replicates using the Erytra at each site.

The study demonstrates that Erytra consistently obtained the expected results in all the repetitions.

In summary the results of this clinical evaluation support a conclusion that the Erytra is safe and effective for its routine use in immunohematology laboratories.

Erytra

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